DETERGENT

FORMULATIONS

ENCYCLOPEDIA

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178 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS – 2 (OPAQUE AND HIGH QUALITY)

179 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS – 1 (OPAQUE AND QUALITY)

180 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS – 2 (OPAQUE AND QUALITY)

181 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS – 1 (OPAQUE AND ECONOMICAL)

182 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS – 2 (OPAQUE AND ECONOMICAL)

183 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS (CLEAR AND HIGH QUALITY)

184 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO FORMULATION AND MANUFACTURING PROCESS (CLEAR AND QUALITY)

185 – CONCENTRATED LIQUID CARPET CLEANER SHAMPOO

FORMULATION AND MANUFACTURING PROCESS

(CLEAR AND ECONOMICAL)

186 – CONCENTRATED LIQUID CARPET DISINFECTANT AGENT

FORMULATION AND MANUFACTURING PROCESS

187 – CONCENTRATED POWDER CARPET DISINFECTANT AGENT

FORMULATION AND MANUFACTURING PROCESS

188 – CONCENTRATED LIQUID CARPET RINSE AGENT

FORMULATION AND MANUFACTURING PROCESS

189 – CONCENTRATED LIQUID SPOT REMOVER FOR CARPETS

FORMULATION AND MANUFACTURING PROCESS

190 – CONCENTRATED POWDER SPOT REMOVER FOR CARPETS

FORMULATION AND MANUFACTURING PROCESS

191 – CONCENTRATED LIQUID OIL REMOVER FOR CARPETS

FORMULATION AND MANUFACTURING PROCESS

LIQUID HAND SOAP

FORMULATIONS AND PRODUCTION METHODS

192 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS

(CLEAR AND HIGH QUALITY)

193 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS

(CLEAR AND QUALITY)

194 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS

(CLEAR AND ECONOMICAL)

195 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS

(OPAQUE AND HIGH QUALITY)

196 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS

(OPAQUE AND QUALITY)

197 – LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (OPAQUE AND ECONOMICAL)

198 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (CLEAR AND HIGH QUALITY)

199 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (CLEAR AND QUALITY)

200 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (CLEAR AND ECONOMICAL)

201 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (OPAQUE AND HIGH QUALITY)

202 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (OPAQUE AND QUALITY)

203 – ANTIBACTERIAL LIQUID HAND SOAP FORMULATION AND MANUFACTURING PROCESS (OPAQUE AND ECONOMICAL)

FOAM HANDWASH SOAPS FORMULATIONS AND PRODUCTION METHODS

204 – FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

205 – FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (QUALITY)

206 – FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

207 – FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

208 – FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (QUALITY)

209 – FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

210 – ANTIBACTERIAL FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

211 – ANTIBACTERIAL FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (QUALITY)

212 – ANTIBACTERIAL FOAM HANDWASH SOAP FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

213 – ANTIBACTERIAL FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

214 – ANTIBACTERIAL FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (QUALITY)

215 – ANTIBACTERIAL FOAM HANDWASH SOAP WITH MOISTURIZERS FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

ANTIBACTERIAL GEL HAND

FORMULATIONS AND PRODUCTION METHODS

216 – ANTIBACTERIAL GEL HAND FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

217 – ANTIBACTERIAL GEL HAND FORMULATION AND MANUFACTURING PROCESS (QUALITY)

218 – ANTIBACTERIAL GEL HAND FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

219 – ANTIBACTERIAL GEL HAND ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

220 – ANTIBACTERIAL GEL HAND ALCOHOL FREE

FORMULATION AND MANUFACTURING PROCESS

(QUALITY)

221 – ANTIBACTERIAL GEL HAND ALCOHOL FREE

FORMULATION AND MANUFACTURING PROCESS

(ECONOMICAL

GENERAL SURFACE CLEANER DETERGENT

FORMULATION AND PRODUCTION TECHNIQUES

222 – GENERAL SURFACE CLEANER DETERGENT

FORMULATION AND MANUFACTURING PROCESS

(HIGH QUALITY)

223 – GENERAL SURFACE CLEANER DETERGENT

FORMULATION AND MANUFACTURING PROCESS

(QUALITY)

224 – GENERAL SURFACE CLEANER DETERGENT

FORMULATION AND MANUFACTURING PROCESS

(ECONOMICAL)

225 – GENERAL SURFACE CLEANER DETERGENT ADDED CHLORINE

FORMULATION AND MANUFACTURING PROCESS

(HIGH QUALITY)

226 – GENERAL SURFACE CLEANER DETERGENT ADDED CHLORINE

FORMULATION AND MANUFACTURING PROCESS

(QUALITY)

227 – GENERAL SURFACE CLEANER DETERGENT ADDED CHLORINE

FORMULATION AND MANUFACTURING PROCESS

(ECONOMICAL)

228 – GENERAL SURFACE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

229 – GENERAL SURFACE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (QUALITY)

230 – GENERAL SURFACE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

GENERAL PURPOSE CLEANER DETERGENT FORMULATION AND PRODUCTION TECHNIQUES

231 – GENERAL PURPOSE CLEANER DETERGENT FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND HIGH QUALITY)

232 – GENERAL PURPOSE CLEANER DETERGENT FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND QUALITY)

233 – GENERAL PURPOSE CLEANER DETERGENT FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND ECONOMICAL)

234 – GENERAL PURPOSE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND HIGH QUALITY)

235 – GENERAL PURPOSE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND QUALITY)

236 – GENERAL PURPOSE CLEANER DETERGENT ALCOHOL FREE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND ECONOMICAL)

237 – GENERAL PURPOSE CLEANER DETERGENT ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND HIGH QUALITY)

238 – GENERAL PURPOSE CLEANER DETERGENT ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND QUALITY)

239 – GENERAL PURPOSE CLEANER DETERGENT ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND ECONOMICAL)

GLASSES AND WINDOWS CLEANER FORMULATIONS AND PRODUCTION METHODS

240 – GLASSES AND WINDOWS CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

241 – GLASSES AND WINDOWS CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

242 – GLASSES AND WINDOWS CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

SOAP SCUM REMOVER FORMULATIONS AND PRODUCTION TECHNIQUES

243 – SOAP SCUM REMOVER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

244 – SOAP SCUM REMOVER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

245 – SOAP SCUM REMOVER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

GENERAL DEGREASER FORMULATIONS AND PRODUCTION METHODS

246 – CONCENTRATED DEGREASER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

247 – CONCENTRATED DEGREASER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

248 – CONCENTRATED DEGREASER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

249 – DENSE DEGREASER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

250 – DENSE DEGREASER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

251 – DENSE DEGREASER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

252 – CONCENTRATED DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

253 – CONCENTRATED DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (QUALITY)

254 – CONCENTRATED DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

255 – DENSE DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

256 – DENSE DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (QUALITY)

257 – DENSE DEGREASER ADDED CHLORINE FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

GENERAL RUST REMOVER

FORMULATIONS AND PRODUCTION TECHNIQUES

258 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND HIGH QUALITY)

259 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND QUALITY)

260 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ALKALINE AND ECONOMICAL)

261 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ACIDIC AND HIGH QUALITY)

262 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ACIDIC AND QUALITY)

263 – CONCENTRATED RUST REMOVER FORMULATION AND MANUFACTURING PROCESS (ACIDIC AND ECONOMICAL)

264 – RUST REMOVER GEL FORMULATION AND MANUFACTURING PROCESS (ACIDIC AND HIGH QUALITY)

265 – RUST REMOVER GEL FORMULATION

AND MANUFACTURING PROCESS

(ACIDIC AND QUALITY)

266 – RUST REMOVER GEL FORMULATION

AND MANUFACTURING PROCESS

(ACIDIC AND ECONOMICAL)

ACIDIC WC CLEANERS

FORMULATION AND PRODUCTION METHODS

267 – ACIDIC WC CLEANER FORMULATION AND

MANUFACTURING PROCESS (HIGH QUALITY)

268 – ACIDIC WC CLEANER FORMULATION AND

MANUFACTURING PROCESS (QUALITY)

269 – ACIDIC WC CLEANER FORMULATION AND

MANUFACTURING PROCESS (ECONOMICAL)

270 – DENSE LIQUID ACIDIC WC CLEANER FORMULATION

AND MANUFACTURING PROCESS (HIGH QUALITY)

271 – DENSE LIQUID ACIDIC WC CLEANER FORMULATION

AND MANUFACTURING PROCESS (QUALITY)

272 – DENSE LIQUID ACIDIC WC CLEANER FORMULATION

AND MANUFACTURING PROCESS (ECONOMICAL)

DRAIN OPENER PRODUCTS

FORMULATION AND PRODUCTION PROCESS

273 – ACIDIC DRAINER OPENER FORMULATION AND

MANUFACTURING PROCESS (HIGH QUALITY)

274 – ACIDIC DRAINER OPENER FORMULATION AND

MANUFACTURING PROCESS (QUALITY)

275 – ACIDIC DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

276 – ALKALINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

277 – ALKALINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

278 – ALKANLINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

279 – DENSE ALKALINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

280 – DENSE ALKALINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

281 – DENSE ALKALINE DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

282 – DENSE ACIDIC DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

283 – DENSE ACIDIC DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

284 – DENSE ACIDIC DRAINER OPENER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

HOUSEHOLD DISINFECTANT PRODUCTS FORMULATION AND PRODUCTION TECHNIQUES

285 – HOUSEHOLD DISINFECTANT SPRAY CHLORINE BASED FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

286 – HOUSEHOLD DISINFECTANT SPRAY CHLORINE BASED FORMULATION AND MANUFACTURING PROCESS (QUALITY)

287 – HOUSEHOLD DISINFECTANT SPRAY CHLORINE BASED FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

288 – HOUSEHOLD DISINFECTANT SPRAY OXYGENE BASED FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

289 – HOUSEHOLD DISINFECTANT SPRAY OXYGENE BASED FORMULATION AND MANUFACTURING PROCESS (QUALITY)

290 – HOUSEHOLD DISINFECTANT SPRAY OXYGENE BASED FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

291 – HOUSEHOLD DISINFECTANT SPRAY GERMICIDE BASED FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

292 – HOUSEHOLD DISINFECTANT SPRAY GERMICIDE BASED FORMULATION AND MANUFACTURING PROCESS (QUALITY)

293 – HOUSEHOLD DISINFECTANT SPRAY GERMICIDE BASED FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

HOUSEHOLD METAL CLEANER AND POLISHER

FORMULATION AND PRODUCTION METHODS

294 – HOUSEHOLD METAL CLEANER AND POLISHER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(HIGH QUALITY)

295 – HOUSEHOLD METAL CLEANER AND POLISHER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(QUALITY)

296 – HOUSEHOLD METAL CLEANER AND POLISHER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(ECONOMICAL)

297 – HOUSEHOLD METAL CLEANER AND POLISHER CREAM
FORMULATION AND MANUFACTURING PROCESS
(HIGH QUALITY)

298 – HOUSEHOLD METAL CLEANER AND POLISHER CREAM
FORMULATION AND MANUFACTURING PROCESS
(QUALITY)

299 – HOUSEHOLD METAL CLEANER AND POLISHER CREAM
FORMULATION AND MANUFACTURING PROCESS
(ECONOMICAL)

HOUSEHOLD FLOOR CLEANER

FORMULATIONS AND PRODUCTION METHODS

300 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER
FORMULATION ANS MANUFACTURING PROCESS
(HIGH QUALITY)

301 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (QUALITY)

302 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (ECONOMICAL)

303 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (CHLORINE BASED AND HIGH QUALITY)

304 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (CHLORINE BASED AND QUALITY)

305 – CONCENTRATED LIQUID HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (CHLORINE BASED AND ECONOMICAL)

306 – CONCENTRATED POWDER HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (HIGH QUALITY)

307 – CONCENTRATED POWDER HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (QUALITY)

308 – CONCENTRATED POWDER HOUSEHOLD FLOOR CLEANER FORMULATION ANS MANUFACTURING PROCESS (ECONOMICAL)

HOUSEHOLD SHOWER CLEANER

FORMULATIONS AND PRODUCTION METHODS

309 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ACIDIC AND HIGH QUALITY)

310 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ACIDIC AND QUALITY)

311 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ACIDIC AND ECONOMICAL)

312 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ALKALINE AND HIGH QUALITY)

313 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ALKALINE AND QUALITY)

314 – CONCENTRATED HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ALKALINE AND ECONOMICAL)

315 – CONCENTRATED POWDER HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ACIDIC AND HIGH QUALITY)

316 – CONCENTRATED POWDER HOUSEHOLD SHOWER CLEANER
FORMULATION AND MANUFACTURING PROCESS
(ACIDIC AND QUALITY)

317 – CONCENTRATED POWDER HOUSEHOLD SHOWER CLEANER FORMULATION AND MANUFACTURING PROCESS (ACIDIC AND ECONOMICAL)

TOILET BOWL CLEANER

FORMULATIONS AND PRODUCTION TECHNIQUES

318 – TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

319 – TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

320 – TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

321 – THICKENED TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

322 – THICKENED TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

323 – THICKENED TOILET BOWL CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

324 – TOILET BOWL CLEANER AND GLOSS ENHANCER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

325 – TOILET BOWL CLEANER AND GLOSS ENHANCER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

326 – TOILET BOWL CLEANER AND GLOSS ENHANCER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

LIME AND RUST CLEANER

FORMULATIONS AND PRODUCTION METHODS

327 – LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

328 – LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

329 – LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

330 – THICKENED LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

331 – THICKENED LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

332 – THICKENED LIME AND RUST CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

TILE AND TUB CLEANER

FORMULATION AND PRODUCTION TECHNIQUES

333 – ACİDİC TILE AND TUB CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

334 – ACİDİC TILE AND TUB CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

335 – ACİDİC TILE AND TUB CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

336 – ALKALINE TILE AND TUB CLEANER FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

337 – ALKALINE TILE AND TUB CLEANER FORMULATION AND MANUFACTURING PROCESS (QUALITY)

338 – ALKALINE TILE AND TUB CLEANER FORMULATION
AND MANUFACTURING PROCESS (ECONOMICAL)

COMPUTER AND TV SCREEN CLEANER SPRAY FORMULATIONS AND PRODUCTION METHODS

339 – COMPUTER AND TV SCREEN CLEANER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(HIGH QUALITY)
340 – COMPUTER AND TV SCREEN CLEANER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(QUALITY)
341 – COMPUTER AND TV SCREEN CLEANER SPRAY
FORMULATION AND MANUFACTURING PROCESS
(ECONOMICAL)

AIR CONDITIONER CLEANER AND DISINFECTANT AGENT FORMULATION AND PRODUCTION METHODS

342 – ALKALINE AIR CONDITIONER CLEANER FORMULATION
AND MANUFACTURING PROCESS (HIGH QUALITY)
343 – ALKALINE AIR CONDITIONER CLEANER FORMULATION
AND MANUFACTURING PROCESS (QUALITY)
344 – ALKALINE AIR CONDITIONER CLEANER FORMULATION
AND MANUFACTURING PROCESS (ECONOMICAL)
345 – ACIDIC AIR CONDITIONER CLEANER FORMULATION
AND MANUFACTURING PROCESS (HIGH QUALITY)
346 – ACIDIC AIR CONDITIONER CLEANER FORMULATION
AND MANUFACTURING PROCESS (QUALITY)

347 – ACIDIC AIR CONDITIONER CLEANER FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

348 – AIR CONDITIONER DISINFECTANT AGENT FORMULATION AND MANUFACTURING PROCESS (HIGH QUALITY)

349 – AIR CONDITIONER DISINFECTANT AGENT FORMULATION AND MANUFACTURING PROCESS (QUALITY)

350 – AIR CONDITIONER DISINFECTANT AGENT FORMULATION AND MANUFACTURING PROCESS (ECONOMICAL)

SOME INGREDIENTS USED IN DETERGENT PRODUCTION

A

351 – ACETYLATED LANOLIN

353 – ACNIBIO AC

355 – ACNIBIO AP

357 – ALKYL POLYGLUCOSIDE

359 – AMARANTH (E 123)

361 – AMINO TRIMETHYLENE PHOSPHORIC ACID (ATMP)

363 – AMMONIUM HYDROXIDE (Aqueous Ammonia)

365 – AMMONIUM BICARBONATE

367 – AMMONIUM CHLORIDE

369 – AMMONIUM LAURYL ETHER SULPHATE (ALES)

371 – AMMONIUM LAURYL SULFATE (% 28)

373 – ACETIC ACID

375 – ACETONE

B

377 – BARITE (MINERAL)

379 – BEESWAX

381 – BENZALDEHYDE

383 – BENZALKONIUM CHLORIDE

385 – BENZYL ALCOHOL

387 – BENZOTRIAZOLE

389 – BIS PEG 1 DIMETHICONE

391 – BIS PEG 20 DIMETHICONE

393 – BORAX

395 – BORIC ACID

397 – BUTYLENE GLYCOL

399 – BUTYL HYDROXY TOLUENE (BHT)

C

401 – CALCITE

403 - CALCIUM HYDROXIDE

405 - CALCIUM HYPOCHLORIDE

407 – CALCIUMCHLORIDE

409 - CALCIUM SILICATE

411 - CALCIUM SULFATE

413 – CAPRYLYL CAPRYL GLUCOSIDE

415 – CARRAGEENAN

417 – CARBOXYMETHYL CELLULOSE (CMC)

419 – CARMOISINE

421 – CARNAUBA WAX

423 – CETEARETH 11

425 – CETEARETH 20

427 – CETRIMONIUM CHLORIDE

429 – CHOCOLATE BROWN HT

431 – CINNAMYL ALCOHOL

433 – CLIMBAZOLE

435 – COCAMIDOPROPYL BETAINE

437 – COCAMIDE DEA

439 – COCAMIDE MEA

441 – CYCLOMETHICONE

443 – CYCLOPENTASILOXANE

445 – C10 – 8 EO ALCOHOL ETHOXYLATE

447 – C10 – 3 EO ALCOHOL ETHOXYLATE

449 – C13 – 8 EO ALCOHOL ETHOXYLATE

451 – C 30 – 45 ALKYL METHICONE

D

453 – DECYL GLUCOSIDE

455 – DECYL OLEATE

457 – DECYLAMINE OXIDE

459 – DI C12 – 13 ALKYL MALATE

461 – DIDECYL DIMETHYL AMMONIUM CHLORIDE

463 – DIETHANOLAMINE DEA

465 – DIETHYLENE GLYCOL

467 – DIETHYLHYDROXYLAMINE (DEHA)

469 – DIMETHICONE

471 – DIOCTYL ETHER

473 – DIOCTYL DIMETHYL AMMONIUM CHLORIDE

475 – DIOCTYL SODIUM SULFOSUCCINATE

477 – DIPROPYLENE GLYCOL

479 – DISODIUM COCOAMPHODIACETATE

481 – DISODIUM LAURETH SULFOSUCCINATE

483 – DMDM HYDANTOIN

485 – D – PANTHENOL

487 – DOLOMITE

E

489 – EDTA

491 – ESTERQUAT

493 – ETHYL ALCOHOL

495 – ETHYL ACETATE

497 – ETHYL PARABEN

F

499 – FORMALDEHYDE

G

501 – GLYCERINE

503 – GLYCERYL MONOOLEATE

505 – GLYCERYL MONOSTEARATE

507 – GUAR GUM

H

509 – HYDRAZINE

511 – HYDROFLUORIC ACID

513 – HYDROCHLORIC ACID

515 – HYDROGEN PEROXIDE

517 – HYDROXYPROPYL METHYLCELLULOSE

I

519 - ISOPROPYL ALCOHOL (IPA)

521 – ISOBUTYL ALCOHOL

523 – ISOPROPYL ACETATE

L

525 - LINEAR ALKYLBENZENE SULFONIC ACID (LABSA)

527 - LAKE ALLURA RED AC

529 – LAKE AMARANTH

531 – LAKE PONCEAU 4R

533 – LAURYL BETAINE (POWDER)

535 – LAURYL BETAINE (LIQUID)

537 – LAURYL GLUCOSIDE

539 – LAURYLDIMETHYLAMINE OXIDE

541 – LIQUID PARAFFIN

M

543 – MAGNESIUM HYDROXIDE (POWDER)

545 – MAGNESIUM HYDROXIDE (LIQUID)

547 – MAGNESIUM CARBONATE

549 – MAGNESIUM CHLORIDE

551 – MAGNESIUM LAURETH SULFATE

553 – MAGNESIUM OXIDE

555 – MAGNESIUM SILICATE

557 – MAGNESIUM SULFATE MONOHYDRATE

559 – MENTHOL

561 – METHYLISOTHIAZOLINONE

563 – METHYL ALCOHOL

565 – METHYL BENZOATE

567 – METHYL ETHYL KETONE

569 – METHYL PARABEN

571 – MONOETHANOLAMINE (MEA)

573 – MONOETHYLENE GLYCOL (MEG)

575 – MONOPROPYLENE GLYCOL

N

577 – NITRIC ACID

579 – NONYLPHENOL ETHOXYLATE (NP: 6)

581 – NONYLPHENOL ETHOXYLATE (NP: 10)

O

583 – OCTYDODECANOL

585 – OLEIC ACID

P

587 – PEA GREEN

589 – PEG 400 POLYETHYLENE GLYCOL

591 – PEG 40 HYDROGENATED CASTOR OIL

593 – PEG 40 STEARATE

595 – PEG 75 – ETOXYLATED LANOLIN

597 – PHOSPHORIC ACID

599 – POLYSORBATE 20

601 – POLYQUATERNIUM 7

603 – POTASSIUM ACETATE

605 – POTASSIUM HYDROXIDE

607 – POTASSIUM CHLORIDE

609 – POTASSIUM SULFATE

611 – POTASSIUM TRIPOLYPHOSPHATE

S

613 – SODIUM ALPHA OLEFIN SULFONATE

615 – SODIUM HEGZAMETAPHOSPHATE (SHMP)

617 – SODIUM HYDROXIDE

619 – SODIUM HYPOCHLORIDE

621 – SODIUM CARBONATE

623 – SODIUM LAURYL ETHER SULFATE (SLES)

625 – SODIUM LAURYL SULFATE (SLS)

627 – SODIUM METASILICATE

629 – SODIUM PERCARBONATE

631 – SODIUM SULFATE

633 – SODIUM TRIPOLYPHOSPHATE (STPP)

635 – STEARIC ACID

637 – SULFAMIC ACID

639 – SULPHURIC ACID

641 – TARTARIC ACID

643 – TITANIUM DIOXIDE

645 – TETRAPOTASSIUM PYROPHOSPHATE

647 – TRIETHANOLAMINE (TEA)

U

649 – URE

DETERGENT FORMULATIONS ENCYCLOPEDIA

AUTOMATIC

DISHWASHER

PRODUCTS

FORMULATIONS

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(LIQUID AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCAMIDE DEA	2
2	TETRAPOTASSIUM PYROPHOSPHATE	20
3	EDTA	5
5	SODIUM HYPOCHLORIDE	7
6	SODIUM HYDROXIDE	5
7	PRESERVATIVE	0.100
8	WATER	60.990
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add cocamide DEA and edta. Continue to stirr. Then, add tetrapotassium pyrophosphate, sodium hydroxide, sodium hypochloride and preservative while stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(LIQUID AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CAPRYLYL SULFONATE	3
2	TETRAPOTASSIUM PYROPHOSPHATE	15
3	EDTA	5
5	SODIUM HYPOCHLORIDE	7
6	SODIUM HYDROXIDE	5
7	POTASSIUM TRIPOLYPHOSPHATE	4.900
8	PRESERVATIVE	0.100
9	WATER	60
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add sodium caprylyl sulfonate and edta. Continue to stirr. Then, add tetrapotassium pyrophosphate, potassium tripolyphosphate, sodium hydroxide, sodium hypochloride and preservative while stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(LIQUID AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CAPRYLYL SULFONATE	2
2	TETRAPOTASSIUM PYROPHOSPHATE	12
3	EDTA	5
5	SODIUM HYPOCHLORIDE	7
6	SODIUM HYDROXIDE	6
7	POTASSIUM TRIPOLYPHOSPHATE	4.900
8	PRESERVATIVE	0.100
9	WATER	63
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add sodium caprylyl sulfonate and edta. Continue to stirr. Then, add tetrapotassium pyrophosphate, potassium tripolyphosphate, sodium hydroxide, sodium hypochloride and preservative during constant stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(LIQUID AND QUALITY)

NO	INGREDIENTS	W/W
1	COCAMIDE MEA	2
2	TETRAPOTASSIUM PYROPHOSPHATE	10
3	EDTA	5
5	SODIUM HYPOCHLORIDE	5
6	SODIUM HYDROXIDE	5
7	POTASSIUM TRIPOLYPHOSPHATE	4.900
8	PRESERVATIVE	0.100
9	WATER	68
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add cocamide mea and edta. Continue to stirr. Then, add tetrapotassium pyrophosphate, potassium tripolyphosphate, sodium hydroxide, sodium hypochloride and preservative while stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(LIQUID AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM CARBONATE	5
2	TETRAPOTASSIUM PYROPHOSPHATE	5
3	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	SODIUM HYPOCHLORIDE	5
6	SODIUM HYDROXIDE	5
7	POTASSIUM TRIPOLYPHOSPHATE	2.900
8	PRESERVATIVE	0.100
9	WATER	75
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add sodium carbonate and STPP. Continue to stirr. Then, add tetrapotassium pyrophosphate, potassium tripolyphosphate, sodium hydroxide, sodium hypochloride and preservative while stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(LIQUID AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	POTASSIUM TRIPOLYPHOSPHATE	5
2	TETRAPOTASSIUM PYROPHOSPHATE	5
3	SODIUM HYDROXIDE	5
5	SODIUM HYPOCHLORIDE	5
6	PRESERVATIVE	0.100
7	WATER	79.900
	TOTAL	100

PROCESS: First, Put water into a stirred tank. Start stirring. Add sodium hydroxide and sodium hypochloride. Continue to stirr. Then, add rest of ingredients one by one during constant stirring. Dissolve all ingredients in water completely. If homogenous is enough. Mixing is stopped and process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(POWDER AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	3
2	TETRAPOTASSIUM PYROPHOSPHATE	30
3	SODIUM HYDROXIDE (SMALL GRANULAR)	15
5	SODIUM CUMENESULFONATE (POWDER)	2
6	EDTA	20
7	SODIUM META SILICATE	2
8	SODIUM CARBONATE (GRANULAR)	10
9	SODIUM XYLENE SULFONATE	3
10	SODIUM TRIPOLYPHOSPHATE (STPP)	15
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(POWDER AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	3
2	TETRAPOTASSIUM PYROPHOSPHATE	34
3	SODIUM HYDROXIDE (SMALL GRANULAR)	15
5	SODIUM CUMENESULFONATE (POWDER)	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	30
7	SODIUM META SILICATE	3
8	SODIUM CARBONATE (GRANULAR)	10
9	SODIUM XYLENE SULFONATE	3
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(POWDER AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	5
2	SODIUM XYLENE SULFONATE (POWDER)	3
3	SODIUM HYDROXIDE (SMALL GRANULAR)	20
5	SODIUM CUMENESULFONATE (POWDER)	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	36
7	SODIUM META SILICATE	2
8	SODIUM CARBONATE (GRANULAR)	10
9	EDTA	22
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(POWDER AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	2
2	SODIUM XYLENE SULFONATE (POWDER)	3
3	SODIUM HYDROXIDE (SMALL GRANULAR)	20
5	SODIUM CUMENESULFONATE (POWDER)	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	28
7	SODIUM META SILICATE	2
8	SODIUM CARBONATE (GRANULAR)	10
9	SODIUM HEGZA META PHOSPHATE	33
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 1
(POWDER AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	3
2	SODIUM XYLENE SULFONATE (POWDER)	3
3	SODIUM HYDROXIDE (SMALL GRANULAR)	20
5	SODIUM HEGZA META PHOSPHATE	20
6	SODIUM TRIPOLYPHOSPHATE (STPP)	20
7	SODIUM META SILICATE	2
8	SODIUM CARBONATE (GRANULAR)	10
9	SODIUM SULFATE	22
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

INDUSTRIAL
AUTOMATIC DISHWASHER DETERGENT – 2
(POWDER AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM DICHLOROISOCYANURATE	2
2	SODIUM XYLENE SULFONATE (POWDER)	3
3	SODIUM HYDROXIDE (SMALL GRANULAR)	20
5	SODIUM HEGZA META PHOSPHATE	15
6	SODIUM TRIPOLYPHOSPHATE (STPP)	28
7	SODIUM META SILICATE	2
8	SODIUM CARBONATE (GRANULAR)	10
9	SODIUM SULFATE	20
	TOTAL	100

PROCESS: First, All ingredients are solid. So, all ingredients put a mixer tank and stirr. Mixing is continued until mixture is homogenous. If mixture in the tank is homogenous, process is completed and mixture produced is packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

DETERGENT FORMULATIONS ENCYCLOPEDIA

AUTOMATIC

DISHWASHER

RINSE AIDS

FORMULATIONS

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CUMENESULFONATE	3
2	LAURYL GLUCOSIDE	1
3	ISOPROPYL ALCOHOL	10
5	CITRIC ACID	8
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	77.500
	TOTAL	100

PROCESS: Dissolve citric acid and lauryl glucoside in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CUMENESULFONATE	2
2	LAURYL GLUCOSIDE	1
3	ISOPROPYL ALCOHOL	8
5	CITRIC ACID	5
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	83.500
	TOTAL	100

PROCESS: Dissolve citric acid and lauryl glucoside in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	PHOSPHORIC ACID	5
2	LAURYL GLUCOSIDE	1
3	ISOPROPYL ALCOHOL	5
5	CITRIC ACID	3
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	85.500
	TOTAL	100

PROCESS: Dissolve citric acid in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(HIGH QUALITY AND ALCOHOL FREE)

NO	INGREDIENTS	W/W
1	SODIUM CUMENESULFONATE	7
2	LAURYL GLUCOSIDE	2
3	C11 OXO ALCOHOL 8 EO	7
5	CITRIC ACID	8
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	75.500
	TOTAL	100

PROCESS: Dissolve citric acid and lauryl glucoside in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(QUALITY AND ALCOHOL FREE)

NO	INGREDIENTS	W/W
1	SODIUM CUMENESULFONATE	5
2	LAURYL GLUCOSIDE	1
3	C11 OXO ALCOHOL 8 EO	7
5	CITRIC ACID	7
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	79.500
	TOTAL	100

PROCESS: Dissolve citric acid and lauryl glucoside in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

INDUSTRIAL
AUTOMATIC DISHWASHER RINSE AID
(ECONOMICAL AND ALCOHOL FREE)

NO	INGREDIENTS	W/W
1	SODIUM CUMENESULFONATE	3
2	LAURYL GLUCOSIDE	1
3	C11 OXO ALCOHOL 3 EO	3
5	CITRIC ACID	5
6	DYE (% 1 CONC.)	0.200
7	FRAGRANCE	0.200
8	PRESERVATIVE	0.100
9	WATER	87.500
	TOTAL	100

PROCESS: Dissolve citric acid and lauryl glucoside in water. Continue to stirr. Add rest of ingredients during constant stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

DETERGENT FORMULATIONS ENCYCLOPEDIA

AUTOMATIC

DISHWASHER

CLEANER

FORMULATIONS

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(LIQUID AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM METASILICATE	2
2	POTASSIUM TRIPOLYPHOSPHATE	20
3	SODIUM HYDROXIDE	5
5	SODIUM ALKANE SULFONATE	2
6	SODIUM HYPOCHLORIDE	11
7	PRESERVATIVE	0.100
8	WATER	59.900
	TOTAL	100

PROCESS: Dissolve sodium metasilicate and sodium alkane sulfonate in water. Continue to stirr. Add potassium tripolyphosphate, sodium hypochloride, preservative and sodium hydroxide while stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher.

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(LIQUID AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM METASILICATE	3
2	POTASSIUM TRIPOLYPHOSPHATE	12
3	SODIUM HYDROXIDE	5
5	SODIUM ALKANE SULFONATE	1
6	SODIUM HYPOCHLORIDE	5
7	PRESERVATIVE	0.100
8	WATER	73.900
	TOTAL	100

PROCESS: Dissolve sodium metasilicate and sodium alkane sulfonate in water. Continue to stirr. Add potassium tripolyphosphate, sodium hypochloride, preservative and sodium hydroxide while stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher.

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(LIQUID AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM CARBONATE	6
2	POTASSIUM TRIPOLYPHOSPHATE	8
3	SODIUM HYDROXIDE (% 50 SOL.)	5
4	SODIUM METASILICATE	2
5	PRESERVATIVE	0.100
6	WATER	78.900
	TOTAL	100

PROCESS: Dissolve sodium metasilicate and sodium carbonate in water. Continue to stirr. Add potassium tripolyphosphate, preservative and sodium hydroxide while stirring. If homogenous is enough, stirring is stopped and process is completed. The product producted can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher.

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(POWDER AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CARBONATE	10
2	SODIUM TRIPOLYPHOSPHATE (STPP)	8
3	SODIUM HYDROXIDE (GRANULAR)	20
4	SODIUM METASILICATE	2
5	TETRAPOTASSIUM PYROPHOSPHATE	20
6	EDTA	40
	TOTAL	100

PROCESS: Dissolve All ingredients are solid. So, add all ingredients in the order given and mix until homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher. This product is diluted and used like 3 – 5 %.

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(POWDER AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CARBONATE	15
2	EDTA	30
3	SODIUM HYDROXIDE (GRANULAR)	25
4	SODIUM METASILICATE	2
5	TETRAPOTASSIUM PYROPHOSPHATE	18
6	SODIUM BICARBONATE	10
	TOTAL	100

PROCESS: Dissolve All ingredients are solid. So, add all ingredients in the order given and mix until homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher. This product is diluted and used like 3 – 5 %.

INDUSTRIAL
AUTOMATIC DISHWASHER CLEANER
(POWDER AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM CARBONATE	10
2	EDTA	10
3	SODIUM HYDROXIDE (GRANULAR)	25
4	SODIUM METASILICATE	2
5	TETRAPOTASSIUM PYROPHOSPHATE	5
6	SODIUM HEGZA META PHOSPHATE	38
7	SODIUM SULFATE	10
	TOTAL	100

PROCESS: Dissolve All ingredients are solid. So, add all ingredients in the order given and mix until homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

This product is used for cleaning inside of industrial automatic dishwasher. This product is diluted and used like 3 – 5 %.

DETERGENT FORMULATIONS ENCYCLOPEDIA

HOUSEHOLD

DISHWASHER

PRODUCTS

FORMULATIONS

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT
(GEL AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	1
2	SODIUM BICARBONATE	5
3	SODIUM CARBONATE	5
4	SODIUM CAPRYLYL SULFONATE	2
5	SODIUM HYPOCHLORIDE	3
6	DEFOAMER	0.150
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.250
9	EDTA	7
10	BORAX PENTAHYDRATE	2
11	CALCIUM CHLORIDE	1
12	WATER	73.500
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Different preservative ingredients can be used.

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT
(GEL AND QUALITY)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	1
2	SODIUM BICARBONATE	5
3	MONOPROPYLENE GLYCOL	2
4	SODIUM CAPRYLYL SULFONATE	2
5	SODIUM HYPOCHLORIDE	3
6	DEFOAMER	0.150
7	PRESERVATIVE	0.100
8	FRANGRANCE	0.250
9	EDTA	5
10	BORAX PENTAHYDRATE	2
11	WATER	79.500
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Preservative and frangrance can be used different amount and quality.

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT
(GEL AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	0.75
2	SODIUM CARBONATE	3
3	MONOPROPYLENE GLYCOL	2
4	SODIUM HYDROXIDE (SMALL GRANULAR)	3
5	SODIUM BICARBONATE	3
6	SODIUM HYPOCHLORIDE	3
7	DYE (% 1 SOLN.)	0.150
8	PRESERVATIVE	0.100
9	FRANGRANCE	0.250
10	TETRAPOTASSIUM PYROPHOSPHATE	3
11	BORAX PENTAHYDRATE	2
12	WATER	79.750
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Preservative and frangrance can be used different amount and quality.

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT ADDED ENZYME
(GEL AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	0.75
2	SODIUM BICARBONATE	5
3	SODIUM CARBONATE	5
4	SODIUM CAPRYLYL SULFONATE	2
5	PROTEASE ENZYME	1
6	DEFOAMER	0.150
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.250
9	EDTA	7
10	BORAX PENTAHYDRATE	2
11	AMYLASE ENZYME	0.250
12	WATER	76.500
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Preservative, frangrance and dye can be used different amount and quality.

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT ADDED ENZYME
(GEL AND QUALITY)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	1
2	SODIUM BICARBONATE	5
3	MONOPROPYLENE GLYCOL	2
4	SODIUM CAPRYLYL SULFONATE	2
5	PROTEASE ENZYME	0.750
6	DEFOAMER	0.150
7	PRESERVATIVE	0.100
8	FRANGRANCE	0.250
9	EDTA	5
10	BORAX PENTAHYDRATE	1
11	AMYLASE ENZYME	0.250
12	WATER	82.500
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Preservative, dye and frangrance can be used different amount and quality.

HOUSEHOLD
AUTOMATIC DISHWASHER DETERGENT ADDED ENZYME
(GEL AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	HYDROXYETHYLCELLULOSE (HEC)	0.75
2	SODIUM CARBONATE	3
3	MONOPROPYLENE GLYCOL	2
4	SODIUM HYDROXIDE (SMALL GRANULAR)	3
5	SODIUM BICARBONATE	3
6	PROTEASE ENZYME	0.500
7	DYE (% 1 SOLN.)	0.150
8	PRESERVATIVE	0.100
9	FRANGRANCE	0.250
10	TETRAPOTASSIUM PYROPHOSPHATE	4
11	BORAX PENTAHYDRATE	1
12	AMYLASE ENZYME	0.150
13	WATER	82.100
	TOTAL	100

PROCESS: Put water into tank and heat 40 C. Add hydroxyethylcellulose and stirr until lumps is disappear. Add all ingredients in the order listed above during constant stirring. Continue mixing until mixture is homogenous. If homogenous is enough. Process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

Preservative, dye and frangrance can be used different amount and quality.

HOUSEHOLD
AUTOMATIC DISHWASHER SALT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CHLORIDE (LARGE GRANULAR)	90
2	TRISODYUM SITRATE	5
3	SODIUM TRIPOLYPHOSPHATE (STPP)	5
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. İf mixture is homogenous, process is completed. Mixture can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

HOUSEHOLD
AUTOMATIC DISHWASHER SALT
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM CHLORIDE (LARGE GRANULAR)	95
2	TRISODYUM SITRATE	5
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. İf mixture is homogenous, process is completed. Mixture can be packed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

HOUSEHOLD
AUTOMATIC DISHWASHER SALT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM CHLORIDE (LARGE GRANULAR)	100
	TOTAL	100

PROCESS: This ingredient is solıd above. So, This ingredient is packed directly. Process is completed.

NOTE: Different sodium chloride can be used for household automatic dishwasher as salt.

HOUSEHOLD
TABLET DISHWASHER DETERGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	2
2	TRISODIUM CITRATE	2
3	SODIUM DICHLOROISOCYANURATE	3
4	SODIUM CHLORIDE	10
5	TETRAPOTASSIUM PYROPHOSPHATE	10
6	SODIUM CARBONATE	10
7	SODIUM BICARBONATE	17.900
8	XANTHAN GUM	0.100
9	EDTA	20
10	SODIUM TRIPOLYPHOSPHATE	20
11	SODIUM HYDROXIDE	5
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

HOUSEHOLD
TABLET DISHWASHER DETERGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	1
2	TRISODIUM CITRATE	2
3	SODIUM DICHLOROISOCYANURATE	2
4	SODIUM CHLORIDE	10
5	TETRAPOTASSIUM PYROPHOSPHATE	10
6	SODIUM CARBONATE	10
7	SODIUM BICARBONATE	34.900
8	XANTHAN GUM	0.100
9	EDTA	10
10	SODIUM TRIPOLYPHOSPHATE	15
11	SODIUM HYDROXIDE	5
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

HOUSEHOLD
TABLET DISHWASHER DETERGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	1
2	TRISODIUM CITRATE	1
3	SODIUM DICHLOROISOCYANURATE	1
4	SODIUM SULFATE	5
5	SODIUM CHLORIDE	15
6	TETRAPOTASSIUM PYROPHOSPHATE	10
7	SODIUM CARBONATE	10
8	SODIUM BICARBONATE	31.900
9	XANTHAN GUM	0.100
10	EDTA	5
11	SODIUM TRIPOLYPHOSPHATE	10
12	SODIUM HYDROXIDE	10
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

HOUSEHOLD
TABLET DISHWASHER DETERGENT ADDED ENZYME
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	2
2	TRISODIUM CITRATE	2
3	SODIUM DICHLOROISOCYANURATE	3
4	SODIUM CHLORIDE	8.700
5	TETRAPOTASSIUM PYROPHOSPHATE	10
6	SODIUM CARBONATE	10
7	SODIUM BICARBONATE	17.900
8	XANTHAN GUM	0.100
9	EDTA	20
10	SODIUM TRIPOLYPHOSPHATE	20
11	SODIUM HYDROXIDE	5
12	PROTEASE ENZYME	1
13	AMYLASE ENZYME	0.300
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

HOUSEHOLD
TABLET DISHWASHER DETERGENT ADDED ENZYME
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	1
2	TRISODIUM CITRATE	2
3	SODIUM DICHLOROISOCYANURATE	2
4	SODIUM CHLORIDE	10
5	TETRAPOTASSIUM PYROPHOSPHATE	10
6	SODIUM CARBONATE	10
7	SODIUM BICARBONATE	34.900
8	XANTHAN GUM	0.100
9	EDTA	10
10	SODIUM TRIPOLYPHOSPHATE	15
11	SODIUM HYDROXIDE	4
12	PROTEASE ENZYME	0.750
13	AMYLASE ENZYME	0.250
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

HOUSEHOLD
TABLET DISHWASHER DETERGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM GLUCONATE	1
2	TRISODIUM CITRATE	1
3	SODIUM DICHLOROISOCYANURATE	1
4	SODIUM SULFATE	5
5	SODIUM CHLORIDE	15
6	TETRAPOTASSIUM PYROPHOSPHATE	10
7	SODIUM CARBONATE	10
8	SODIUM BICARBONATE	31.100
9	XANTHAN GUM	0.100
10	EDTA	5
11	SODIUM TRIPOLYPHOSPHATE	10
12	SODIUM HYDROXIDE	10
13	PROTEASE ENZYME	0.700
14	AMYLASE ENZYME	0.100
	TOTAL	100

PROCESS: All ingredients are solıd above. So, all ingredients are put ınto tank and mix. If mixture is homogenous, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market.

The above solid mixture is brought into a pressed tablet form by machines.

DISHWASHING

LIQUID

DETERGENT

FORMULATIONS

DISHWASHING
LIQUID DETERGENT – 1
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	13
2	COCAMIDOPROPYL BETAINE	5
3	AMMONIUM LAURYL SULFATE	5
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.5
6	TRIETHANOLAMINE	1
7	DYE (% 1 SOLN.)	0.100
8	FRAGRANGE	0.200
9	PRESERVATIVE	0.100
10	MAGNESIUM CHLORIDE	1.5 - 2
11	WATER	70.600
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide and triethanolamine for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT – 2
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	12
2	COCAMIDOPROPYL BETAINE	3
3	AMMONIUM LAURYL SULFATE	5
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.3
6	LAURAMINE OXIDE	2
7	DYE (% 1 SOLN.)	0.100
8	FRAGRANGE	0.200
9	PRESERVATIVE	0.100
10	MAGNESIUM CHLORIDE	1.6
11	WATER	73.700
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT – 1
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	10
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	6
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.3
6	DECYL GLUCOSIDE	1
7	DYE (% 1 SOLN.)	0.100
8	FRAGRANGE	0.200
9	PRESERVATIVE	0.100
10	MAGNESIUM CHLORIDE	1.6
11	WATER	75.700
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT – 2
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	10
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.25
6	DYE (% 1 SOL.)	0.100
7	FRAGRANGE	0.250
8	PRESERVATIVE	0.150
9	SODIUM CHLORIDE	1.5
10	WATER	75.700
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT – 1
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	COCAMIDE DEA	1
5	SODIUM HYDROXIDE	1
6	DYE (% 1 SOL.)	0.100
7	FRAGRANGE	0.250
8	PRESERVATIVE	0.150
9	SODIUM CHLORIDE	2.5
10	WATER	81
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust pH of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT – 2
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	6
2	COCAMIDE DEA	1
3	SODIUM LAURYL ETHER SULFATE (SLES)	5
4	SODIUM CHLORIDE	2.5
5	SODIUM HYDROXIDE	1
6	DYE (% 1 SOL.)	0.100
7	FRAGRANGE	0.250
8	PRESERVATIVE	0.150
10	WATER	84
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust pH of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT ADDED GLYCERINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	12
2	COCAMIDOPROPYL BETAINE	3
3	AMMONIUM LAURYL SULFATE	5
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.3
6	GLYCERINE	2
7	LAURAMINE OXIDE	2
8	DYE (% 1 SOLN.)	0.100
9	FRAGRANGE	0.200
10	PRESERVATIVE	0.100
11	MAGNESIUM CHLORIDE	1.6
12	WATER	71.700
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT ADDED GLYCERINE
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	10
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	6
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.3
6	GLYGERINE	2
7	DECYL GLUCOSIDE	1
8	DYE (% 1 SOLN.)	0.100
9	FRAGRANGE	0.200
10	PRESERVATIVE	0.100
11	MAGNESIUM CHLORIDE	1.6
12	WATER	73.700
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

DISHWASHING
LIQUID DETERGENT ADDED GLYCERINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	COCAMIDE DEA	1
5	SODIUM HYDROXIDE	1
6	GLYCERINE	1
7	DYE (% 1 SOL.)	0.100
8	FRAGRANGE	0.250
9	PRESERVATIVE	0.150
10	SODIUM CHLORIDE	2.5
11	WATER	80
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust pH of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

ANTIBACTERIAL
DISHWASHING LIQUID DETERGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	13
2	COCAMIDOPROPYL BETAINE	5
3	AMMONIUM LAURYL SULFATE	5
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.5
6	SODIUM CAPROYL / LAUROYL LACTYLATE	0.500
7	TRIETHANOLAMINE	1
8	DYE (% 1 SOLN.)	0.100
9	FRAGRANGE	0.200
10	PRESERVATIVE	0.100
11	MAGNESIUM CHLORIDE	1.5 - 2
12	WATER	70.100
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide and triethanolamine for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

ANTIBACTERIAL
DISHWASHING LIQUID DETERGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	10
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	6
4	COCAMIDE DEA	2
5	SODIUM HYDROXIDE	1.350
6	SODIUM CAPROYL / LAUROYL LACTYLATE	0.350
7	DECYL GLUCOSIDE	1
8	DYE (% 1 SOLN.)	0.100
9	FRAGRANGE	0.200
10	PRESERVATIVE	0.100
11	MAGNESIUM CHLORIDE	1.5
12	WATER	75.400
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

Sodium caproyl / lauroyl lactylate is an antibacterial agent.

ANTIBACTERIAL
DISWASHING LIQUID DETERGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	COCAMIDE DEA	1
5	SODIUM HYDROXIDE	1
6	SODIUM CAPROYL / LAUROYL LACTYLATE	0.250
7	DYE (% 1 SOL.)	0.100
8	FRAGRANGE	0.250
9	PRESERVATIVE	0.150
10	SODIUM CHLORIDE	2.250
11	WATER	81
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust pH of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

Sodium caproyl / lauroyl lactylate is an antibacterial agent.

ANTIBACTERIAL
DISHWASHING LIQUID DETERGENT ADDED GLYCERINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	12
2	COCAMIDOPROPYL BETAINE	3
3	AMMONIUM LAURYL SULFATE	5
4	COCAMIDE DEA	2
5	SODIUM CAPROYL / LAUROYL LACTYLATE	0.500
6	SODIUM HYDROXIDE	1.250
7	GLYCERINE	2
8	LAURAMINE OXIDE	2
9	DYE (% 1 SOLN.)	0.100
10	FRAGRANGE	0.200
11	PRESERVATIVE	0.100
12	MAGNESIUM CHLORIDE	1.5
13	WATER	70.350
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

Sodium caproyl / lauroyl lactylate is an antibacterial agent.

ANTIBACTERIAL
DISHWASHING LIQUID DETERGENT ADDED GLYCERINE
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	10
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	6
4	COCAMIDE DEA	2
5	SODIUM CAPROYL / LAUROYL LACTYLATE	0.350
6	SODIUM HYDROXIDE	1.250
7	GLYGERINE	2
8	DECYL GLUCOSIDE	1
9	DYE (% 1 SOLN.)	0.100
10	FRAGRANGE	0.200
11	PRESERVATIVE	0.100
12	MAGNESIUM CHLORIDE	2
13	WATER	73
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except magnesium chloride).continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust Ph of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

Sodium caproyl / lauroyl lactylate is an antibacterial agent.

ANTIBACTERIAL
DISHWASHING LIQUID DETERGENT ADDED GLYCERINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	COCAMIDOPROPYL BETAINE	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	COCAMIDE DEA	1
5	SODIUM CAPROYL / LAUROYL LACTYLATE	0.200
6	SODIUM HYDROXIDE	1
7	GLYCERINE	1
8	DYE (% 1 SOL.)	0.100
9	FRAGRANGE	0.250
10	PRESERVATIVE	0.150
11	SODIUM CHLORIDE	2
12	WATER	80.300
	TOTAL	100

PROCESS: Charge tank with deionized water. Add 1, 2, 3, 4 ingredients and mix. Add sodium hydroxide for adjusting Ph: 7. If Ph is right, add the rest of ingredients (except sodium chloride) . Continue to stirr until homogenous. Add magnesium chloride part to part for viscosity desired. Viscosity is enough, process is completed.

NOTE: In the above manufacturing process, the amounts of all ingredients can be changed depending on condition of market and values desired.

Sodium hydroxide is used to adjust pH of the mixture in the tank. So, the amount of sodium hydroxide is changed to Ph of mixture.

Sodium caproyl / lauroyl lactylate is an antibacterial agent.

DETERGENT FORMULATIONS ENCYCLOPEDIA

NATURAL

KITCHEN COUNTERTOP

CLEANER AND DISINFECTANT

PRODUCTS

FORMULATIONS

CONCENTRATED
KITCHEN COUNTERTOP DISINFECTANT AND CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	2
2	ISOPROPYL ALCOHOL (IPA)	20
3	ACETIC ACID	2
4	BENZALKONIUM CHLORIDE	5
5	WATER	71
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and alkyl polyglucoside. Continue to mix. Add acetic acid and benzalkonium chloride. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 20 or 1 / 25 for using

CONCENTRATED
KITCHEN COUNTERTOP DISINFECTANT AND CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	1
2	ETHYL ALCOHOL	15
3	ACETIC ACID	2
4	BENZALKONIUM CHLORIDE	5
5	WATER	77
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add ethyl alkol and alkyl polyglucoside. Continue to mix. Add acetic acid and benzalkonium chloride. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 15 or 1 / 20 for using

CONCENTRATED
KITCHEN COUNTERTOP DISINFECTANT AND CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	1
2	ETHYL ALCOHOL	10
3	ACETIC ACID	1
4	BENZALKONIUM CHLORIDE	3
5	WATER	85
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add ethyl alkol and alkyl polyglucoside. Continue to mix. Add acetic acid and benzalkonium chloride. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 10 or 1 / 15 for using

KITCHEN
COUNTERTOP DISINFECTANT AND CLEANER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	0.5
2	ISOPROPYL ALCOHOL (IPA)	5
3	ACETIC ACID	1
4	BENZALKONIUM CHLORIDE	2
5	WATER	91.5
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and lauramine oxide. Continue to mix. Add acetic acid and benzalkonium chloride. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly

KITCHEN
COUNTERTOP DISINFECTANT AND CLEANER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	1
2	ISOPROPYL ALCOHOL (IPA)	1
3	ACETIC ACID	1
4	BENZALKONIUM CHLORIDE	1
5	WATER	96
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and lauramine oxide. Continue to mix. Add acetic acid and benzalkonium chloride. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly

KITCHEN
COUNTERTOP DISINFECTANT AND CLEANER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	1
2	ISOPROPYL ALCOHOL (IPA)	1
3	ACETIC ACID	1
4	WATER	93
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and lauramine oxide. Continue to mix. Add acetic acid and mix well. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly.

DETERGENT FORMULATIONS ENCYCLOPEDIA

REFRIGERATOR SURFACE SPRAY CLEANER FORMULATIONS

REFRIGERATOR
SURFACE SPRAY CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	EDTA	3
2	ISOPROPYL ALCOHOL (IPA)	5
3	NONYLPHENOL ETOXYLATE 10	2
4	TRIETHANOLAMINE	1
5	WATER	89
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and nonylphenol etoxylate. Continue to mix. Add edta and triethanolamine. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

REFRIGERATOR
SURFACE SPRAY CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM TRIPOLYPHOSPHATE (STPP)	2
2	ISOPROPYL ALCOHOL (IPA)	3
3	NONYLPHENOL ETOXYLATE 10	2
4	TRIETHANOLAMINE	1
5	WATER	92
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and nonylphenol etoxylate. Continue to mix. Add sodium tripolyphosphate and triethanolamine. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

REFRIGERATOR
SURFACE SPRAY CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM TRIPOLYPHOSPHATE (STPP)	1
2	ISOPROPYL ALCOHOL (IPA)	3
3	NONYLPHENOL ETOXYLATE 10	1
4	SODIUM HYDROXIDE (% 50 SOLN)	1
5	WATER	94
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add isopropyl alkol and nonylphenol etoxylate. Continue to mix. Add sodium tripolyphosphate and sodium hydroxide. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

DETERGENT FORMULATIONS ENCYCLOPEDIA

OWEN

CLEANER

PRODUCTS

FORMULATIONS

NATURAL
OWEN CLEANER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	CASTOR OIL	10
2	COCOA OIL	1
3	SODIUM HYDROXIDE (SOLID)	2.5
4	NONYLPHENOL ETOXYLATE 10	0.5
5	GLYCERINE	30
6	WATER	56.5
	TOTAL	100

PROCESS: Put half of the water to tank. Add castor oil, sodium hydroxide, cocoa oil, glycerine and nonylphenol etoxylate. Start to stirr and heat to 70 – 80 C. Continue to mix during constant temperature. The mixture in the tank starts foaming. After that, foaming ends. At this point, mixture is removed from heat and add rest of water during allowing to cool. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 5 or 1 / 10 for using

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

NATURAL
OWEN CLEANER SPRAY
(QUALITY)

NO	INGREDIENTS	W/W
1	CASTOR OIL	10
2	SODIUM HYDROXIDE (SOLID)	2.5
3	GLYCERINE	25
4	WATER	62.5
	TOTAL	100

PROCESS: Put half of the water to tank. Add castor oil, sodium hydroxide and glycerine. Start to stirr and heat to 70 – 80 C. Continue to mix during constant temperature. The mixture in the tank starts foaming. After that, foaming ends. At this point, mixture is removed from heat and add rest of water during allowing to cool. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 5 or 1 / 10 for using

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

NATURAL
OWEN CLEANER SPRAY
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	CASTOR OIL	8
2	SODIUM HYDROXIDE (SOLID)	2
3	GLYCERINE	20
4	WATER	70
	TOTAL	100

PROCESS: Put half of the water to tank. Add castor oil, sodium hydroxide and glycerine. Start to stirr and heat to 70 – 80 C. Continue to mix during constant temperature. The mixture in the tank starts foaming. After that, foaming ends. At this point, mixture is removed from heat and add rest of water during allowing to cool. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The product produced above is concentrated. It must dilute as 1 / 3 or 1 / 5 for using

This product is used to be filled spray bottle and using directly. Also, this product can be used multi - purpose cleaner.

QWEN
CLEANER CREAM
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	5
2	SODIUM HYDROXIDE (SOLID)	1
3	SODIUM CARBONATE	5
4	SODIUM LAURYL ETHER SULFATE (SLES)	5
5	COCAMIDE DEA	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	5
7	CALCITE	37
8	WATER	40
	TOTAL	100

PROCESS: Put deionized water to tank. Start to mix. Add 1, 2, 4, 5 one by one. Continue to stirr. Check to mixture in the tank homogenous or not. Add sodium hydroxide and sodium carbonate in the order given and mix well. Add sodium tripolyphosphate and calcite during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER CREAM
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
2	SODIUM HYDROXIDE (SOLID)	1
3	SODIUM CARBONATE	6
4	SODIUM LAURYL ETHER SULFATE (SLES)	4
5	COCAMIDE DEA	1
6	SODIUM TRIPOLYPHOSPHATE (STPP)	3
7	CALCITE	41
8	WATER	40
	TOTAL	100

PROCESS: Put deionized water to tank. Start to mix. Add 1, 2, 4, 5 one by one. Continue to stirr. Check to mixture in the tank homogenous or not. Add sodium hydroxide and sodium carbonate in the order given and mix well. Add sodium tripolyphosphate and calcite during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER CREAM
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	3
2	SODIUM HYDROXIDE (SOLID)	1
3	SODIUM CARBONATE	4
4	SODIUM LAURYL ETHER SULFATE (SLES)	4
5	COCAMIDE DEA	1
6	SODIUM TRIPOLYPHOSPHATE (STPP)	2
7	CALCITE	45
8	WATER	40
	TOTAL	100

PROCESS: Put deionized water to tank. Start to mix. Add 1, 2, 4, 5 one by one. Continue to stirr. Check to mixture in the tank homogenous or not. Add sodium hydroxide and sodium carbonate in the order given and mix well. Add sodium tripolyphosphate and calcite during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER GEL
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	12
2	SODIUM HYDROXIDE (SOLID)	3
3	EDTA	5
4	SODIUM CARBONATE	2
5	OXO ALCOHOL C11 EO / PO	1
6	SODIUM TRIPOLYPHOSPHATE (STPP)	3
7	WATER	74
	TOTAL	100

PROCESS: Put deionized water to tank. Add sodium lauryl ether sulfate and mix. Dissolve sodium lauryl ether sulfate in water completely. Add oxo alcohol during constant stirring. Add rest of the ingredients in order given. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER GEL
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	SODIUM HYDROXIDE (SOLID)	3
3	EDTA	3
4	SODIUM CARBONATE	2
5	OXO ALCOHOL C11 EO / PO	1
6	SODIUM TRIPOLYPHOSPHATE (STPP)	2
7	WATER	79
	TOTAL	100

PROCESS: Put deionized water to tank. Add sodium lauryl ether sulfate and mix. Dissolve sodium lauryl ether sulfate in water completely. Add oxo alcohol during constant stirring. Add rest of the ingredients in order given. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER GEL
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	SODIUM HYDROXIDE (SOLID)	3
3	EDTA	3
4	SODIUM CARBONATE	2
5	WATER	82
	TOTAL	100

PROCESS: Put deionized water to tank. Add sodium lauryl ether sulfate and mix. Dissolve sodium lauryl ether sulfate in water completely. Add rest of the ingredients in order given. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	4
2	BUTYL GLYCOL	3
3	EDTA	2
4	SODIUM HYDROXIDE (SOLID)	1
5	WATER	90
	TOTAL	100

PROCESS: Put deionized water to tank. Add decyl glucoside and mix. Dissolve decyl glucoside in water completely. Add butyl glycol, sodium hydroxide and edta while mixing. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER SPRAY
(QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	3
2	BUTYL GLYCOL	2
3	SODIUM HEGZAMETAPHOSPHATE	2
4	SODIUM HYDROXIDE (SOLID)	1
5	WATER	92
	TOTAL	100

PROCESS: Put deionized water to tank. Add decyl glucoside and mix. Dissolve decyl glucoside in water completely. Add butyl glycol, sodium hydroxide and sodium hegzametaphosphate while mixing. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

QWEN
CLEANER SPRAY
(QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	2
2	ISOPROPYL ALCOHOL	2
3	SODIUM HYDROXIDE (GRANULAR)	2
4	WATER	94
	TOTAL	100

PROCESS: Put deionized water to tank. Add decyl glucoside and mix. Dissolve decyl glucoside in water completely. Add isopropyl and sodium hydroxide while mixing. Continue to mix until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is used to remover oil and dirty on surface of owen.

DETERGENT FORMULATIONS ENCYCLOPEDIA

LIQUID

LAUNDRY DETERGENT

FORMULATIONS

LIQUID
LAUNDRY DETERGENT FOR WHITE
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	C11 OXO ALCOHOL 3 EO	3
3	C 12 – 15 LINEAR ALCOHOL 9 EO	2
4	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	4
5	SODIUM HYPOCHLORIDE	2
6	SODIUM HYDROXIDE (SOLID)	0.5
7	ANTIFOAM AGENT	0.100
8	SODIUM TRIPOLYPHOSPHATE (STPP)	2
9	SODIUM METASILICATE	1
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	76.900
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR WHITE
(CLEAR AND QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	C11 OXO ALCOHOL 3 EO	2
3	EDTA	2
4	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	3
5	SODIUM HYPOCHLORIDE	2
6	SODIUM HYDROXIDE (SOLID)	0.5
7	ANTIFOAM AGENT	0.100
8	SODIUM TRIPOLYPHOSPHATE (STPP)	2
9	SODIUM METASILICATE	1
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	78.900
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR WHITE
(CLEAR AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHATE (LAS)	5
2	C11 OXO ALCOHOL 3 EO	2
3	C 12 – 15 LINEAR ALCOHOL 9 EO	2
4	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	2
5	SODIUM HYPOCHLORIDE	2
6	SODIUM HYDROXIDE (SOLID)	0.5
7	ANTIFOAM AGENT	0.100
8	SODIUM TRIPOLYPHOSPHATE (STPP)	2
9	SODIUM METASILICATE	1
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	82.900
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHATE (LAS)	8
2	C11 OXO ALCOHOL 8 EO	5
3	ANTIFOAM AGENT	0.250
4	SODIUM TRIPOLYPHOSPHATE (STPP)	3
5	SODIUM HYDROXIDE (SOLID)	0.5
6	SODIUM METASILICATE	1
7	PRESERVATIVE	0.150
8	DYE (% 1 SOL.)	0.150
9	FRAGRANCE	0.200
10	WATER	81.75
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(CLEAR AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULPHATE (SLES)	6
2	C11 OXO ALCOHOL 8 EO	5
3	ANTIFOAM AGENT	0.500
4	POTASSIUM TRIPOLYPHOSPHATE (STPP)	3
5	POTASSIUM HYDROXIDE (SOLID)	1
6	POTASSIUM METASILICATE	1
7	PRESERVATIVE	0.150
8	DYE (% 1 SOL.)	0.150
9	FRAGRANCE	0.200
10	WATER	83
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium hydroxide and potassium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(CLEAR AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHATE (LAS)	3
2	C11 OXO ALCOHOL 8 EO	5
3	ANTIFOAM AGENT	0.200
4	POTASSIUM TRIPOLYPHOSPHATE (STPP)	3
5	POTASSIUM HYDROXIDE (SOLID)	0.5
6	POTASSIUM METASILICATE	0.5
7	PRESERVATIVE	0.150
8	DYE (% 1 SOL.)	0.150
9	FRAGRANCE	0.200
10	WATER	87.300
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add linear alkylbenzene sulfate and oxo alcohol. Continue to mix. Add potassium hydroxide and potassium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR WHITE
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C12 – C13 LINEAR ALCOHOL 7 EO	4
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	14
3	TRIETHANOLAMINE (TEA)	1
4	SODIUM CITRATE	4
5	POTASSIUM METASILICATE	0.850
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	8
7	SODIUM HYPOCHLORIDE	2
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	65
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR WHITE
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	4
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	10
3	TRIETHANOLAMINE (TEA)	1
4	SODIUM CITRATE	4
5	POTASSIUM METASILICATE	0.850
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	6
7	SODIUM HYPOCHLORIDE	2
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	71
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR WHITE
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	4
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	6
3	POTASSIUM HYDROXIDE	0.5
4	SODIUM CARBONATE	1
5	POTASSIUM METASILICATE	0.350
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	6
7	SODIUM HYPOCHLORIDE	2
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	79
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C12 – C13 LINEAR ALCOHOL 7 EO	4
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	12
3	TRIETHANOLAMINE (TEA)	1
4	SODIUM CITRATE	2
5	POTASSIUM METASILICATE	0.850
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	8
7	SODIUM CARBONATE	2
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	69
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium carbonate while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	3
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	7
3	TRIETHANOLAMINE (TEA)	1
4	SODIUM CITRATE	2
5	POTASSIUM METASILICATE	0.850
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	4
7	SODIUM CARBONATE	1
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.200
13	WATER	80
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium carbonate while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT FOR COLOR
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	3
2	LINEAR DODECYL BENZENE SULFONATE (% 60)	5
3	POTASSIUM HYDROXIDE	0.5
4	SODIUM CARBONATE	1
5	POTASSIUM METASILICATE	0.450
6	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	3
7	SODIUM TRIPOLYPHOSPHATE (STPP)	2
8	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
9	ANTIFOAM AGENT	0.150
10	PRESERVATIVE	0.150
11	DYE (% 1 SOL.)	0.150
12	FRAGRANCE	0.100
13	WATER	84
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium tripolyphosphate while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	C12 – C15 LINEAR ALCOHOL 9 EO	2
3	C11 OXO ALCOHOL 3 EO	3
4	PROTEASE ENZYME	0.800
5	AMYLASE ENZYME	0.300
6	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	4
6	SODIUM HYPOCHLORIDE	2
8	SODIUM HYDROXIDE (SOLID)	0.5
9	ANTIFOAM AGENT	0.100
10	SODIUM TRIPOLYPHOSPHATE (STPP)	2
11	SODIUM METASILICATE	1
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	75.800
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(CLEAR AND QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	8
2	EDTA	2
3	C11 OXO ALCOHOL 3 EO	2
4	PROTEASE ENZYME	0.650
5	AMYLASE ENZYME	0.250
6	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	3
7	SODIUM HYPOCHLORIDE	2
8	SODIUM HYDROXIDE (SOLID)	0.5
9	ANTIFOAM AGENT	0.100
10	SODIUM TRIPOLYPHOSPHATE (STPP)	2
11	SODIUM METASILICATE	1
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	78
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(CLEAR AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHATE (LAS)	5
2	C12 – C15 LINEAR ALCOHOL 9 EO	2
3	C11 OXO ALCOHOL 3 EO	2
4	PROTEASE ENZYME	0.500
5	AMYLASE ENZYME	0.200
6	TRIPOTASSIUM POLYPHOSPHATE (TKPP)	2
7	SODIUM HYPOCHLORIDE	2
8	SODIUM HYDROXIDE (SOLID)	0.5
9	ANTIFOAM AGENT	0.100
10	SODIUM TRIPOLYPHOSPHATE (STPP)	2
11	SODIUM METASILICATE	1
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	82.200
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add sodium hydroxide and sodium metasilicate and mix well. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C12 – C13 LINEAR ALCOHOL 7 EO	4
2	TRIETHANOLAMINE (TEA)	1
3	LINEAR DODECYL BENZENE SULFONATE (% 60)	14
4	PROTEASE ENZYME	0.700
5	AMYLASE ENZYME	0.300
6	SODIUM CITRATE	4
7	POTASSIUM METASILICATE	0.850
8	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	8
9	SODIUM HYPOCHLORIDE	2
10	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
11	ANTIFOAM AGENT	0.150
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	64
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	4
2	MONOETHANOLAMINE (MEA)	1
3	LINEAR DODECYL BENZENE SULFONATE (% 60)	10
4	PROTEASE ENZYME	0.600
5	AMYLASE ENZYME	0.200
6	SODIUM CITRATE	4
7	POTASSIUM METASILICATE	0.850
8	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	6
9	SODIUM HYPOCHLORIDE	2
10	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
11	ANTIFOAM AGENT	0.150
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	70.200
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

LIQUID
LAUNDRY DETERGENT ADDED ENZYME FOR WHITE
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	C9 – C11 OXO ALCOHOL 8 EO	4
2	POTASSIUM HYDROXIDE	0.5
3	LINEAR DODECYL BENZENE SULFONATE (% 60)	6
4	PROTEASE ENZYME	0.500
5	AMYLASE ENZYME	0.200
6	SODIUM CARBONATE	1
7	POTASSIUM METASILICATE	0.350
8	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	6
9	SODIUM HYPOCHLORIDE	2
10	STYRENE/ACRYLİCCOPOLYMER (OPAQUE AGENT)	0.5
11	ANTIFOAM AGENT	0.150
12	PRESERVATIVE	0.150
13	DYE (% 1 SOL.)	0.150
14	FRAGRANCE	0.200
15	WATER	78.300
	TOTAL	100

PROCESS: Put deionized water to tank. Start to stirr. Add 1, 2, 3 ingredients one by one. Continue to mix. Add potassium tripolyphosphate and potassium metasilicate and mix well. Add sodium hypochloride while mixing. Add remainder of ingredients in order given one by one during constant stirring. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

The quality and amount of dye, fragrance and preservative depend on producer.

DETERGENT FORMULATIONS ENCYCLOPEDIA

POWDER

LAUNDRY DETERGENT

FORMULATIONS

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	SODIUM TRIPOLYPHOSPHATE	25
3	EDTA	5
4	SODIUM METASILICATE PENTAHYDRATE	5
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	SODIUM CARBONATE	10
9	SODIUM SULFATE	14
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	SODIUM TRIPOLYPHOSPHATE	20
3	EDTA	2
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	SODIUM CARBONATE	15
9	SODIUM SULFATE	19
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	20
2	SODIUM TRIPOLYPHOSPHATE	10
3	EDTA	3
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	8
8	SODIUM CARBONATE	20
9	SODIUM SULFATE	35
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	13
2	SODIUM TRIPOLYPHOSPHATE	20
3	EDTA	7
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	15
8	SODIUM CARBONATE	10
9	SODIUM SULFATE	19
10	FRAGRANCE – OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	10
2	SODIUM HEGZAMETAPHOSPHATE	15
3	POTASSIUM SILICATE	10
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	SODIUM CARBONATE	15
9	SODIUM SULFATE	24
10	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	8
2	SODIUM HEGZAMETAPHOSPHATE	10
3	POTASSIUM SILICATE	10
4	SODIUM METASILICATE PENTAHYDRATE	10
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	SODIUM CARBONATE	25
9	SODIUM SULFATE	26
10	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	60
2	SODIUM TRIPOLYPHOSPHATE	24
3	EDTA	2
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	OPTICAL BRIGHTENER (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED LAS

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	35
2	SODIUM TRIPOLYPHOSPHATE	25
3	EDTA	10
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	14
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	OPTICAL BRIGHTENER (QUANTITY BRIGHTENER)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(PHOSPHOROUS FREE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	ZEOLITE PROCESSED	28
3	SODIUM POLYACRYLATE (MW 4500 – 5500)	2
4	SODIUM METASILICATE PENTAHYDRATE	5
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	SODIUM CARBONATE	10
9	SODIUM SULFATE	14
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(PHOSPHOROUS FREE AND QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	ZEOLITE PROCESSED	19
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	SODIUM CARBONATE	15
9	SODIUM SULFATE	19
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED TOWER DETERGENT
(PHOSPHOROUS FREE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	20
2	ZEOLITE PROCESSED	9.5
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	8
8	SODIUM CARBONATE	20
9	SODIUM SULFATE	38
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(PHOSPHOROUS FREE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	14
2	ZEOLITE PROCESSED	30
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	10
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	14
8	SODIUM CARBONATE	10
9	SODIUM SULFATE	20
10	FRAGRANCE – OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(PHOSPHOROUS FREE AND QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	10
2	ZEOLITE PROCESSED	20
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	SODIUM CARBONATE	20.5
9	SODIUM SULFATE	23
10	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER
LAUNDRY DETERGENT BASED LAS
(PHOSPHOROUS FREE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	8
2	ZEOLITE PROCESSED	25
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	8
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	8
8	SODIUM CARBONATE	23.5
9	SODIUM SULFATE	26
10	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT
PHOSPHOROUS FREE

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	55
2	ZEOLITE PROCESSED	22
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM CITRATE	3
8	SODIUM PERCARBONATE	15
9	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
10	OPTICAL BRIGHTENER (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT BASED LAS
PHOSPHOROUS FREE

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	30
2	ZEOLITE PROCESSED	30
3	SODIUM POLYACRYLATE (MW 4500)	2
4	SODIUM METASILICATE PENTAHYDRATE	20
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	15
8	SODIUM SILICATE	2
9	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
10	OPTICAL BRIGHTENER (QUANTITY BRIGHTENER)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	SODIUM TRIPOLYPHOSPHATE	25
3	EDTA	5
4	SODIUM METASILICATE PENTAHYDRATE	4
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM CITRATE	4.9
8	SODIUM PERCARBONATE	10
9	PROTEASE ENZYME (POWDER)	0.07
10	SODIUM CARBONATE	10
11	AMYLASE ENZYME (POWDER)	0.02
12	SODIUM SULFATE	10
13	LIPASE ENZYME (POWDER)	0.01
14	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	SODIUM TRIPOLYPHOSPHATE	20
3	EDTA	2
4	SODIUM METASILICATE PENTAHYDRATE	2.880
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	PROTEASE ENZYME (POWDER)	0.06
9	SODIUM CARBONATE	15
10	AMYLASE ENZYME	0.01
11	SODIUM SULFATE	19
12	LIPASE ENZYME (POWDER)	0.050
13	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	20
2	SODIUM TRIPOLYPHOSPHATE	10
3	EDTA	3
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	8
8	PROTEASE ENZYME (POWDER)	0.05
9	SODIUM CARBONATE	20
10	AMYLASE ENZYME (POWDER)	0.01
11	SODIUM SULFATE	34.890
12	LIPASE ENZYME (POWDER)	0.050
13	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	13
2	SODIUM TRIPOLYPHOSPHATE	20
3	EDTA	7
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM CITRATE	3.9
8	SODIUM PERBORATE TETRAHYDRATE	15
9	PROTEASE ENZYME (POWDER)	0.08
10	SODIUM CARBONATE	10
11	AMYLASE ENZYME (POWDER)	0.01
12	SODIUM SULFATE	15
13	LIPASE ENZYME (POWDER)	0.01
14	FRAGRANCE – OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	10
2	SODIUM HEGZAMETAPHOSPHATE	15
3	POTASSIUM SILICATE	10
4	SODIUM METASILICATE PENTAHYDRATE	14.920
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM CITRATE	4
8	SODIUM PERCARBONATE	10
9	PROTEASE ENZYME (POWDER)	0.06
10	SODIUM CARBONATE	15
11	AMYLASE ENZYME (POWDER)	0.01
12	SODIUM SULFATE	20
13	LIPASE ENZYME (POWDER)	0.01
14	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	8
2	SODIUM HEGZAMETAPHOSPHATE	10
3	POTASSIUM SILICATE	10
4	SODIUM METASILICATE PENTAHYDRATE	10
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	PROTEASE ENZYME (POWDER)	0.02
9	SODIUM CARBONATE	25
10	AMYLASE ENZYME (POWDER)	0.01
11	SODIUM SULFATE	25.920
12	LIPASE ENZYME (POWDER)	0.05
13	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	60
2	SODIUM TRIPOLYPHOSPHATE	22
3	EDTA	2
4	SODIUM METASILICATE PENTAHYDRATE	2.5
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	SODIUM CITRATE	2.450
10	OPTICAL BRIGHTENER (QUANTITY SUFFICIENT)	Q.S.
11	PROTEASE ENYZME (POWDER)	0.04
12	AMYLASE ENZYME (POWDER)	0.01
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	35
2	SODIUM TRIPOLYPHOSPHATE	25
3	EDTA	10
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	13.950
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	OPTICAL BRIGHTENER (QUANTITY BRIGHTENER)	Q.S.
10	PROTEASE ENZYME (POWDER)	0.035
11	AMYLASE ENZYME (POWDER)	0.015
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(PHOSPHOROUS FREE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	ZEOLITE PROCESSED	28
3	SODIUM POLYACRYLATE (MW 4500 – 5500)	2
4	SODIUM METASILICATE PENTAHYDRATE	5
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	10
8	PROTEASE ENZYME (POWDER)	0.035
9	SODIUM CARBONATE	10
10	AMYLASE ENZYME (POWDER)	0.01
11	SODIUM SULFATE	13.935
12	LIPASE ENZYME (POWDER)	0.020
13	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(PHOSPHOROUS FREE AND QUALITY)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	30
2	ZEOLITE PROCESSED	19
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	11.950
8	SODIUM CARBONATE	15
9	PROTEASE ENZYME	0.025
10	SODIUM SULFATE	19
11	AMYLASE ENZYME	0.025
12	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDERLAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
(PHOSPHOROUS FREE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	20
2	ZEOLITE PROCESSED	9.5
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	8
	PROTEASE ENZYME	0.03
8	SODIUM CARBONATE	20
	AMYLASE ENZYME	0.01
9	SODIUM SULFATE	37.950
	LIPASE ENZYME	0.01
10	FRAGRANCE (SUFFICIENT QUANTITY)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(PHOSPHOROUS FREE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	14
2	ZEOLITE PROCESSED	30
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	10
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERBORATE TETRAHYDRATE	14
8	SODIUM CARBONATE	9.925
9	PROTEASE ENZYME	0.06
10	SODIUM SULFATE	20
11	AMYLASE ENZYME	0.015
12	FRAGRANCE – OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(PHOSPHOROUS FREE AND QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	10
2	ZEOLITE PROCESSED	20
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	15
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	10
8	PROTEASE ENZYME	0.05
9	SODIUM CARBONATE	20.5
10	AMYLASE ENZYME	0.02
11	SODIUM SULFATE	22.920
12	LIPASE ENZYME	0.010
13	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

POWDER LAUNDRY DETERGENT
BASED LAS WITH ENZYME
(PHOSPHOROUS FREE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	8
2	ZEOLITE PROCESSED	25
3	SODIUM POLYACRYLATE (MW 4500)	0.5
4	SODIUM METASILICATE PENTAHYDRATE	8.45
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	8
8	SODIUM CARBONATE	23
9	PROTEASE ENZYME	0.045
10	SODIUM SULFATE	26
11	AMYLASE ENZYME	0.050
12	FRAGRANCE, OPTICAL BRIGHTENER	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT
BASED TOWER DETERGENT WITH ENZYME
PHOSPHOROUS FREE

NO	INGREDIENTS	W/W
1	TOWER DETERGENT POWDER (% 40 ACTIVE)	55
2	ZEOLITE PROCESSED	25
3	SODIUM POLYACRYLATE (MW 4500)	1
4	SODIUM METASILICATE PENTAHYDRATE	3
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	14.925
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	OPTICAL BRIGHTENER (QUANTITY SUFFICIENT)	Q.S.
10	PROTEASE ENZYME	0.05
11	AMYLASE ENZYME	0.025
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance and antifoam agent are used to quantity sufficient.

CONCENTRATED
POWDER LAUNDRY DETERGENT BASED LAS WITH ENZYME
PHOSPHOROUS FREE

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	30
2	ZEOLITE PROCESSED	30
3	SODIUM POLYACRYLATE (MW 4500)	2
4	SODIUM METASILICATE PENTAHYDRATE	20
5	CARBOXYMETHYLCELLULOSE (CMC)	0.5
6	ANTIFOAM AGENT (POWDER)	0.5
7	SODIUM PERCARBONATE	16.950
8	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
9	OPTICAL BRIGHTENER (QUANTITY BRIGHTENER)	Q.S.
10	PROTEASE ENYZME	0.035
11	AMYLASE ENZYME	0.015
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Fragrance, optical brightener and antifoam agent are used to quantity sufficient.

DETERGENT FORMULATIONS ENCYCLOPEDIA

AUXILIARY

WASHING

PRODUCTS

FORMULATIONS

POWDER
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM METASILICATE	20
2	SODIUM CARBONATE	24
3	CARBOXYMETHYLCELLULOSE (CMC)	0.900
4	SODIUM TRIPOLYPHOSPHATE (STPP)	30
5	SODIUM HYDROXIDE (BEADS)	10
6	SODIUM PERCARBONATE	15
7	PROTEASE ENZYME	0.075
8	AMYLASE ENZYME	0.025
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM METASILICATE	20
2	SODIUM CARBONATE	23
3	CARBOXYMETHYLCELLULOSE (CMC)	0.420
4	SODIUM TRIPOLYPHOSPHATE (STPP)	25
5	SODIUM HYDROXIDE (BEADS)	15
6	SODIUM PERCARBONATE	16.5
7	PROTEASE ENZYME	0.050
8	AMYLASE ENZYME	0.030
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM METASILICATE	29
2	SODIUM CARBONATE	30
3	CARBOXYMETHYLCELLULOSE (CMC)	0.5
4	SODIUM TRIPOLYPHOSPHATE (STPP)	15
5	SODIUM HYDROXIDE (BEADS)	15
6	SODIUM PERCARBONATE	10.430
7	PROTEASE ENZYME	0.050
8	AMYLASE ENZYME	0.020
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	POTASSIUM HYDROXIDE	10
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	30
3	POTASSIUM SILICATE	5
4	CARBOXYMETHYLCELLULOSE (CMC)	0.200
5	PROTEASE ENZYME	0.100
6	AMYLASE ENZYME	0.050
7	WATER	64.650
	TOTAL	100

PROCESS: Put water and potassium tripolyphosphate in the tank. Start to stirr. Add the rest of ingredients while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(QUALITY)

NO	INGREDIENTS	W/W
1	POTASSIUM HYDROXIDE	10
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	20
3	POTASSIUM SILICATE	3
4	CARBOXYMETHYLCELLULOSE (CMC)	0.500
5	PROTEASE ENZYME	0.0700
6	AMYLASE ENZYME	0.0300
7	WATER	66.400
	TOTAL	100

PROCESS: Put water and potassium tripolyphosphate in the tank. Start to stirr. Add the rest of ingredients while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
AUXILIARY LAUNDRY WASHING
FOR SOLVING SOIL, OIL AND BLOOD
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	POTASSIUM HYDROXIDE	10
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	10
3	POTASSIUM SILICATE	3
4	CARBOXYMETHYLCELLULOSE (CMC)	0.300
5	PROTEASE ENZYME	0.050
6	AMYLASE ENZYME	0.050
7	SODIUM TRIPOLYPHOSPHATE (STPP)	3
8	WATER	73.600
	TOTAL	100

PROCESS: Put water and potassium tripolyphosphate in the tank. Start to stirr. Add the rest of ingredients while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

POWDER

LAUNDRY

BLEACHES

FORMULATIONS

POWDER
LAUNDRY BLEACHES CHLORINE BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	CALCIUM HYPOCHLORITE	60
2	SODIUM METASILICATE	20
3	SODIUM TRIPOLYPHOSPHATE (STPP)	10
4	SODIUM SULFATE	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
LAUNDRY BLEACHES CHLORINE BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	CALCIUM HYPOCHLORITE	45
2	SODIUM METASILICATE	25
3	SODIUM CARBONATE	20
4	SODIUM SULFATE	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
LAUNDRY BLEACHES CHLORINE BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	CALCIUM HYPOCHLORITE	35
2	SODIUM METASILICATE	35
3	SODIUM CARBONATE	30
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
LAUNDRY BLEACHES OXYGENE BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM PERCARBONATE	80
2	SODIUM METASILICATE	20
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
LAUNDRY BLEACHES OXYGENE BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM PERCARBONATE	60
2	SODIUM METASILICATE	20
3	SODIUM CARBONATE (GRANULAR)	20
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER
LAUNDRY BLEACHES OXYGENE BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM PERCARBONATE	50
2	SODIUM METASILICATE	10
3	SODIUM CARBONATE (GRANULAR)	20
4	SODIUM SULFATE	20
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
CHLORINE BLEACH FOR WASHING MACHINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	45
2	SODIUM CARBONATE	2
3	WATER	53
	TOTAL	100

PROCESS: Put water and sodium hypochlorite in the tank. Start to mix. Add sodium carbonate while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
CHLORINE BLEACH FOR WASHING MACHINE
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	35
2	SODIUM CARBONATE	3
3	WATER	62
	TOTAL	100

PROCESS: Put water and sodium hypochlorite in the tank. Start to mix. Add sodium carbonate while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
CHLORINE BLEACH FOR WASHING MACHINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	20
2	SODIUM CARBONATE	5
3	WATER	75
	TOTAL	100

PROCESS: Put water and sodium hypochlorite in the tank. Start to mix. Add sodium carbonate while mixing. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 1
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	40
2	SODIUM HYDROXIDE (% 50)	2
3	SODIUM LAURYL ETHER SULFATE (SLES)	3
4	WATER	55
	TOTAL	100

PROCESS: Put water and sles in the tank. Start to stirr. Dissolve sles in the water completely. Add sodium hydroxide and mix. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 2
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	38
2	SODIUM HYDROXIDE (% 50)	4
3	LAURAMIN OXIDE	3.5
4	EDTA	0.5
5	WATER	54
	TOTAL	100

PROCESS: Put water and lauramine oxide in the tank. Start to stirr. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and edta while stirring. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 1
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	32
2	SODIUM HYDROXIDE (% 50)	4
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	WATER	60
	TOTAL	100

PROCESS: Put water and sles in the tank. Start to stirr. Dissolve sles in the water completely. Add sodium hydroxide and mix. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 2
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	30
2	SODIUM HYDROXIDE (% 50)	4
3	LAURAMIN OXIDE	3.5
4	EDTA	1.5
5	WATER	61
	TOTAL	100

PROCESS: Put water and lauramine oxide in the tank. Start to stirr. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and edta while stirring. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 1
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	25
2	SODIUM HYDROXIDE (% 50)	4
3	SODIUM LAURYL ETHER SULFATE (SLES)	4
4	WATER	67
	TOTAL	100

PROCESS: Put water and sles in the tank. Start to stirr. Dissolve sles in the water completely. Add sodium hydroxide and mix. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

DENSE
CHLORINE BLEACH FOR WASHING MACHINE - 2
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORITE	25
2	SODIUM HYDROXIDE (% 50)	4.5
3	LAURAMIN OXIDE	4
4	EDTA	0.5
5	WATER	66
	TOTAL	100

PROCESS: Put water and lauramine oxide in the tank. Start to stirr. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and edta while stirring. Add sodium hypochlorite during constant mixing. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition and desired dense.

LIQUID
OXYGENE BLEACH FOR WASHING MACHINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	20
2	PHOSPHONATE HEDP	2
3	TRIETHANOLAMINE (TEA)	0.5
4	WATER	77.5
	TOTAL	100

PROCESS: Put water in the mixing tank. Add hydrogen peroxide Start to stirr. Add phosphonate hedp while mixing. Add triethanolamine for adjustment PH: 3,5 – 3,8. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
OXYGENE BLEACH FOR WASHING MACHINE
(QUALITY)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	15
2	PHOSPHONATE HEDP	1
3	TRIETHANOLAMINE (TEA)	0.5
4	WATER	83.5
	TOTAL	100

PROCESS: Put water in the mixing tank. Add hydrogen peroxide Start to stirr. Add phosphonate hedp while mixing. Add triethanolamine for adjustment PH: 3,5 – 3,8. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
OXYGENE BLEACH FOR WASHING MACHINE
(QUALITY)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	12
2	PHOSPHONATE HEDP	1
3	SODIUM HYDROXIDE (% 50 SOLN)	0.5
4	WATER	86.5
	TOTAL	100

PROCESS: Put water in the mixing tank. Add hydrogen peroxide Start to stirr. Add phosphonate hedp while mixing. Add sodium hydroxide for adjustment PH: 3,5 – 3,8. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
OXYGENE BLEACH FOR WASHING MACHINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	20
2	PHOSPHONATE HEDP	2
3	TRIETHANOLAMINE (TEA)	0.5
4	SODIUM LAURYL ETHER SULFATE (SLES)	3
5	SODIUM CHLORIDE	1.5
6	WATER	73
	TOTAL	100

PROCESS: Put water in the mixing tank. Add sodium lauryl ether sulfate. Start to stirr. Dissolve sles in the water completely. Add hydrogen peroxide and stirr. Add triethanolamine for adjustment PH: 3,5 – 3,8. Add sodium chloride part to part and slowly for dense. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
OXYGENE BLEACH FOR WASHING MACHINE
(QUALITY)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	15
2	PHOSPHONATE HEDP	2
3	TRIETHANOLAMINE (TEA)	0.5
4	SODIUM LAURYL ETHER SULFATE (SLES)	4
5	SODIUM CHLORIDE	1.5
6	WATER	77
	TOTAL	100

PROCESS: Put water in the mixing tank. Add sodium lauryl ether sulfate. Start to stirr. Dissolve sles in the water completely. Add hydrogen peroxide and stirr. Add triethanolamine for adjustment PH: 3,5 – 3,8. Add sodium chloride part to part and slowly for dense. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
OXYGENE BLEACH FOR WASHING MACHINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE	12
2	PHOSPHONATE HEDP	2
3	SODIUM HYDROXIDE (% 50 SOLN.)	0.5
4	SODIUM LAURYL ETHER SULFATE (SLES)	5
5	SODIUM CHLORIDE	1.5
6	WATER	79
	TOTAL	100

PROCESS: Put water in the mixing tank. Add sodium lauryl ether sulfate. Start to stirr. Dissolve sles in the water completely. Add hydrogen peroxide and stirr. Add sodium hydroxide for adjustment PH: 3,5 – 3,8. Add sodium chloride part to part and slowly for dense. Continue to stirr until homogenous and dense. Homogenous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

POWDER AND LIQUID

LIMESCALE

PREVENTER

FORMULATIONS

POWDER

LIMESCALE PREVENTER FOR WASHING MACHINE

(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HEGZA META PHOSPHATE	80
2	SODIUM METASILICATE	5
3	SODIUM TRIPOLYPHOSPHATE	5
4	SODIUM CARBONATE (GRANULAR)	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER

LIMESCALE PREVENTER FOR WASHING MACHINE

(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HEGZA META PHOSPHATE	60
2	SODIUM CARBONATE (GRANULAR)	30
3	SODIUM TRIPOLYPHOSPHATE	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

POWDER

LIMESCALE PREVENTER FOR WASHING MACHINE

(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM HEGZA META PHOSPHATE	50
2	SODIUM CARBONATE (GRANULAR)	50
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
LIMESCALE PREVENTER FOR WASHING MACHINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	70
2	SODIUM METASILICATE	1
3	EDTA	5
4	WATER	24
	TOTAL	100

PROCESS: Put the water in the mixing tank. Start to stirr. Add the rest of ingredients. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
LIMESCALE PREVENTER FOR WASHING MACHINE
(QUALITY)

NO	INGREDIENTS	W/W
1	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	60
2	SODIUM METASILICATE	2
3	WATER	38
	TOTAL	100

PROCESS: Put the water in the mixing tank. Start to stirr. Add the rest of ingredients. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
LIMESCALE PREVENTER FOR WASHING MACHINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	50
2	SODIUM METASILICATE	1
3	WATER	49
	TOTAL	100

PROCESS: Put the water in the mixing tank. Start to stirr. Add the rest of ingredients. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

FABRIC

SOFTENER

FORMULATIONS

FABRIC SOFTENER
QUARTAMIN BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	DISTEARYL DIMETHYL AMMONIUM CHLORIDE (QUARTAMIN) (CATIONIC SURFACTANT)	8
2	FRAGRANCE	8
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	83.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add quartamin and mix. Dissolve quartamin in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

FABRIC SOFTENER
ESTERQUAT BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	ESTERQUAT (CATIONIC SURFACTANT)	7
2	FRAGRANCE	8
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	83.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add esterquat and mix. Dissolve esterquat in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

FABRIC SOFTENER
QUARTAMIN BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	DISTEARYL DIMETHYL AMMONIUM CHLORIDE (QUARTAMIN) (CATIONIC SURFACTANT)	6
2	FRAGRANCE	7
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	86.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add quartamin and mix. Dissolve quartamin in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

FABRIC SOFTENER
ESTERQUAT BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	ESTERQUAT (CATIONIC SURFACTANT)	5
2	FRAGRANCE	7
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	87.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add esterquat and mix. Dissolve esterquat in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

FABRIC SOFTENER
QUARTAMIN BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	DISTEARYL DIMETHYL AMMONIUM CHLORIDE (QUARTAMIN) (CATIONIC SURFACTANT)	5
2	FRAGRANCE	5
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	89.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add quartamin and mix. Dissolve quartamin in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

FABRIC SOFTENER
ESTERQUAT BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	ESTERQUAT (CATIONIC SURFACTANT)	5
2	FRAGRANCE	5
3	PRESERVATIVE AGENT	0.150
4	DYE (% 1 SOLUTION)	0.100
5	WATER	89.75
	TOTAL	100

PROCESS: Charge tank with dionized water and heat 40 – 45 C. Add esterquat and mix. Dissolve esterquat in the water completely. Add the rest of ingredients during constant mixing and heating. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Preservative, fragrance and dye can be used different amount and quality by producer.

DETERGENT FORMULATIONS ENCYCLOPEDIA

CARPET

CLEANER

DETERGENT

FORMULATIONS

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 1
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	12
2	EDTA	10
3	SODIUM TRIPOLYPHOSPHATE	5
4	SODIUM CARBONATE	15
5	SODIUM SULFATE	58
6	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpet directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 2
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	25
2	EDTA	15
3	SODIUM HEGZA META PHOSPHATE	10
4	SODIUM CARBONATE	50
5	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add edta and sodium hegza meta phosphate while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpet directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 1
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	10
2	EDTA	6
3	SODIUM TRIPOLYPHOSPHATE	5
4	SODIUM CARBONATE	15
5	SODIUM SULFATE	64
6	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carper directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 2
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	20
2	EDTA	10
3	SODIUM HEGZA META PHOSPHATE	10
4	SODIUM CARBONATE	45
5	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
6	SODIUM SULFATE	15
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add edta, sodium sulfate and sodium hegza meta phosphate while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpet directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 1
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	6
2	EDTA	5
3	SODIUM TRIPOLYPHOSPHATE	3
4	SODIUM CARBONATE	26
5	SODIUM SULFATE	60
6	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpet directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT – 2
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	EDTA	10
3	SODIUM HEGZA META PHOSPHATE	10
4	SODIUM CARBONATE	35
5	FRAGRANCE (QUANTITY SUFFICIENT)	Q.S.
6	SODIUM SULFATE	30
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add edta, sodium sulfate and sodium hegza meta phosphate while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpet directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(CONTROLLED FOAM AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	SODIUM CARBONATE	35
3	SODIUM TRIPOLYPHOSPHATE (STPP)	15
4	SODIUM METASILICATE	15
5	SODIUM EDTA	10
6	SODIUM SULFATE	10
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add remainder of ingredients in order given while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(CONTROLLED FOAM AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	SODIUM CARBONATE	35
3	SODIUM TRIPOLYPHOSPHATE (STPP)	10
4	SODIUM METASILICATE	10
5	SODIUM EDTA	10
6	SODIUM SULFATE	20
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add remainder of ingredients in order given while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(CONTROLLED FOAM AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	SODIUM CARBONATE	30
3	SODIUM TRIPOLYPHOSPHATE (STPP)	10
4	SODIUM METASILICATE	10
5	SODIUM SULFATE	35
	TOTAL	100

PROCESS: Put sodium carbonate and sles a mixing tank. Start to stirr until mixture is granular. At this point. Making of mixing process is high speed and used special mixter (ribbon type blender) to make granular desired. Add remainder of ingredients in order given while stirring. Continue to stirr until homogenous and granular. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(OXYGEN BASED AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	35
2	SODIUM PERCARBONATE	25
3	SODIUM PERBORATE TETRAHYDRATE	10
4	SODIUM METASILICATE	10
5	SODIUM CARBONATE (GRANULAR)	10
6	SODIUM SULFATE	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(OXYGEN BASED AND QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	30
2	SODIUM PERCARBONATE	20
3	SODIUM PERBORATE TETRAHYDRATE	10
4	SODIUM METASILICATE	10
5	SODIUM CARBONATE (GRANULAR)	15
6	SODIUM SULFATE	15
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
POWDER CARPET CLEANER DETERGENT
(OXYGEN BASED AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULFONATE (LAS)	25
2	SODIUM PERCARBONATE	15
3	SODIUM PERBORATE TETRAHYDRATE	5
4	SODIUM METASILICATE	15
5	SODIUM CARBONATE (GRANULAR)	15
6	SODIUM SULFATE	25
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 1
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	TITAN DIOXIDE	0.250
3	SODIUM CARBONATE	5
4	TRIETHANOLAMINE	1
5	PHOSPHONATE HEDP (% 60)	10
6	SODIUM TRIPOLYPHOSPHATE (STPP)	2
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	WATER	71.5
	TOTAL	100

PROCESS: Put sles and titan dioxide in tank and mix until homogenous. Add the water while stirring. Take another cup and put phosphonate hedp. Then, add sodium carbonate and triethanolamine over phosphonate hedp part to part and slowly and carefully. Stirr until homogeneous. Then, pour the second cup into first tank and mix. Add the rest of ingredients one by one. Continue to stirr until homogenous mixture in the tank. Process is completed. PH of this mixture is 7 – 9 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 2
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	COCAMIDE DEA	2
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	6
4	SODIUM HYDROXIDE (SOLID)	0.650
5	TITAN DIOXIDE	0.250
6	EDTA	5
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	1.5
10	WATER	74.5
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 1
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	8
2	TITAN DIOXIDE	0.250
3	SODIUM CARBONATE	5
4	TRIETHANOLAMINE	1
5	PHOSPHONATE HEDP (% 60)	8
6	SODIUM TRIPOLYPHOSPHATE (STPP)	2
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	WATER	75.5
	TOTAL	100

PROCESS: Put sles and titan dioxide in tank and mix until homogenous. Add the water while stirring. Take another cup and put phosphonate hedp. Then, add sodium carbonate and triethanolamine over phosphonate hedp part to part and slowly and carefully. Stirr until homogeneous. Then, pour the second cup into first tank and mix. Add the rest of ingredients one by one. Continue to stirr until homogenous mixture in the tank. Process is completed. PH of this mixture is 7 – 9 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 2
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	8
2	COCAMIDE DEA	2
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
4	SODIUM HYDROXIDE (SOLID)	0.500
5	TITAN DIOXIDE	0.250
6	EDTA	4
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	1.5
10	WATER	79.650
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 1
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	TITAN DIOXIDE	0.250
3	SODIUM CARBONATE	3
4	TRIETHANOLAMINE	1
5	PHOSPHONATE HEDP (% 60)	5
6	SODIUM TRIPOLYPHOSPHATE (STPP)	1
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	WATER	83.150
	TOTAL	100

PROCESS: Put sles and titan dioxide in tank and mix until homogenous. Add the water while stirring. Take another cup and put phosphonate hedp. Then, add sodium carbonate and triethanolamine over phosphonate hedp part to part and slowly and carefully. Stirr until homogeneous. Then, pour the second cup into first tank and mix. Add the rest of ingredients one by one. Continue to stirr until homogenous mixture in the tank. Process is completed. PH of this mixture is 7 – 9 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO – 2
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	COCAMIDE DEA	1
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
4	SODIUM HYDROXIDE (SOLID)	0.500
5	TITAN DIOXIDE	0.250
6	EDTA	2
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	2
10	WATER	84.150
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	COCAMIDE DEA	2
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	6
4	SODIUM HYDROXIDE (SOLID)	0.650
5	DYE (%1 SOLUTION)	0.250
6	EDTA	5
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	1.5
10	WATER	74.5
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	8
2	COCAMIDE DEA	2
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
4	SODIUM HYDROXIDE (SOLID)	0.500
5	DYE (% 1 SOLUTION)	0.250
6	EDTA	4
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	1.5
10	WATER	79.650
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 2 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET CLEANER SHAMPOO
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	COCAMIDE DEA	1
3	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
4	SODIUM HYDROXIDE (SOLID)	0.500
5	DYE (% 1 SOLUTION)	0.250
6	EDTA	2
7	PRESERVATIVE	0.050
8	FRAGRANCE	0.050
9	SODIUM CHLORIDE	2
10	WATER	84.150
	TOTAL	100

PROCESS: Charge tank with deionized water. Start to stirr. Add 1, 2, 3, 4 ingredients in the tank during constant stirring. Dissolve all ingredients in water completely. Add the rest of ingredients one by one (excepted sodium chloride). Continue to mix. Add sodium chloride part to part for viscosity of mixture. İf homogeneous is enough. Process is completed. PH of this mixture is 7 – 8 approximately.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 3 / 100 with dionized water and using of washing carpets directly.

CONCENTRATED
LIQUID CARPET DISINFECTANT
AGENT

NO	INGREDIENTS	W/W
1	GERMICIDE (QUATERNARY COMPOUND)	5
2	ISOPROPYL ALCOHOL (IPA)	5
3	EDTA	5
4	WATER	85
	TOTAL	100

PROCESS: Charge tank with deionized water. Add all ingredients to tank and mix. Continue to stirr until mixture is homogeneous in the tank. Homogeneous is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 20 with dionized water and is filled to spray bottle for using.

CONCENTRATED
POWDER CARPET DISINFECTANT
AGENT

NO	INGREDIENTS	W/W
1	GERMICIDE (QUATERNARY COMPOUND)	5
2	LINEAR ALKYLBENZENE SULPHONATE (LAS)	5
3	EDTA	10
4	SODIUM META SILICATE	10
5	SODIUM CARBONATE (GRANULAR)	40
6	SODIUM TRIPOLYPHOSPHATE (STPP)	30
	TOTAL	100

PROCESS: All ingredients are powder or granular. So, all ingredients are put to mixing tank and mix well. Continue to stirr until mixture is homogeneous. Homogeneous is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 30 with dionized water and is filled to spray bottle for using.

CONCENTRATED
LIQUID CARPET RINSE
AGENT

NO	INGREDIENTS	W/W
1	CITRIC ACID	3
2	ISOPROPYL ALCOHOL (IPA)	2
3	PHOSPHONATE HEDP	3
4	ASETIC ACID	2
5	WATER	90
	TOTAL	100

PROCESS: Charge tank with deionized water. Add all ingredients to tank and mix. Continue to stirr until mixture is homogeneous in the tank. Homogeneous is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 25 with dionized water and is filled to spray bottle for using.

CONCENTRATED
LIQUID SPOT REMOVER
FOR CARPETS

NO	INGREDIENTS	W/W
1	HYDROGEN PEROXIDE (% 50)	8
2	NONYLPHENOL ETOXYLATE 10 EO (NP: 10)	2
3	PHOSPHONATE HEDP (% 60)	3
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	WATER	85
	TOTAL	100

PROCESS: Charge tank with deionized water. Add sodium tripolyphosphate and nonylphenol etoxylate. Start to stirr. Dissolve two ingredients in the water completely. Add hydrogen peroxide and phosphonate hedp order given while mixing. Continue to stirr until mixture is homogeneous in the tank. Homogeneous is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 10 with dionized water and is filled to spray bottle for using.

CONCENTRATED
LIQUID SPOT REMOVER
FOR CARPETS

NO	INGREDIENTS	W/W
1	SODIUM PERCARBONATE	50
2	SODIUM CARBONATE (GRANULAR)	20
3	SODIUM TRIPOLYPHOSPHATE (STPP)	20
4	EDTA	10
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 20 with dionized water and is filled to spray bottle for using.

CONCENTRATED
LIQUID OIL REMOVER
FOR CARPETS

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	10
2	SODIUM HYDROXIDE (% 50)	5
3	BUTYL GLYCOL	10
4	EDTA	5
5	WATER	70
	TOTAL	100

PROCESS: Charge tank with deionized water. Add to edta. Start to stirr. Add the rest of ingredients during constant mixing. Continue to stirr until mixture is homogeneous in the tank. Homogeneous is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 15 with dionized water and is filled to spray bottle for using.

DETERGENT FORMULATIONS ENCYCLOPEDIA

LIQUID

HAND SOAP

PRODUCTS

FORMULATIONS

LIQUID
HAND SOAP
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	LAURYL BETAINE	3
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2
5	CLYCERINE	1
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PRESERVATIVE AGENT	0.100
9	WATER	76.900
	TOTAL	100

PROCESS: Charge water. Add sles, glycerine, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5.5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
HAND SOAP
(CLEAR AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	LAURYL BETAINE	3
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2.5
5	GLYCERINE	0.900
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PRESERVATIVE AGENT	0.100
9	WATER	81.500
	TOTAL	100

PROCESS: Charge water. Add sles, glycerine, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
HAND SOAP
(CLEAR AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	7
2	LAURYL BETAINE	2
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2.5
5	PRESERVATIVE AGENT	0.100
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	WATER	86.400
	TOTAL	100

PROCESS: Charge water. Add sles, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIQUID
HAND SOAP
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	SODIUM LAURYL SULFATE (SLS)	3
3	COCAMİDOPROPYL BETAİNE	3
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (suffıcıently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
9	OPAQUE AGENT	0.250
10	SODIUM CHLORIDE (NaCl)	2
11	WATER	78
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

LIQUID
HAND SOAP
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	7
2	SODIUM LAURYL SULFATE (SLS)	3
3	COCAMİDOPROPYL BETAİNE	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (suffıcıently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
9	OPAQUE AGENT	0.250
10	SODIUM CHLORIDE (NaCl)	2.5
11	WATER	81.5
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

LIQUID
HAND SOAP
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	SODIUM LAURYL SULFATE (SLS)	2
3	COCAMİDOPROPYL BETAİNE	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (suffıcıently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
9	OPAQUE AGENT	0.250
10	SODIUM CHLORIDE (NaCl)	3
11	WATER	83
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

ANTIBACTERIAL
LIQUID HAND SOAP
(CLEAR AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	15
2	LAURYL BETAINE	3
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2
5	CLYCERINE	1
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PRESERVATIVE AGENT	0.100
9	TRICLOSAN	0.500
10	WATER	76.400
	TOTAL	100

PROCESS: Charge water. Add sles, glycerine, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, triclosan, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5.5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
LIQUID HAND SOAP
(CLEAR AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	LAURYL BETAINE	3
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2.5
5	GLYCERINE	0.900
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	PRESERVATIVE AGENT	0.100
9	TRICLOSAN	0.500
10	WATER	81
	TOTAL	100

PROCESS: Charge water. Add sles, glycerine, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, triclosan, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
LIQUID HAND SOAP
(CLEAR AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	7
2	LAURYL BETAINE	2
3	COCAMIDE DEA	2
4	SODIUM CHLORIDE (NaCl)	2.5
5	PRESERVATIVE AGENT	0.100
6	DYE, FRAGRANCE (QUANTITY SUFFICIENT)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	TRICLOSAN	0.400
9	WATER	86
	TOTAL	100

PROCESS: Charge water. Add sles, cocamide dea and lauryl betaine and mix. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. Add dye, triclosan, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
LIQUID HAND SOAP
(OPAQUE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	SODIUM LAURYL SULFATE (SLS)	3
3	COCAMİDOPROPYL BETAİNE	3
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (sufficiently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	TRICLOSAN	0.500
9	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
10	OPAQUE AGENT	0.250
11	SODIUM CHLORIDE (NaCl)	2
12	WATER	77.500
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine and triclosan. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

ANTIBACTERIAL
LIQUID HAND SOAP
(OPAQUE AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	7
2	SODIUM LAURYL SULFATE (SLS)	3
3	COCAMİDOPROPYL BETAİNE	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (sufficiently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	TRICLOSAN	0.500
9	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
10	OPAQUE AGENT	0.250
11	SODIUM CHLORIDE (NaCl)	2.5
12	WATER	81
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine and triclosan. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

ANTIBACTERIAL
LIQUID HAND SOAP
(OPAQUE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	SODIUM LAURYL SULFATE (SLS)	2
3	COCAMİDOPROPYL BETAİNE	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	PRESERVATIVE, DYE, FRAGRANCE (sufficiently)	q.s.
7	CITRIC ACID (FOR PH: 5,5) (QUALITY SUFFICIENT)	q.s.
8	TRICLOSAN	0.250
9	PEARLIZING AGENT (ZOHARPEARL 771)	0.750
10	OPAQUE AGENT	0.250
11	SODIUM CHLORIDE (NaCl)	3
12	WATER	82.750
	TOTAL	100

PROCESS: Charge water. Add 1.2.3.4 ingredients and mix. Dissolve all ingredients in water completely. Add glycerine and triclosan. Continue to stirr. Add dye, pearlizing agent, opaque agent, preservative and fragrance as desired and stirr well. Add citric acid and adjust Ph: 5,5. Add sodium chloride for viscosity. More can be added if higher viscosity is desired. Continue to stirr until homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

NOTE: Pearlizing agent and opaque agent sell different name, i.e. trade names

DETERGENT FORMULATIONS ENCYCLOPEDIA

FOAM

HANDWASH SOAP

PRODUCTS

FORMULATIONS

FOAM
HANDWASH SOAP
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	12
2	COCAMİDOPROPYL BETAİNE	10
3	COCAMIDE DEA	3
4	PRESERVATIVE	0.100
5	FRAGRANCE	0.150
6	WATER	74.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP
(QUALITY)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	10
2	COCAMİDOPROPYL BETAİNE	8
3	COCAMIDE DEA	3
4	PRESERVATIVE	0.100
5	FRAGRANCE	0.150
6	WATER	78.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	7
2	COCAMİDOPROPYL BETAİNE	8
3	COCAMIDE DEA	1
4	PRESERVATIVE	0.100
5	FRAGRANCE	0.150
6	WATER	83.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP WITH MOISTURIZERS
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	LAURYL BETAİNE	12
3	SODIUM LAURYL SULFATE (SLS)	3
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	MONOPROPYLENE GLYCOL	1
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	WATER	70.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP WITH MOISTURIZERS
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	8
2	LAURYL BETAİNE	8
3	SODIUM LAURYL SULFATE (SLS)	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	MONOPROPYLENE GLYCOL	1
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	WATER	77.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP WITH MOISTURIZERS
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	LAURYL BETAİNE	8
3	COCAMIDE DEA	2
4	MONOPROPYLENE GLYCOL	1
5	GLYCERINE	1
6	FRAGRANCE	0.100
7	PRESERVATIVE	0.100
8	WATER	81.80
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

ANTIBACTERIAL
FOAM HANDWASH SOAP
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	12
2	COCAMİDOPROPYL BETAİNE	10
3	COCAMIDE DEA	2
4	TRICLOSAN	0.500
5	FRAGRANCE	0.150
6	PRESERVATIVE	0.100
7	WATER	75.25
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Antibacterial foam handwash soap is used with special pomps.

ANTIBACTERIAL
FOAM HANDWASH SOAP
(QUALITY)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	10
2	COCAMİDOPROPYL BETAİNE	8
3	COCAMIDE DEA	1.750
4	TRICLOSAN	0.500
5	FRAGRANCE	0.150
6	PRESERVATIVE AGENT	0.100
7	WATER	79.5
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Antibacterial foam handwash soap is used with special pomps.

ANTIBACTERIAL
FOAM HANDWASH SOAP
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	AMMONIUM LAURYL ETHER SULFATE (ALES)	7
2	COCAMİDOPROPYL BETAİNE	7
3	COCAMIDE DEA	1
4	TRICLOSAN	0.250
5	FRAGRANCE	0.150
6	PRESERVATIVE	0.100
7	WATER	84.5
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Antibacterial foam handwash soap is used with special pomps.

ANTIBACTERIAL
FOAM HANDWASH SOAP WITH MOISTURIZERS
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	10
2	LAURYL BETAİNE	12
3	SODIUM LAURYL SULFATE (SLS)	3
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	MONOPROPYLENE GLYCOL	1
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	TRICLOSAN	0.500
10	WATER	68.25
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Antibacterial foam handwash soap is used with special pomps.

ANTIBACTERIAL
FOAM HANDWASH SOAP WITH MOISTURIZERS
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	8
2	LAURYL BETAİNE	8
3	SODIUM LAURYL SULFATE (SLS)	2
4	COCAMIDE DEA	2
5	GLYCERINE	1
6	MONOPROPYLENE GLYCOL	1
7	PRESERVATIVE	0.100
8	FRAGRANCE	0.150
9	TRICLOSAN	0.500
10	WATER	77.25
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

FOAM
HANDWASH SOAP WITH MOISTURIZERS
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	LAURYL BETAİNE	7
3	COCAMIDE DEA	2
4	MONOPROPYLENE GLYCOL	1
5	GLYCERINE	1
6	FRAGRANCE	0.100
7	PRESERVATIVE	0.100
8	TRICLOSAN	0.300
9	WATER	82.500
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients in the order given and stirr. Dissolve all ingredients in water completely. Continue to mix until mixture is clear and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Foam handwash soap is used with special pomps.

DETERGENT FORMULATIONS ENCYCLOPEDIA

ANTIBACTERIAL

GEL HAND

FORMULATIONS

ANTIBACTERIAL
GEL HAND
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	CARBOMER	0.75
2	TRIETHANOLAMINE (TEA)	0.75
3	TRICLOSAN	0.500
4	MONOPROPYLENE GLYCOL	2
5	ETHYL ALCOHOL	36
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	60
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
GEL HAND
(QUALITY)

NO	INGREDIENTS	W/W
1	CARBOMER	0.65
2	TRIETHANOLAMINE (TEA)	0.65
3	TRICLOSAN	0.200
4	MONOPROPYLENE GLYCOL	2
5	ETHYL ALCOHOL	25
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	71.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
GEL HAND
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	CARBOMER	0.600
2	TRIETHANOLAMINE (TEA)	0.600
3	TRICLOSAN	0.300
4	MONOPROPYLENE GLYCOL	1
5	ETHYL ALCOHOL	20
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	77.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
GEL HAND ALCOHOL FREE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	CARBOMER	0.85
2	TRIETHANOLAMINE (TEA)	0.85
3	TRICLOSAN	0.300
4	MONOPROPYLENE GLYCOL	5
5	GLYCERINE	3
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	90
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
GEL HAND ALCOHOL FREE
(QUALITY)

NO	INGREDIENTS	W/W
1	CARBOMER	0.75
2	TRIETHANOLAMINE (TEA)	0.75
3	TRICLOSAN	0.250
4	MONOPROPYLENE GLYCOL	3
5	GLYCERINE	2
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	93.25
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ANTIBACTERIAL
GEL HAND ALCOHOL FREE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	CARBOMER	0.65
2	TRIETHANOLAMINE (TEA)	0.65
3	TRICLOSAN	0.200
4	MONOPROPYLENE GLYCOL	2
5	GLYCERINE	2
6	FRAGRANCE, DYE, PRESERVATIVE	q.s.
7	WATER	94.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Add carbomer and stirr. Dissolve carbomer in the water completely. Wait the mixture one day. Next day, Add all ingredients one by one in the mixing tank (excepted triethanolamine). Continue to stirr until homogeneous and clear. Add triethanolamine as desired gelation.
If homogeneous, clear and gelation is enough. Process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

SURFACE CLEANER

DETERGENTS

FORMULATIONS

GENERAL
SURFACE CLEANER DETERGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	4
2	SODIUM LAURYL ETHER SULFATE (SLES)	4
3	SODIUM HYDROXIDE (% 48 SOLUTION)	1
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	ISOPROPYL ALCOHOL (IPA)	5
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	84
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add labsa, sles and sodium hydroxide. Continue to stirr. Add stpp and ipa while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
SURFACE CLEANER DETERGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONIC ACID (LABSA)	3
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	SODIUM HYDROXIDE (% 48 SOLUTION)	0.75
4	EDTA	2
5	ISOPROPYL ALCOHOL (IPA)	4
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	88.25
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add labsa, sles and sodium hydroxide. Continue to stirr. Add edta and ipa while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
SURFACE CLEANER DETERGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	EDTA	2
3	ISOPROPYL ALCOHOL (IPA)	2
4	DYE, FRAGRANCE, PRESERVATIVE	q.s.
5	WATER	93
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve sles in the water completely. Continue to stirr. Add edta and ipa while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
SURFACE CLEANER ADDED CHLORINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	5
2	SODIUM HYDROXIDE (% 48 SOLUTION)	2
3	COCO GLUCOSIDE	2
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	SODIUM HYPOCHLORIDE	3
6	WATER	87
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve sles in the water completely. Continue to stirr. Add stpp and sodium hydroxide while mixing. Add sodium hypochloride and coco glucoside and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL
SURFACE CLEANER ADDED CHLORINE
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	2
3	COCO GLUCOSIDE	1
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	SODIUM HYPOCHLORIDE	3
6	WATER	89
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve sles in the water completely. Continue to stirr. Add stpp and sodium hydroxide while mixing. Add sodium hypochloride and coco glucoside and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL
SURFACE CLEANER ADDED CHLORINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	2
3	SODIUM HYPOCHLORIDE	1
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	WATER	92
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve sles in the water completely. Continue to stirr. Add stpp and sodium hydroxide while mixing. Add sodium hypochloride and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL
SURFACE CLEANER ALCOHOL FREE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C 13 ALCOHOL ETHOXYLATE 8EO	6
2	SODIUM HYDROXIDE (% 48 SOLUTION)	0.5
3	EDTA	2
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	SODIUM XYLENE SULFONATE	2
6	DYE, PRESERVATIVE, FRAGRANCE	q.s.
7	WATER	87.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium xylene sulfonate and alcohol ethoxylate. Dissolve two ingredients in the water completely. Continue to stirr. Add stpp and edta while mixing. Add sodium hydroxide and stpp and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL
SURFACE CLEANER ALCOHOL FREE
(QUALITY)

NO	INGREDIENTS	W/W
1	C 13 ALCOHOL ETHOXYLATE 8EO	4
2	SODIUM HYDROXIDE (% 48 SOLUTION)	0.5
3	EDTA	2
4	SODIUM TRIPOLYPHOSPHATE (STPP)	2
5	SODIUM XYLENE SULFONATE	1
6	DYE, PRESERVATIVE, FRAGRANCE	q.s.
7	WATER	90.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add sodium xylene sulfonate and alcohol ethoxylate. Dissolve two ingredients in the water completely. Continue to stirr. Add stpp and edta while mixing. Add sodium hydroxide and stpp and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL
SURFACE CLEANER ALCOHOL FREE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	C 13 ALCOHOL ETHOXYLATE 8EO	4
2	SODIUM HYDROXIDE (% 48 SOLUTION)	1
3	SODIUM TRIPOLYPHOPHATE (STPP)	2
4	DYE, PRESERVATIVE, FRAGRANCE	q.s.
5	WATER	93
		.
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add alcohol ethoxylate. Continue to stirr. Add stpp while mixing. Add sodium hydroxide and mix well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

GENERAL

PURPOSE

CLEANERS

FORMULATIONS

GENERAL PURPOSE CLEANER DETERGENT (HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL GLUCOSIDE	5
2	LAURYL BETAINE	5
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	5
5	BUTYL GLYCOL	4
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	80
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add lauryl glucoside and lauryl betaine. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
PURPOSE CLEANER DETERGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	C12 – 15 ALCOHOL ETHOXYLATE 7EO	3
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	4
5	BUTYL GLYCOL	4
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	86
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add alcohol ethoxylate and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL PURPOSE CLEANER DETERGENT (ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	2
5	ISOPROPYL ALCOHOL (IPA)	2
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	90
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add coco glucoside and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL PURPOSE CLEANER ALCOHOL FREE (HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL GLUCOSIDE	5
2	LAURYL BETAINE	5
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	5
5	SODIUM METASILICATE	2
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	82
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add lauryl glucoside and lauryl betaine. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
PURPOSE CLEANER ALCOHOL FREE
(QUALITY)

NO	INGREDIENTS	W/W
1	C12 – 15 ALCOHOL ETHOXYLATE 7EO	3
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	4
5	SODIUM METASILICATE	1
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	89
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add alcohol ethoxylate and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL PURPOSE CLEANER ALCOHOL FREE (ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	2
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	2
5	POTASSIUM METASILICATE	1
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	92
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add coco glucoside and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL PURPOSE CLEANER ADDED CHLORINE (HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL GLUCOSIDE	5
2	LAURYL BETAINE	5
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	5
5	SODIUM HYPOCHLORIDE	4
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	80
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add lauryl glucoside and lauryl betaine. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
PURPOSE CLEANER ADDED CHLORINE
(QUALITY)

NO	INGREDIENTS	W/W
1	C12 – 15 ALCOHOL ETHOXYLATE 7EO	3
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	4
5	SODIUM HYPOCHLORIDE	3
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	87
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add alcohol ethoxylate and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

GENERAL
PURPOSE CLEANER ADDED CHLORINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	2
2	SODIUM LAURYL ETHER SULFATE (SLES)	2
3	POTASSIUM HYDROXIDE (SOLID)	1
4	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	2
5	SODIUM HYPOCHLORIDE	2
6	DYE, FRAGRANCE, PRESERVATIVE	q.s.
7	WATER	91
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add coco glucoside and sodium lauryl ether sulfate. Dissolve two ingredients in the water completely. Continue to stirr. Add potassium hydroxide and potassium tripolyphosphate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye, preservative and fragrance are changable by manufacturer.

DETERGENT FORMULATIONS ENCYCLOPEDIA

GLASS

&

WINDOWS

CLEANERS

FORMULATIONS

GLASSES AND WINDOWS CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP 10	1
2	SODIUM LAURYL ETHER SULFATE (SLES)	0.500
3	İSOPROPYL ALCOHOL	25
4	DYE (% 1 SOLUTION)	q.s.
5	WATER	73.5
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add the all ingredients one by one during constant mixing. Dissolve all ingredients in the water completely. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye is changable by manufacturer.

GLASSES AND WINDOWS CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP 10	0.500
2	SODIUM LAURYL ETHER SULFATE (SLES)	0.500
3	İSOPROPYL ALCOHOL	20
4	DYE (% 1 SOLUTION)	q.s.
5	WATER	79
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add the all ingredients one by one during constant mixing. Dissolve all ingredients in the water completely. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye is changable by manufacturer.

GLASSES AND WINDOWS CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE	1
2	ISOPROPYL ALCOHOL (IPA)	10
3	DYE (% 1 SOLUTION)	q.s.
4	WATER	89
	TOTAL	100

PROCESS: Charge water in mixing tank. Start to stirr. Add the all ingredients one by one during constant mixing. Dissolve all ingredients in the water completely. Mixing until mixture is homogenous and clear. If homogeneous and clear is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Used amounts and quality of dye is changable by manufacturer.

DETERGENT FORMULATIONS ENCYCLOPEDIA

SOAP SCUM

CLEANERS

FORMULATIONS

SOAP
SCUM REMOVER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP 6	3
2	ISOPROPYL ALCOHOL	2
3	PHOSPHONATE HEDP (% 60)	2
4	GLYCOLIC ACID	4
5	SULFAMIC ACID	5
6	WATER	84
	TOTAL	100

PROCESS: Take water in the mixing tank. Add isopropyl alcohol and nonylphenol etoxylate while mixing. Add water and stirr. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

SOAP
SCUM REMOVER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP 6	2
2	ISOPROPYL ALCOHOL	1
3	PHOSPHONATE HEDP (% 60)	2
4	GLYCOLIC ACID	4
5	SULFAMIC ACID	4
6	WATER	87
	TOTAL	100

PROCESS: Take water in the mixing tank. Add isopropyl alcohol and nonylphenol etoxylate while mixing. Add water and stirr. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

SOAP
SCUM REMOVER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP 6	1
2	ISOPROPYL ALCOHOL (IPA)	0.5
3	PHOSPHONATE HEDP (% 60)	1
4	GLYCOLIC ACID	4
5	WATER	93.5
	TOTAL	100

PROCESS: Take water in the mixing tank. Add isopropyl alcohol and nonylphenol etoxylate while mixing. Add water and stirr. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

GENERAL

DEGREASER

FORMULATIONS

CONCENTRATED
DEGREASER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	5
2	ISOPROPYL ALCOHOL	5
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	TETRAPOTASSIUM PYROPHOSPHATE	3
5	SODIUM XYLENE PHOSPHONATE	2
6	WATER	80
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 15 with dionized water and using directly.

CONCENTRATED
DEGREASER
(QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	3
2	ISOPROPYL ALCOHOL	3
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	TETRAPOTASSIUM PYROPHOSPHATE	2
5	SODIUM XYLENE PHOSPHONATE	2
6	WATER	85
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 10 with dionized water and using directly.

CONCENTRATED
DEGREASER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	ISOPROPYL ALCOHOL	3
3	SODIUM HYDROXIDE (% 48 SOLUTION)	6
4	TETRAPOTASSIUM PYROPHOSPHATE	2
5	SODIUM METASILICATE	1
6	WATER	85
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 5 with dionized water and using directly.

DENSE
DEGREASER AGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6.5
2	SODIUM CUMEN SULFONATE	2
3	SODIUM HYDROXIDE (% 48 SOLUTION)	10
4	EDTA	5
5	SODIUM METASILICATE	1
6	WATER	75.5
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
DEGREASER AGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	5
2	C11 ALCOHOL ETOXYLATE EO8	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	7
4	EDTA	3
5	SODIUM METASILICATE	1
6	WATER	83
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
DEGREASER AGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	SODIUM LAURYL SULFATE (SLS)	2
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	EDTA	2
5	SODIUM METASILICATE	1
6	WATER	87
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

CONCENTRATED
DEGREASER ADDED CHLORINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	5
2	ISOPROPYL ALCOHOL	5
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	TETRAPOTASSIUM PYROPHOSPHATE	3
5	SODIUM XYLENE PHOSPHONATE	2
6	SODIUM HYPOCHLORIDE	3
7	WATER	77
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add sodium hydroxide and stirr well. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 15 with dionized water and using directly.

CONCENTRATED
DEGREASER ADDED CHLORINE
(QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	3
2	ISOPROPYL ALCOHOL	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	4
4	TETRAPOTASSIUM PYROPHOSPHATE	2
5	SODIUM XYLENE PHOSPHONATE	2
6	SODIUM HYPOCHLORIDE	2
7	WATER	86
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add sodium hydroxide and mix. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 10 with dionized water and using directly.

CONCENTRATED
DEGREASER ADDED CHLORINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	ISOPROPYL ALCOHOL	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	4
4	TETRAPOTASSIUM PYROPHOSPHATE	2
5	SODIUM METASILICATE	1
6	SODIUM HYPOCHLORIDE	2
7	WATER	87
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and isopropyl alcohol while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 5 with dionized water and using directly.

DENSE
DEGREASER ADDED CHLORINE
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	6
2	COCO GLUCOSIDE	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	EDTA	5
5	SODIUM METASILICATE	1
6	SODIUM HYPOCHLORIDE	5
7	WATER	77
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add sodium hydroxide and stirr well. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
DEGREASER ADDED CHLORINE
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	4
2	C11 ALCOHOL ETOXYLATE EO8	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	5
4	SODIUM HYPOCHLORIDE	3
5	SODIUM METASILICATE	1
6	WATER	86
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add sodium hydroxide and mix well. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DENSE
DEGREASER AGENT ADDED CHLORINE
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	SODIUM LAURYL SULFATE (SLS)	2
3	SODIUM HYDROXIDE (% 48 SOLUTION)	4
4	SODIUM HYPOCHLORIDE	3
5	SODIUM METASILICATE	1
6	WATER	87
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve this ingredient in the water completely. Add sodium hydroxide while mixing. Add the rest of ingredients one by one while mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

RUST

REMOVER

FORMULATIONS

CONCENTRATED
RUST REMOVER
(ALKALINE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	2
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	25
3	POTASSIUM HYDROXIDE (SOLID)	20
4	TRIETHANOLAMINE (TEA)	5
5	SODIUM XYLENE PHOSPHONATE	3
6	POTASSIUM SILICATE	5
7	WATER	40
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and sodium xylene phosphonate while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 20 with dionized water and using directly.

CONCENTRATED
RUST REMOVER
(ALKALINE AND QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	2
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	20
3	POTASSIUM HYDROXIDE (SOLID)	15
4	TRIETHANOLAMINE (TEA)	5
5	C13 ALCOHOL ETOXYLATE 8EO	1
6	POTASSIUM SILICATE	5
7	WATER	52
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and alcohol etoxylate while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 15 with dionized water and using directly.

CONCENTRATED
RUST REMOVER
(ALKALINE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURYL GLUCOSIDE	1
2	POTASSIUM TRIPOLYPHOSPHATE (KTPP)	10
3	POTASSIUM HYDROXIDE (SOLID)	10
4	TRIETHANOLAMINE (TEA)	5
5	C13 ALCOHOL ETOXYLATE 8EO	1
6	POTASSIUM SILICATE	5
7	WATER	68
	TOTAL	100

PROCESS: Put water in the mixing tank. Add lauryl glucoside and alcohol etoxylate while mixing. Continue to stirr. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 10 with dionized water and using directly.

CONCENTRATED
RUST REMOVER
(ACIDIC AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETHOXYLATE NP 10	2
2	GLYCOLIC ACID	5
3	PHOSPHORIC ACID	30
4	PHOSPHONATE HEDP (% 60)	5
5	WATER	58
	TOTAL	100

PROCESS: Put water in the mixing tank. Add nonylphenol etoxylate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 15 with dionized water and using directly.

CONCENTRATED
RUST REMOVER
(ACIDIC AND QUALITY)

NO	INGREDIENTS	W/W
1	C 11 ALCOHOL ETOXYLATE 8 EO	2
2	GLYCOLIC ACID	3
3	PHOSPHORIC ACID	25
4	PHOSPHONATE HEDP (% 60)	5
5	WATER	65
	TOTAL	100

PROCESS: Put water in the mixing tank. Add alcohol etoxylate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 10 with dionized water and using directly.

CONCENTRATED
RUST REMOVER
(ACIDIC AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	C 11 ALCOHOL ETOXYLATE 8 EO	2
2	GLYCOLIC ACID	2
3	PHOSPHORIC ACID	20
4	PHOSPHONATE HEDP (% 60)	3
5	WATER	73
	TOTAL	100

PROCESS: Put water in the mixing tank. Add alcohol etoxylate while mixing. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated. So, it is diluted to 1 / 5 with dionized water and using directly.

CONCENTRATED
RUST REMOVER GEL
(ACIDIC AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETHOXYLATE NP 10	2
2	XANTHAN GUM	0.60
3	PHOSPHORIC ACID	35
4	PHOSPHONATE HEDP (% 60)	3
5	WATER	59.400
	TOTAL	100

PROCESS: Put water in the mixing tank. Add xanthan gum while mixing. Dissolve xanthan gum in the water completely. Add the rest of ingredients during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

More can be added **xanthan gum**, if higher gel is desired.

CONCENTRATED
RUST REMOVER GEL
(ACIDIC AND QUALITY)

NO	INGREDIENTS	W/W
1	C 11 ALCOHOL ETOXYLATE 8EO	2
2	XANTHAN GUM	0.70
3	PHOSPHORIC ACID	20
4	NITRIC ACID	5
5	WATER	72.30
	TOTAL	100

PROCESS: Put water in the mixing tank. Add xanthan gum while mixing. Dissolve xanthan gum in the water completely. Add the rest of ingredients one by one during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

More can be added **xanthan gum**, if higher gel is desired.

CONCENTRATED
RUST REMOVER GEL
(ACIDIC AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	C 11 ALCOHOL ETOXYLATE 8EO	2
2	XANTHAN GUM	0.85
3	PHOSPHORIC ACID	10
4	NITRIC ACID	5
5	WATER	82.15
	TOTAL	100

PROCESS: Put water in the mixing tank. Add xanthan gum while mixing. Dissolve xanthan gum in the water completely. Add the rest of ingredients one by one during constant mixing. Mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

More can be added **xanthan gum**, if higher gel is desired.

ACIDIC

WC CLEANER

FORMULATIONS

ACIDIC
WC CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	2.5
2	ISOPROPYL ALCOHOL	2.5
3	HYDROCHLORIC ACID	20
4	WATER	75
	TOTAL	100

PROCESS: Put nonylphenol etoxylate and isopropyl alcohol in the cup and mix. Dissolve nonylphenol etoxylate in the isopropyl alcohol. Add water while stirring. Add hydrochloric acid part to part and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ACIDIC
WC CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	2
2	ISOPROPYL ALCOHOL	1
3	HYDROCHLORIC ACID	15
4	WATER	82
	TOTAL	100

PROCESS: Put alcohol etoxylate and isopropyl alcohol in the cup and mix. Dissolve alcohol etoxylate in the isopropyl alcohol. Add water while stirring. Add hydrochloric acid part to part and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ACIDIC
WC CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	1
2	ISOPROPYL ALCOHOL	1
3	HYDROCHLORIC ACID	10
4	WATER	88
	TOTAL	100

PROCESS: Put alcohol etoxylate and isopropyl alcohol in the cup and mix. Dissolve alcohol etoxylate in the isopropyl alcohol. Add water while stirring. Add hydrochloric acid part to part and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ACIDIC LIQUID WC CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	2.5
2	ISOPROPYL ALCOHOL	2
3	HYDROCHLORIC ACID	20
4	ACIDIC THICKENER	0.500
5	WATER	75
	TOTAL	100

PROCESS: Put nonylphenol etoxylate and isopropyl alcohol in the cup and mix. Dissolve nonylphenol etoxylate in the isopropyl alcohol completely. Add water while stirring. Add hydrochloric acid part to part and mix well. Add acidic thickener for thickener. Continue to mixing until mixture is homogenous and thickener. Homogeneous and thickener is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener is sold as trade name. Therefore it can be acidic thickener from market.

THICKENED
ACIDIC LIQUID WC CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM ALKAN SULFONATE	1
2	ISOPROPYL ALCOHOL	1
3	HYDROCHLORIC ACID	15
4	ACIDIC THICKENER	0.500
5	WATER	82.5
	TOTAL	100

PROCESS: Put sodium alkane sulfonate and isopropyl alcohol in the cup and mix. Dissolve sodium alkane sulphonate in the isopropyl alcohol completely. Add water while stirring. Add hydrochloric acid part to part and mix well. Add acidic thickener. Continue to mixing until mixture is homogenous and thickener. Homogeneous and thickener is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener is sold as trade name. Therefore it can be acidic thickener from market.

THICKENED
ACIDIC LIQUID WC CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM ALKAN SULFONATE	1
2	ISOPROPYL ALCOHOL	0.5
3	HYDROCHLORIC ACID	10
4	ACIDIC THICKENER	0.500
5	WATER	88
	TOTAL	100

PROCESS: Put sodium alkane sulfonate and isopropyl alcohol in the cup and mix. Dissolve sodium alkane sulphonate in the isopropyl alcohol completely. Add water while stirring. Add hydrochloric acid part to part and mix well. Add acidic thickener. Continue to mixing until mixture is homogenous and thickener. Homogeneous and thickener is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener is sold as trade name. Therefore it can be acidic thickener from market.

DETERGENT FORMULATIONS ENCYCLOPEDIA

DRAIN

OPENER

FORMULATIONS

ACIDIC
DRAIN OPENER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	5
2	NITRIC ACID	25
3	HYDROCHLORIC ACID	5
4	WATER	65
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add nitric acid and hydrochloric acid order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ACIDIC
DRAIN OPENER
(QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	2
2	NITRIC ACID	15
3	HYDROCHLORIC ACID	5
4	WATER	78
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add nitric acid and hydrochloric acid order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ACIDIC
DRAIN OPENER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	HYDROCHLORIC ACID	10
2	NITRIC ACID	10
3	WATER	80
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nitric acid and hydrochloric acid order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ALKALINE
DRAIN OPENER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORIDE	5
2	SODIUM HYDROXIDE (% 48 SOLUTION)	65
3	WATER	30
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add the rest of ingredients one by one and mix. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ALKALINE
DRAIN OPENER
(QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM HYPOCHLORIDE	5
2	SODIUM HYDROXIDE (% 48 SOLUTION)	40
3	WATER	55
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add the rest of ingredients one by one and mix. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

ALKALINE
DRAIN OPENER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	WATER	50
2	SODIUM HYDROXIDE (% 48 SOLUTION)	50
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium hydroxide and mix. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ALKALINE DRAIN OPENER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	5
2	SODIUM HYDROXIDE (% 48 SOLUTION)	40
3	SODIUM HYPOCHLORIDE	5
4	WATER	50
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add lauramine oxide. Dissolve lauramine oxide in the water completely. Add sodium hypochloride and sodium hydroxide and mix well. Continue to mixing until mixture is homogenous and dense. If homogeneous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ALKALINE DRAIN OPENER
(QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	30
3	SODIUM HYPOCHLORIDE	5
4	WATER	62
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add lauramine oxide. Dissolve lauramine oxide in the water completely. Add sodium hypochloride and sodium hydroxide and mix well. Continue to mixing until mixture is homogenous and dense. If homogeneous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ALKALINE DRAIN OPENER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL ETHER SULFATE (SLES)	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	30
3	SODIUM HYPOCHLORIDE	3
4	WATER	64
	TOTAL	100

PROCESS: Charge water in the mixing tank. Start to stirr. Add sodium lauryl ether sulfate. Dissolve sodium lauryl ether sulfate in the water completely. Add sodium hypochloride and sodium hydroxide and mix well. Continue to mixing until mixture is homogenous and dense. If homogeneous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ACIDIC DRAIN OPENER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	5
2	NITRIC ACID	25
3	HYDROCHLORIC ACID	5
4	XANTHAN GUM	0.65
5	WATER	64.35
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum while stirring. Dissolve xanthan gum in the water completely. Add alcohol etoxylate. Continue to stirr. Add nitric acid and hydrochloric acid order given during constant mixing. Continue to mixing until mixture is dense and homogenous. If homogeneous and dense desired is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ACIDIC DRAIN OPENER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1
2	NITRIC ACID	20
3	HYDROCHLORIC ACID	5
4	XANTHAN GUM	0.75
5	WATER	73.25
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum while stirring. Dissolve xanthan gum in the water completely. Add nonylphenol etoxylate. Continue to stirr. Add nitric acid and hydrochloric acid order given during constant mixing. Continue to mixing until mixture is homogenous and dense desired. If homogeneous and dense is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
ACIDIC DRAIN OPENER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	WATER	74.25
2	NITRIC ACID	25
3	XANTHAN GUM	0.75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum while stirring. Dissolve xanthan gum in the water completely. Continue to stirr. Add nitric acid mix well. Continue to mixing until mixture is homogenous and dense desired. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

HOUSEHOLD

DISINFECTANTS

FORMULATIONS

HOUSEHOLD
DISINFECTANT SPRAY CHLORINE BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	1
2	SODIUM CARBONATE	5
3	SODIUM HYPOCHLORIDE	10
4	WATER	84
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add sodium carbonate and mix well. Add sodium hypochloride. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces.

HOUSEHOLD
DISINFECTANT SPRAY CHLORINE BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	1
2	SODIUM HYDROXIDE (% 48 SOLUTION)	3
3	SODIUM HYPOCHLORIDE	7
4	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add sodium hydroxide and mix well. Add sodium hypochloride. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces.

HOUSEHOLD
DISINFECTANT SPRAY CHLORINE BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE 8 EO	1
2	SODIUM HYDROXIDE (% 48 SOLUTION)	5
3	SODIUM HYPOCHLORIDE	5
4	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add sodium hydroxide and mix well. Add sodium hypochloride. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces.

HOUSEHOLD
DISINFECTANT SPRAY OXYGENE BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1
2	HEDP PHOSPHONATE	1
3	HYDROGEN PEROXIDE (% 50)	7
4	EDTA	1
5	WATER	
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nonylphenol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add hedp phosphonate and edta and mix well. Add hydrogen peroxide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

HOUSEHOLD
DISINFECTANT SPRAY OXYGENE BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1
2	HEDP PHOSPHONATE	1
3	HYDROGEN PEROXIDE (% 50)	5
4	WATER	93
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nonylphenol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add hedp phosphonate and mix well. Add hydrogen peroxide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

HOUSEHOLD
DISINFECTANT SPRAY OXYGENE BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	0.5
2	HEDP PHOSPHONATE	0.5
3	HYDROGEN PEROXIDE (% 50)	3
4	WATER	96
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nonylphenol etoxylate while stirring. Dissolve alcohol etoxylate in the water completely. Add hedp phosphonate and mix well. Add hydrogen peroxide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

HOUSEHOLD
DISINFECTANT SPRAY GERMICIDE BASED
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	1
2	TRIETHANOLAMINE	3
3	GERMICIDE	7
4	EDTA	1
5	WATER	88
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add triethanolamine and edta. Continue to stirr. Add germicide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

HOUSEHOLD
DISINFECTANT SPRAY GERMICIDE BASED
(QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	1
2	POTASSIUM HYDROXIDE (% 48 SOLUTION)	1
3	GERMICIDE	5
4	POTASIUM TRIPOLYPHOSPHATE	1
5	WATER	92
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add potassium tripolyphosphate and potassium hydroxide. Continue to stirr. Add germicide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

HOUSEHOLD
DISINFECTANT SPRAY GERMICIDE BASED
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	1
2	POTASSIUM HYDROXIDE (% 48 SOLUTION)	1
3	GERMICIDE	3
4	WATER	95
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add potassium hydroxide. Continue to stirr. Add germicide. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and disinfectant sinks, bathtubs, tiles, toilet and all hard surfaces. In addition to used in cleaning and disinfectant of all household goods.

DETERGENT FORMULATIONS ENCYCLOPEDIA

HOUSEHOLD

METAL

CLEANER AND POLISHER

FORMULATIONS

HOUSEHOLD
METAL CLEANER AND POLISHER SPRAY
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL BETAINE	3
2	İSOPROPYL ALCOHOL (IPA)	3
3	MONOETHANOLAMINE (MEA)	2
4	GLYCERINE	3
5	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients order given to tank during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

HOUSEHOLD
METAL CLEANER AND POLISHER SPRAY
(QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL BETAINE	2
2	İSOPROPYL ALCOHOL (IPA)	2
3	MONOETHANOLAMINE (MEA)	2
4	GLYCERINE	2
5	WATER	92
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients order given to tank during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

HOUSEHOLD
METAL CLEANER AND POLISHER SPRAY
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURYL BETAINE	1
2	İSOPROPYL ALCOHOL (IPA)	1
3	MONOETHANOLAMINE (MEA)	2
4	GLYCERINE	2
5	WATER	94
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add all ingredients order given to tank during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

HOUSEHOLD
METAL CLEANER AND POLISHER CREAM
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	2
2	HYDROXYETHYLCELLULOSE (HEC)	1
3	TRIETHANOLAMINE (TEA)	2
4	ALUMINUM OXIDE	5
5	SODIUM CARBONATE	1
6	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add hydroxyethylcellulose. Dissolve hydroxyethylcellulose in the water completely. Add aluminium oxide and mix. Add rest of ingredients during consant mixing. Continue to mixing until mixture is homogenous and cream form. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

HOUSEHOLD
METAL CLEANER AND POLISHER CREAM
(QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	1
2	HYDROXYETHYLCELLULOSE (HEC)	1
3	TRIETHANOLAMINE (TEA)	2
4	ALUMINUM OXIDE	3
5	SODIUM CARBONATE	1
6	WATER	91
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add hydroxyethylcellulose. Dissolve hydroxyethylcellulose in the water completely. Add aluminium oxide and mix. Add rest of ingredients during consant mixing. Continue to mixing until mixture is homogenous and cream form. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

HOUSEHOLD
METAL CLEANER AND POLISHER CREAM
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	1
2	HYDROXYETHYLCELLULOSE (HEC)	1
3	SODIUM HYDROXIDE (% 48 SOLUTION)	1
4	ALUMINUM OXIDE	3
5	SODIUM CARBONATE	1
6	WATER	93
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add hydroxyethylcellulose. Dissolve hydroxyethylcellulose in the water completely. Add aluminium oxide and mix. Add rest of ingredients during consant mixing. Continue to mixing until mixture is homogenous and cream form. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are used to filling a spray bottle. Generally, used in cleaning and polishing household metal surface and goods.

DETERGENT FORMULATIONS ENCYCLOPEDIA

HOUSEHOLD

FLOOR CLEANER

FORMULATIONS

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	5
2	TETRAPOTASSIUM PYROPHOSPHATE	10
3	SODIUM HYDROXIDE	5
4	SODIUM METASILICATE	5
5	WATER	75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add lauramine oxide while stirring. Dissolve lauramine oxide in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 25 with water. Generally, used in cleaning of household floor.

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE 10	3
2	POTASSIUM TRIPOLYPHOSPHATE	5
3	SODIUM HYDROXIDE	5
4	SODIUM METASILICATE	2
5	WATER	85
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nonylphenol etoxylate oxide while stirring. Dissolve nonylphenol etoxylate in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 15 with water. Generally, used in cleaning of household floor.

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE 10	3
2	POTASSIUM TRIPOLYPHOSPHATE	5
3	SODIUM HYDROXIDE	5
4	SODIUM METASILICATE	2
5	WATER	85
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add nonylphenol etoxylate oxide while stirring. Dissolve nonylphenol etoxylate in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 10 with water. Generally, used in cleaning of household floor.

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(CHLORINE BASED AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	5
2	TETRAPOTASSIUM PYROPHOSPHATE	10
3	SODIUM HYDROXIDE	5
4	SODIUM METASILICATE	5
5	SODIUM HYPOCHLORIDE	5
6	WATER	70
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add lauramine oxide while stirring. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and stirr well. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 25 with water. Generally, used in cleaning of household floor.

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(CHLORINE BASED AND QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	3
2	TETRAPOTASSIUM PYROPHOSPHATE	5
3	SODIUM HYDROXIDE	5
4	SODIUM METASILICATE	2
5	SODIUM HYPOCHLORIDE	5
6	WATER	80
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add lauramine oxide while stirring. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and stirr well. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 20 with water. Generally, used in cleaning of household floor.

CONCENTRATED
LIQUID HOUSEHOLD FLOOR CLEANER
(CHLORINE BASED AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	2
2	SODIUM TRIPOLYPHOSPHATE (STPP)	3
3	SODIUM HYDROXIDE	3
4	SODIUM METASILICATE	2
5	SODIUM HYPOCHLORIDE	5
6	WATER	85
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add lauramine oxide while stirring. Dissolve lauramine oxide in the water completely. Add sodium hydroxide and stirr well. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 10 with water. Generally, used in cleaning of household floor.

CONCENTRATED
POWDER HOUSEHOLD FLOOR CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONATE (LAS)	15
2	TETRAPOTASSIUM PYROPHOSPHATE	20
3	SODIUM CARBONATE	15
4	SODIUM METASILICATE	10
5	SODIUM PERCARBONATE	20
6	SODIUM SULFATE	20
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 50 with water. Generally, used in cleaning of household floor.

CONCENTRATED
POWDER HOUSEHOLD FLOOR CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONATE (LAS)	10
2	TETRAPOTASSIUM PYROPHOSPHATE	15
3	SODIUM CARBONATE	25
4	SODIUM METASILICATE	10
5	SODIUM PERCARBONATE	20
6	SODIUM SULFATE	20
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 40 with water. Generally, used in cleaning of household floor.

CONCENTRATED
POWDER HOUSEHOLD FLOOR CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LINEAR ALKYLBENZENE SULPHONATE (LAS)	5
2	TETRAPOTASSIUM PYROPHOSPHATE	10
3	SODIUM CARBONATE	20
4	SODIUM METASILICATE	10
5	SODIUM PERCARBONATE	10
6	SODIUM SULFATE	45
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingreddients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

This product is concentrated Also, it is diluted to 1 / 30 with water. Generally, used in cleaning of household floor.

DETERGENT FORMULATIONS ENCYCLOPEDIA

HOUSEHOLD

SHOWER CLEANER

FORMULATIONS

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ACIDIC AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	3
2	LAURAMINE OXIDE	3
3	GLYCOLIC ACID	3
4	CITRIC ACID	1
5	LACTIC ACID	3
6	EDTA	1
7	WATER	86
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add lauramine oxide and alkyl polyglucoside while stirring. Dissolve two ingredients in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ACIDIC AND QUALITY)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	2
2	LACTIC ACID	2
3	GLYCOLIC ACID	3
4	EDTA	0.5
5	WATER	92.5
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alkyl polyglucoside while stirring. Dissolve alkyl polyglucoside in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ACIDIC AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	2
2	LACTIC ACID	2
3	HEDP PHOSPHONATE	1
4	WATER	95
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alauramine oxide while stirring. Dissolve lauramine oxide in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ALKALINE AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	5
2	SODIUM HYDROXIDE (% 48 SOLUTION)	5
3	SODIUM METASILICATE	3
4	EDTA	3
5	BUTYL GLYCOL	5
6	WATER	79
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ALKALINE AND QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	4
3	SODIUM METASILICATE	2
4	EDTA	2
5	ISOPROPYL ALCOHOL	5
6	WATER	84
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
HOUSEHOLD SHOWER CLEANER
(ALKALINE AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	SODIUM HYDROXIDE (% 48 SOLUTION)	3
3	SODIUM METASILICATE	0.5
4	EDTA	0.5
5	ISOPROPYL ALCOHOL	3
6	WATER	90
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add coco glucoside while stirring. Dissolve coco glucoside in the water completely. Add rest of ingredients order given during constant mixing. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
POWDER HOUSEHOLD SHOWER CLEANER
(ACIDIC AND HIGH QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL SULFATE (SLS)	2
2	SULFAMIC ACID	10
3	CITRIC ACID	10
4	SODIUM BICARBONATE	30
5	SODIUM METASILICATE	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	10
7	SODIUM SULFATE	36
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
POWDER HOUSEHOLD SHOWER CLEANER
(ACIDIC AND QUALITY)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL SULFATE (SLS)	2
2	SULFAMIC ACID	5
3	CITRIC ACID	10
4	SODIUM BICARBONATE	35
5	SODIUM METASILICATE	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	5
7	SODIUM SULFATE	41
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

CONCENTRATED
POWDER HOUSEHOLD SHOWER CLEANER
(ACIDIC AND ECONOMICAL)

NO	INGREDIENTS	W/W
1	SODIUM LAURYL SULFATE (SLS)	1
2	SULFAMIC ACID	5
3	CITRIC ACID	5
4	SODIUM BICARBONATE	42
5	SODIUM METASILICATE	2
6	SODIUM TRIPOLYPHOSPHATE (STPP)	5
7	SODIUM SULFATE	40
	TOTAL	100

PROCESS: All ingredients are solid above given. Then, all ingredients are put to mixing tank order given and starting to stirr. Continue to stirr until homogenous. Homogenous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured products are diluted and used to filling a spray bottle. Generally, used in cleaning of shower.

TOILET

BOWL CLEANER

FORMULATIONS

TOILET
BOWL CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	2
2	HYDROCHLORIC ACID	20
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	WATER	78
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add hydrochloric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

TOILET
BOWL CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1
2	PHOSPHORIC ACID	15
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	WATER	84
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add phosphoric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

TOILET
BOWL CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1
2	HYDROCHLORIC ACID	10
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add hydrochloric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
TOILET BOWL CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1.25
2	HYDROCHLORIC ACID	20
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	XANTHAN GUM	0.75
5	WATER	78
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum. Dissolve xanthan gum in water completley. Add hydrochloric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous and gel. If homogeneous and gel is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
TOILET BOWL CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1.25
2	PHOSPHORIC ACID	15
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	XANTHAN GUM	0.75
5	WATER	83
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum. Dissolve xanthan gum in the water completely. Add phosphoric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous and thickened. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
TOILET BOWL CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE NP: 10	1.15
2	HYDROCHLORIC ACID	10
3	DYE, PRESERVATIVE, FRAGRANCE	q.s.
4	XANTHAN GUM	0.85
5	WATER	88
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add xanthan gum. Dissolve xanthan gum in the water well. Add hydrochloric acid while stirring. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is thickened and homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

TOILET
BOWL CLEANER AND GLOSS ENHANCER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	1
2	CITRIC ACID	10
3	QUATERNIUM 80	1
4	PHOSPHORIC ACID	3
5	DYE, PRESERVATIVE AND FRAGRANCE	q.s.
6	WATER	85
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while mixing. Dissolve alcohol etoxylate in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

TOILET
BOWL CLEANER AND GLOSS ENHANCER
(QUALITY)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	1
2	CITRIC ACID	7
3	QUATERNIUM 80	0.5
4	PHOSPHORIC ACID	1
5	DYE, PRESERVATIVE AND FRAGRANCE	q.s.
6	WATER	90.5
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while mixing. Dissolve alcohol etoxylate in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

TOILET
BOWL CLEANER AND GLOSS ENHANCER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	C11 ALCOHOL ETOXYLATE	0.5
2	CITRIC ACID	5
3	QUATERNIUM 80	0.5
4	DYE, PRESERVATIVE AND FRAGRANCE	q.s.
5	WATER	94
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add alcohol etoxylate while mixing. Dissolve alcohol etoxylate in the water completely. Add the rest of ingredients one by one and mix well. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIME AND RUST CLEANER FORMULATIONS

LIME AND RUST
CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	5
2	PHOSPHORIC ACID	15
3	PHOSPHONATE HEDP	5
4	WATER	75
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add phosphoric acid and phosphonate hedp given order while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIME AND RUST
CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	3
2	PHOSPHORIC ACID	10
3	HYDROCHLORIC ACID	5
4	WATER	82
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add phosphoric acid and hydrochloric acid given order while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

LIME AND RUST
CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	1
2	PHOSPHORIC ACID	10
3	WATER	89
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add phosphoric acid while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

THICKENED
LIME AND RUST CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	5
2	PHOSPHORIC ACID	15
3	PHOSPHONATE HEDP	2.5
4	ACIDIC THICKENER AGENT	0.5
5	WATER	77
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add acidic thickener agent and mix. Add phosphoric acid and phosphonate hedp given order while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener agent can be buy from market as special name.

THICKENED
LIME AND RUST CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	3
2	PHOSPHORIC ACID	10
3	HYDROCHLORIC ACID	4.25
4	ACIDIC THICKENER AGENT	0.75
5	WATER	82
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add acidic thickener agent and mix. Add phosphoric acid and hydrochloric acid given order while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener agent can be buy from market as special name.

THICKENED
LIME AND RUST CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	LAURAMINE OXIDE	1
2	PHOSPHORIC ACID	8
3	ACIDIC THICKENER AGENT	1
4	WATER	90
	TOTAL	100

PROCESS: Charge water in the mixing tank. Add acidic thickener agent and stirr. Add phosphoric acid while mixing. Add lauramine oxide and mix. Dissolve lauramine oxide in the mixture completely. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Acidic thickener agent can be buy from market as special name.

TILE AND TUB

CLEANER

FORMULATIONS

ACIDIC
TILE AND TUB CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	LAURYL GLUCOSIDE	2
2	ISOPROPRYL ALCOHOL	3
3	PHOSPHORIC ACID	5
4	CITRIC ACID	5
5	WATER	85
	TOTAL	100

PROCESS: Put water in the mixing tank. Add lauryl glucoside and isopropyl alcohol and mix. Add citric acid and phosphoric acid while stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

ACIDIC
TILE AND TUB CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE	1
2	ISOPROPRYL ALCOHOL	2
3	PHOSPHORIC ACID	4
4	CITRIC ACID	3
5	WATER	90
	TOTAL	100

PROCESS: Put water in the mixing tank. Add nonylphenol etoxylate and isopropyl alcohol and mix. Add citric acid and phosphoric acid while stirring. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

ACIDIC
TILE AND TUB CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE	1
2	ISOPROPRYL ALCOHOL	1
3	PHOSPHORIC ACID	2
4	CITRIC ACID	2
5	WATER	94
	TOTAL	100

PROCESS: Put water in the mixing tank. Add nonylphenol etoxylate and isopropyl alcohol and mix. Add citric acid and phosphoric acid while stirring. Continue to mixing until mixture is homogenous. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

ALKALINE
TILE AND TUB CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	BUTYL GLYCOL	5
3	MONOETHANOLAMINE (MEA)	3
4	SODIUM HYDROXIDE (% 48 SOLUTION)	5
5	SODIUM METASILICATE	2
6	SODIUM EDTA	3
7	WATER	79
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and butyl glycol and mix. Dissolve two ingredients in the water completely. Add remainder of ingredients in order given during constant mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

ALKALINE
TILE AND TUB CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	BUTYL GLYCOL	5
3	SODIUM METASILICATE	1
4	SODIUM HYDROXIDE (% 48 SOLUTION)	5
5	WATER	86
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and butyl glycol and mix. Dissolve two ingredients in the water completely. Add remainder of ingredients in order given during constant mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

ALKALINE
TILE AND TUB CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	1
2	BUTYL GLYCOL	3
3	SODIUM METASILICATE	1
4	SODIUM HYDROXIDE (% 48 SOLUTION)	5
5	WATER	90
	TOTAL	100

PROCESS: Put water in the mixing tank. Add coco glucoside and butyl glycol and mix. Dissolve two ingredients in the water completely. Add remainder of ingredients in order given during constant mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is used to cleaning tile, tub and hard surface.

DETERGENT FORMULATIONS ENCYCLOPEDIA

COMPUTER AND TV

SCREEN CLEANER

FORMULATIONS

COMPUTER AND TV
SCREEN SPRAY CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE	1
2	ISOPROPYL ALCOHOL (IPA)	15
3	SODIUM EDTA	2
4	WATER	82
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add nonylphenol etoxylate and isopropyl alcohol while stirring. Dissolve two ingredients in the water and mix well. Add sodium edta and stirr well. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

COMPUTER AND TV
SCREEN SPRAY CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE	1
2	ISOPROPYL ALCOHOL (IPA)	10
3	SODIUM EDTA	1
4	WATER	88
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add nonylphenol etoxylate and isopropyl alcohol while stirring. Dissolve two ingredients in the water and mix well. Add sodium edta and stirr well. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

COMPUTER AND TV
SCREEN SPRAY CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	NONYLPHENOL ETOXYLATE	1
2	ISOPROPYL ALCOHOL (IPA)	6
3	WATER	93
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add nonylphenol etoxylate and isopropyl alcohol while stirring. Dissolve two ingredients in the water and mix well. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

DETERGENT FORMULATIONS ENCYCLOPEDIA

AIR CONDITIONER CLEANER FORMULATIONS

ALKALINE
AIR CONDITIONER CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	10
2	SODIUM METASILICATE	5
3	SODIUM EDTA	5
4	TRIETHANOLAMINE (TEA)	10
5	WATER	70
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add decyl glucoside while mixing. Dissolve decyl glucoside in the water completely. Add triethanolamine, sodium metasilicate and sodium edta during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

ALKALINE
AIR CONDITIONER CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	8
2	SODIUM METASILICATE	5
3	SODIUM EDTA	5
4	TRIETHANOLAMINE (TEA)	5
5	WATER	77
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add decyl glucoside while mixing. Dissolve decyl glucoside in the water completely. Add triethanolamine, sodium metasilicate and sodium edta during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

ALKALINE
AIR CONDITIONER CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	ALKYL POLYGLUCOSIDE	5
2	SODIUM METASILICATE	3
3	SODIUM EDTA	3
4	TRIETHANOLAMINE (TEA)	5
5	WATER	84
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add alkyl polyglucoside while mixing. Dissolve decyl glucoside in the water completely. Add triethanolamine, sodium metasilicate and sodium edta during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

ACIDIC
AIR CONDITIONER CLEANER
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	5
2	HEDP PHOSPHONATE (% 60)	5
3	SULFAMIC ACID	5
4	CITRIC ACID	10
5	WATER	75
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add decyl glucoside while mixing. Dissolve decyl glucoside in the water completely. Add sulfamic acid, hedp phosphonate and citric acid during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

ACIDIC
AIR CONDITIONER CLEANER
(QUALITY)

NO	INGREDIENTS	W/W
1	DECYL GLUCOSIDE	5
2	HEDP PHOSPHONATE (% 60)	3
3	SULFAMIC ACID	2
4	CITRIC ACID	5
5	WATER	85
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add decyl glucoside while mixing. Dissolve decyl glucoside in the water completely. Add sulfamic acid, hedp phosphonate and citric acid during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

ACIDIC
AIR CONDITIONER CLEANER
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	COCO GLUCOSIDE	3
2	HEDP PHOSPHONATE (% 60)	3
3	SULFAMIC ACID	2
4	PHOSPHORIC ACID	3
5	WATER	89
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add decyl glucoside while mixing. Dissolve decyl glucoside in the water completely. Add sulfamic acid, hedp phosphonate and phosphoric acid during constant stirring. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

AIR CONDITIONER
DISINFECTANT AGENT
(HIGH QUALITY)

NO	INGREDIENTS	W/W
1	BENZALKONIUM CHLORIDE	14
2	ISOPROPYL ALCOHOL	20
3	WATER	66
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add benzalkonium chloride and isopropyl alcohol while mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

AIR CONDITIONER
DISINFECTANT AGENT
(QUALITY)

NO	INGREDIENTS	W/W
1	BENZALKONIUM CHLORIDE	10
2	ISOPROPYL ALCOHOL	15
3	WATER	75
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add benzalkonium chloride and isopropyl alcohol while mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

AIR CONDITIONER
DISINFECTANT AGENT
(ECONOMICAL)

NO	INGREDIENTS	W/W
1	BENZALKONIUM CHLORIDE	5
2	ISOPROPYL ALCOHOL	10
3	WATER	85
	TOTAL	100

PROCESS: Charge water and in the mixing tank. Add benzalkonium chloride and isopropyl alcohol while mixing. Continue to mixing until mixture is homogenous and clear. If homogeneous is enough, process is completed.

NOTE: In the above manufacturing products, the amounts of all ingredients can be changed depending on condition.

Manufactured product is concentrated. So, it is diluted to use.

DETERGENT FORMULATIONS ENCYCLOPEDIA

SOME INGREDIENTS USED IN DETERGENT

SOLVER CHEM PUBLICATIONS

DETERGENT FORMULATIONS ENCYCLOPEDIA

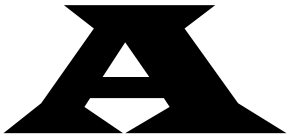
SOLVER

CHEM

PUBLICATIONS

SOLVER CHEM PUBLICATIONS

DETERGENT FORMULATIONS ENCYCLOPEDIA



DETERGENT CHEMICALS PROPERTIES

SOLVER CHEM PUBLICATIONS

ACETYLATED LANOLIN

GENERAL DEFINITIONS	
CAS NUMBER	629 – 70 – 9 + 61788 – 49 – 6
GENERAL CATEGORY	Cosmetic agent
FACTORIAL CATEGORY	Skin softening
TRADE NAME	Acetylated lanolin

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	150
PH RANGE	6 – 9
BOILING POINT (C)	650
DENSITY	0.880 g / cm3
SOLUBILITY	Soluble in hot water and cold water.

ACETYLATED LANOLIN

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid oily
COLOR	Yellow
ODOR	A characteristic odor
AID MEASURES	It does not cause an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Observed that quite often used in the manufacture of cosmetic products. Among these products, hair care products, skin care creams and lotions, we can include the baby products.

ACNIBIO
AC

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic and detergent agent
FACTORIAL CATEGORY	Biodegradable preservative
TRADE NAME	Acnibio AC

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FREEZING POINT (C)	Not specified
PH RANGE	2 – 4
MELTING POINT (C)	Not specified
DENSITY	1.10 g / cm3
SOLUBILITY	Soluble in hot water and cold water.

ACNIBIO
AC

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid viscous
COLOR	Colorless and pale yellow
ODOR	Odorless
AID MEASURES	It does not cause an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**ACNIBIO AC** is suitable for all cosmetics and toiletries (conditioners, gels, shampoos, surfactants, cleansers…). **ACNIBIO AC** is supplied as an aqueous solution for ease of incorporation into cosmetic formulations. It has good compatibility with surfactants and emulsifiers, irrespective of their ionic nature. In hot processes it is advisable to add the preservative during the cooling stage at a temperature no greater than 60ºC, and preferably below 40ºC.

ACNIBIO AP

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic and detergent agent
FACTORIAL CATEGORY	Biodegradable preservative
TRADE NAME	Acnibio AP

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FREEZING POINT (C)	Not specified
PH RANGE	3.5 – 5,5
MELTING POINT (C)	Not specified
DENSITY	1.150 g / cm3
SOLUBILITY	Soluble in glycol. Slightly soluble in water and alcohol.

ACNIBIO AP

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Colorless and pale yellow
ODOR	Odorless
AID MEASURES	It does not cause an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**ACNIBIO AP** is an effective and powerful biocide suitable for a wide range of hydro-alcohol, w/o or o/w emulsion based cosmetics. The product may be applied at different pH ranges, from 3,0 up to 8,0. The advised dosage for these applications goes from 0.25 to 1,0 %. The higher concentrations are generally required only for formulations which, by their nature, are particularly difficult to preserve. Shampoos and foam baths may be preserved with ACNIBIO AP at concentrations typically between 0.25 - 0.65%. Products with a high protein content may require levels from 0,5% - 1,0%. Emulsified systems, both o/w and w/o types, may be effectively preserved by the addition of ACNIBIO AP at 0.4 - 0.7%. **ACNIBIO AP** can also be used to preserve emulsions based on nonionic surfactants, but slightly increased concentrations may be required,

ALKYL POLYGLUCOSIDE

GENERAL DEFINITIONS	
CAS NUMBER	68515 – 73 – 1
GENERAL CATEGORY	Surfactants for detergent
FACTORIAL CATEGORY	Nonionic
TRADE NAME	Nonionic

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FREEZING POINT (C)	10
PH RANGE	6 – 9
MELTING POINT (C)	Not specified
DENSITY	1.150 g / cm3
SOLUBILITY	Soluble in hot water and cold water.

ALKYL POLYGLUCOSIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid viscous
COLOR	Colorless
ODOR	Odorless
AID MEASURES	It does not cause an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Alkyl polyglycosides**, and is a raw material in the form of a colorless liquid with a odorless. **Alkyl polyglucosides**, with ionic properties are actively resistant surface is highly alkaline environment. Including detergents and cosmetics products, find use in many industrial cleaning products.

AMARANTH
E123

GENERAL DEFINITIONS	
CAS NUMBER	915 – 67 – 3
GENERAL CATEGORY	Anionic dye
FACTORIAL CATEGORY	Food dye
TRADE NAME	Amaranth E123

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{20}H_{11}N_2Na_3O_{10}S_3$
MOLECULAR WEIGHT	604.47 gr / mole
ACTIVE (%)	85
PH RANGE	6 – 8
MELTING POINT (C)	Not specified
DENSITY	0.800 g / cm3
SOLUBILITY	Soluble in hot water and cold water.

AMARANTH
E123

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Dark red or brown
ODOR	Odorless
AID MEASURES	It does not cause an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Reddish brow powder. Slightly soluble in water for product red, soluble in 30% ethanol, glycerin and thin syrup, slightly soluble in ethanol and soluble fiber element, insoluble in other organic solvents. The strong sulfuric acid for purple, diluted for pink; In nitric acid solution for colourful red; In the red brown solution for hydrochloric acid, a black precipitation; In the thick of sodium hydroxide solution for red brown.

AMINO TRİMETHYLENE PHOSPHORIC ACID (ATMP)

GENERAL DEFINITIONS	
CAS NUMBER	6419 – 19 - 8
GENERAL CATEGORY	Industrıal agent
FACTORIAL CATEGORY	Chelated agent
TRADE NAME	ATMP

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_3H_{12}NO_9P_3$
MOLECULAR WEIGHT	299.05 gr / mole
BOILING POINT (C)	Not specified
PH RANGE	2
MELTING POINT (C)	200
DENSITY	1.33 g / cm3
SOLUBILITY	610 g / Lt Soluble in water.

AMINO TRİMETHYLENE PHOSPHORIC ACID (ATMP)

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder or granul
COLOR	White
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	ATMP system in hard water and is a stable material which can be hydrolyzed. In high concentrations, ATMP has good corrosion inhibition. ATMP cold water system and oilfield in industrial thermal power plant and oil refinery plants used in circulation in the area of water pipeline. ATMP is a major effect on corrosion of metal equipment and pipeline. Chelating agents for the textile and dyeing industry as ATMP is available and can be utilized as metal surface treatment agent.

AMMONIUM HYDROXIDE
AQUEOUS AMMONIA

GENERAL DEFINITIONS	
CAS NUMBER	7664 – 41 – 7 + 7732 – 18 – 5
GENERAL CATEGORY	Industrıal agent
FACTORIAL CATEGORY	Alkaline agent
TRADE NAME	Aqueous ammonia

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
BOILING POINT (C)	Not specified
PH RANGE	11 - 12
MELTING POINT (C)	-69.2
DENSITY	0.989 g / cm3
SOLUBILITY	Soluble in water easily.

AMMONIUM HYDROXIDE
AQUEOUS AMMONIA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic ammonia odor.
AID MEASURES	It causes an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Ammonium hydroxide**. is used as a cleaning agent and sanitizer in many household and industrial cleaners. Ammonium hydroxide is also used in the manufacture of products such as fertilizer, plastic, rayon and rubber. **Aqueous ammonia** is corrosive to aluminum alloys, copper, copper alloys, and galvanized surfaces. Aqueous ammonia is an excellent acid neutralizer.

AMMONIUM BICARBONATE

GENERAL DEFINITIONS

CAS NUMBER	1066 – 33 – 7
GENERAL CATEGORY	Industrıal agent
FACTORIAL CATEGORY	Industrial and detergent productions
TRADE NAME	Ammonium bicarbonate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	NH_4HCO_3
MOLECULAR WEIGHT	79.06 Gr / Mole
BOILING POINT (C)	Not specified
PH RANGE	7
MELTING POINT (C)	107
DENSITY	1.590 g / cm3
SOLUBILITY	Soluble in water easily.

AMMONIUM BICARBONATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Ammonium bicarbonate**, bakery in (yeast duty), baking powder formulations, in fire extinguishers, conductive plastics, cleaning products, ceramics, paint, leather and fertilizer used in the textile industry. **Ammonium bicarbonate**, shelf life of the product obtained when using ammonium bicarbonate, baking soda is short. It is used for alkalization of cocoa powder and firing to form carbon dioxide.

AMMONIUM CHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	12125 – 02 – 9
GENERAL CATEGORY	Industrıal agent
FACTORIAL CATEGORY	Deoxidising for metal
TRADE NAME	Ammonium Chloride

CHEMICAL PROPERTIES

CHEMICAL FORMULA	NH_4CL
MOLECULAR WEIGHT	53.49 Gr / Mole
BOILING POINT (C)	520
PH RANGE	5.5
MELTING POINT (C)	338
DENSITY	1.530 g / cm3
SOLUBILITY	Soluble in water easily.

AMMONIUM CHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or powder
COLOR	White
ODOR	A characteristic odor
AID MEASURES	It causes an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Ammonium chloride colorless crystalline substance easily dissolves in water to form a liquid slightly acidic property. Evaporating from the melt 340 ° C, to form ammonia and hydrogen chloride in equal volumes. Ammonium chloride, in the production of dry batteries and some explosives, In Galvanized sector, tin plating, as flux raw materials in your hot-dip galvanized, If used as a surfactant ALS shampoo in cosmetics if the construction industry, in the production of cleaner used to thicken Ammonium Chloride and some shampoo, Some noble metals (such as gold) in the refining / purification, Some synthetic adhesives, drying, etc.

AMMONIUM LAURYL ETHER SULFATE

GENERAL DEFINITIONS	
CAS NUMBER	32612 – 48 – 9
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Anionic
TRADE NAME	ALES

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	MIXTURE
MOLECULAR WEIGHT	Mixture
BOILING POINT (C)	100
PH RANGE	5 – 7
FLASH POINT (C)	93.7
DENSITY	0.986 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

AMMONIUM
LAURYL ETHER SULFATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Creamy liquid
COLOR	White or colorless
ODOR	A characteristic odor
AID MEASURES	It causes no irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Ammonium lauryl ether sulphate**, an anionic surface active ingredient, creamy or concentrated liquid form and has a characteristic odor. **Ammonium lauryl ether sulphate** can therefore have a lower irritation than other anionic surfactants raw materials, usually in child care products, in cosmetic products and is used in the detergent industry.

AMMONIUM LAURYL SULFATE (% 28)

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Anionic
TRADE NAME	Ammonium lauryl sulphate (% 28)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	MIXTURE
MOLECULAR WEIGHT	Mixture
BOILING POINT (C)	100
PH RANGE	5 - 7
FLASH POINT (C)	Not specified
DENSITY	1.010 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

AMMONIUM
LAURYL SULFATE (% 28)

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless and pale yellow
ODOR	A characteristic odor
AID MEASURES	It causes no irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Ammonium lauryl sulfate**, diğer yüzey aktif maddelere göre daha az irritasyona sahiptir. Bu nedenle genellikle çocuk bakım ürünlerinde ve kozmetik ürünlerinde kullanılır.

ACETIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	64 – 19 - 7
GENERAL CATEGORY	Food and industrial ingredient
FACTORIAL CATEGORY	Acidic regular
TRADE NAME	E 260

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_2H_4O_2$
MOLECULAR WEIGHT	60.05 Gr / Mole
BOILING POINT (C)	118
PH RANGE	2
MELTING POINT (C)	16.5
DENSITY	1.265 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

ACETIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor
AID MEASURES	It causes an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Vinegar is a household essential often used for cooking, and it even helps unclog sinks. What chemical substance makes vinegar smell the way it does? Vinegar is actually a solution that is made of 4-8% of acetic acid that is dissolved in water and other substances that give vinegar its flavor depending on its type. **Acetic acid**, also known as **ethanoic acid** and **methanecarboxylic acid**, is a colorless liquid that has a strong and distinct pungent and sour smell. Its **chemical formula** is C2H4O2: it has two carbon (C) atoms, four hydrogen (H) atoms and two oxygen (O) atoms.

ACETONE

GENERAL DEFINITIONS	
CAS NUMBER	67 – 64 - 1
GENERAL CATEGORY	industrial ingredient
FACTORIAL CATEGORY	Solvent agent
TRADE NAME	Acetone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_3H_6O
MOLECULAR WEIGHT	58.08 Gr / Mole
BOILING POINT (C)	56.2
PH RANGE	Not specified
MELTING POINT (C)	-95
DENSITY	0.790 g / cm3
SOLUBILITY	Soluble in cold water.

ACETONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor
AID MEASURES	It causes an irritation on contact. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Acetone** is a colorless, flammable liquid that evaporates easily. It is an organic compound because carbon atoms are present in acetone's chemical formula, which is C_3H_6O. It consists of three carbon atoms, six hydrogen atoms, and one oxygen atom. Acetone is a solvent, which is a liquid that can dissolve other substances. Acetone is most commonly known as the solvent that removes nail polish, but it is also found in bath and fragrance products, hair and skin care products, as well as skin-lightening products.

DETERGENT FORMULATIONS ENCYCLOPEDIA

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BARITE

GENERAL DEFINITIONS	
CAS NUMBER	Mineral
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Filler agent
TRADE NAME	Barite

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mineral
MOLECULAR WEIGHT	Mineral
HARDNESS (MOHS)	2.5 – 3,5
MELTING POINT (C)	1580
BOILING POINT (C)	Not specified
DENSITY	4.20 g / cm3
SOLUBILITY	Insoluble in water.

BARITE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder or granular
COLOR	Gray or white
ODOR	Odorless
AID MEASURES	It does not cause an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Barite is the main one of element barium. It is also important in the manufacture of paper and rubber. Barite is also used in radiology for x-rays of the digestive system. When crushed, it is added to mud to form barium mud, which is poured into oil wells during drilling. A rich, white pigment was once made from crushed Barite. Barite is also a very popular and common mineral among collectors.

BEESWAX

GENERAL DEFINITIONS	
CAS NUMBER	8012 – 89 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Wax agent
TRADE NAME	Beeswax

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Natural wax
MOLECULAR WEIGHT	Natural wax
FLASH POINT (C)	Not specified
MELTING POINT (C)	63.5
BOILING POINT (C)	Not specified
DENSITY	0.955 g / cm3
SOLUBILITY	Insoluble in water.

BEESWAX

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Solid,paste and oily
COLOR	Pale yellow
ODOR	A characteristic odor.
AID MEASURES	It does not cause an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Beeswax (Beeswax), in waterproof materials, furniture and floor polish, the polishing of optical lenses, although little is found in chewing gum and ink usage. Yellow wax polishing parquet, sculpture, used in some paints and candle making. Beeswax, a variety of body care creams in cosmetics, face masks, lotions, lipsticks, lip balms and oils and used in production.

BENZALDEHYDE

GENERAL DEFINITIONS	
CAS NUMBER	100 – 52 – 7
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Aniline dye agent
TRADE NAME	Benzaldehyde

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_7H_6O
MOLECULAR WEIGHT	106.13 g / mole
FLASH POINT (C)	65
MELTING POINT (C)	-26.5
BOILING POINT (C)	179
DENSITY	1.05 g / cm3
SOLUBILITY	Slightly soluble in water.

BENZALDEHYDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Benzaldehyde is a color liquid aldehyde with an almond odor. Benzaldehyde is found in many foods and is widely used in the chemical industry. In the chemical industry, this agent is used in the preparation of various aniline dyes, perfumes, flavorings, and pharmaceuticals. In addition, benzaldehyde has been associated with amygdalin, since one of amygdalins breakdown products is benzaldehyde. (NCI) Flash point near 145 °F. More denser than water and insoluble in water. Hence sinks in water. Easily penetrates the soil to contaminate groundwater and nearby waterways. Used in flavoring and perfume making.

BENZALKONIUM CHLORIDE

GENERAL DEFINITIONS	
CAS NUMBER	8001 – 54 – 5
GENERAL CATEGORY	Detergent ingredient
FACTORIAL CATEGORY	Disinfectant agent
TRADE NAME	Benzalkonium chloride

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	Not specified
MELTING POINT (C)	140
BOILING POINT (C)	100
DENSITY	1.010 g / cm3
SOLUBILITY	Soluble in water.

BENZALKONIUM CHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Benzalkonium chloride, itself has a peculiar odor, liquid form, is transparent and colorless raw material. Benzalkonium chloride, detergent industry and in particular disinfectant products, industrial products and disinfectants in particular surgical tools, disinfectant used in swimming pools and many other related products.

BENZYL ALCOHOL

GENERAL DEFINITIONS	
CAS NUMBER	100 – 51 – 6
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	General solvent
TRADE NAME	Benzyl alcohol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_7H_8O
MOLECULAR WEIGHT	108.14 g / Mole
FLASH POINT (C)	Not specified
MELTING POINT (C)	15
BOILING POINT (C)	205
DENSITY	1.045 g / cm3
SOLUBILITY	Slightly soluble in water.Soluble in ether and chloroform.

BENZYL ALCOHOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	Sweet odor.
AID MEASURES	It causes an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Benzyl alcohol**, inks, paints, lacquers, and is used as a general solvent for epoxy resin coatings. Also it esters used soap, perfume, flavor, and is a precursor for a variety of industries. This generally bacteriostatic and intravenous drug solution is added as a preservative, depending on the antipruritic properties. Also it used as a photo developer. **Benzyl alcohol**, quartz wool fibers and has almost the same refractive index. If a clear quartz object is immersed in benzyl alcohol, it is almost invisible. This is an object (see crystal skull) is made of quartz or right is used as a way to recognize as destructive.

BENZOTRIAZOLE

GENERAL DEFINITIONS	
CAS NUMBER	95 – 14 – 7
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Metal inhibitor
TRADE NAME	Benzotriazole

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_6H_5N_3$
MOLECULAR WEIGHT	119.12 g / Mole
FLASH POINT (C)	Not specified
MELTING POINT (C)	100
BOILING POINT (C)	350
DENSITY	1.360 g / cm3
SOLUBILITY	Partially, soluble in water.

BENZOTRIAZOLE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Benzotriazole, no smell, the white color is a solid raw crystal or powder form. Benzotriazole, usually in the metal processing industry and chemical and oil taken up in acidic work used as the metal inhibitor.

BIS PEG 1 DIMETHICONE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Bis peg 1 dimethicone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	Not specified
MELTING POINT (C)	Not specified
BOILING POINT (C)	200
DENSITY	Not specified
SOLUBILITY	Soluble in water.

BIS PEG 1
DIMETHICONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	BIS-PEG-1 DIMETHICONE is classified as surfactant skin conditioning plasticiser hair conditioning. Often used in the production of cosmetics. Protect hair shampoo and hair care products and hair styling used in lotions and creams. It also has to provide softness to the hair and foam also supportive.

BIS PEG 20 DIMETHICONE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Bis peg 20 dimethicone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	Not specified
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
DENSITY	Not specified
SOLUBILITY	Soluble in water.

BIS PEG 20 DIMETHICONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	BIS PEG 20 DIMETHICONE is classified as surfactant skin conditioning plasticiser hair conditioning. Often used in the production of cosmetics. Protect hair shampoo and hair care products and hair styling used in lotions and creams. It also has to provide softness to the hair and foam also supportive.

BORAX

GENERAL DEFINITIONS	
CAS NUMBER	1303 – 96 – 4
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Detergent
TRADE NAME	Borax

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mineral
MOLECULAR WEIGHT	Mineral
PH RANGE	9 – 9.5
MELTING POINT (C)	743
BOILING POINT (C)	Not specified
DENSITY	1.73 g / cm3
SOLUBILITY	Partially, soluble in water.

BORAX

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Borax is used in the process of making boric acid, but there is a tremendous chemical difference between the two. Borax is a naturally occurring mineral, though of course, that doesn't make it inert or safe either. Arsenic is a naturally occurring metalloid but it isn't safe for human use. Natural doesn't always mean safe. At the same time, the studies used to back up the safety (or danger) of borax often use boric acid or are ambiguous about which was used.

BORIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	10043 – 35 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Inorganic acid
TRADE NAME	Boric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	H_3PO_3
MOLECULAR WEIGHT	61.83
PH RANGE	Not specified
MELTING POINT (C)	171
BOILING POINT (C)	300
DENSITY	1.435 g / cm3
SOLUBILITY	Partially, soluble in water.

BORIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Boric acid is a solid raw material that does not smell white and crystalline form. Boric acid, diethanolamine or monoethanolamine in this compound used in the construction of anti-corrosion products in the detergents sector. Boric acid is also found in use in the cosmetic and pharmaceutical industries. Also located on site due to the boron is also used in agriculture.

BUTYLENE GLYCOL

GENERAL DEFINITIONS	
CAS NUMBER	111 – 76 – 2
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Butylene glycol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C6H14O2
MOLECULAR WEIGHT	118.17 g / Mole
FLASH POINT (C)	65
MELTING POINT (C)	-77
BOILING POINT (C)	171
DENSITY	0.900 g / cm3
SOLUBILITY	Soluble in water.

BUTYLENE GLYCOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Butylene glycol is an organic solvent. Also it has a very good solvent and dirt rust remover feature. Tozut to be used on surfaces where anti-surface drying regulatory and anti-feature is also available. Quick-drying lacquer and find use in the manufacture of varnishes. Butylene glycol, detergents and industrial degreasing cleaning products used as soil removal. In hydraulic oil, truffles used as a solvent in the oil and surface cleaner.

BUTYL HYDROXY TOLUENE
BHT

GENERAL DEFINITIONS	
CAS NUMBER	128 – 37 – 0
GENERAL CATEGORY	Food ingredient
FACTORIAL CATEGORY	Antioxidant
TRADE NAME	Butyl hydroxy tolüene (BHT)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{15}H_{24}O$
MOLECULAR WEIGHT	220.36 g / Mole
E CODE	321
MELTING POINT (C)	70
BOILING POINT (C)	265
DENSITY	1.05 g / cm3
SOLUBILITY	Insoluble in water. Soluble in oil.

BUTYL HYDROXY TOLUENE
BHT

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	This material, despite having lower activity in vegetable oils such as BHA, are used in the food protection feature has been added fat when used in combination with other antioxidants. BHA; It showed synergistic effect with BHT can not synergize with gallate. Synthetic antioxidants BHA and BHT entering class; widely used food additives is excessively dissolved in the fat, while insoluble in water. BHA and BHT; solid, liquid containing fats and oils slowing the rancidity of foods; various cereals, chewing gum, the products that use plant-derived oils such as potato chips, color can be commonly arises from exposure to air these products, flavor or texture prevents the change.

DETERGENT FORMULATIONS ENCYCLOPEDIA

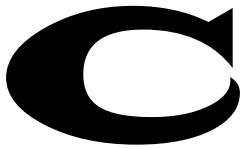
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DETERGENT FORMULATIONS ENCYCLOPEDIA



DETERGENT CHEMICALS PROPERTIES

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CALCITE
MINERAL

GENERAL DEFINITIONS

CAS NUMBER	Mineral
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Filler agent
TRADE NAME	Calcite

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$CaCO_3$
MOLECULAR WEIGHT	100.1
PH RANGE	8 – 9
MELTING POINT (C)	825
BOILING POINT (C)	Not specified
DENSITY	2.700 g / cm3
SOLUBILITY	Insoluble in water.

CALCITE MINERAL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	white
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Calcite, including chemical structure first, insulation mortar, paint production, polyurethane structures, used in detergents and in many industrial areas. Calcite, plastic and rubber manufacturing out of them is used as a filler.

CALCIUM HYDROXIDE

GENERAL DEFINITIONS

CAS NUMBER	1305 – 62 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	slaked lime
TRADE NAME	Calcium hydroxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$Ca(OH)_2$
MOLECULAR WEIGHT	74.1
PH RANGE	14
MELTING POINT (C)	580
BOILING POINT (C)	Not specified
DENSITY	2.200 g / cm3
SOLUBILITY	Slightly soluble in water.

CALCIUM HYDROXIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	white
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Calcium hydroxide, calcium hypochlorite and to obtain industry finds its use in many areas. Among these sectors, the production of calcium silicate and calcium silicate bricks made accordingly, are used in the production of gas concrete and mortar construction. It is also used as the main raw material in the production of calcium-based grease in grease production.

CALCIUM HYPOCHLORIDE
POWDER CHLORINE

GENERAL DEFINITIONS	
CAS NUMBER	7778 – 54 – 3
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	chlorine bleach and disinfectant agent
TRADE NAME	Calcium hypochloride

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$CaCI_2O_2$
MOLECULAR WEIGHT	143
PH RANGE	10 - 11
MELTING POINT (C)	100
BOILING POINT (C)	Not specified
DENSITY	2.350 g / cm3
SOLUBILITY	Slightly soluble in water.

CALCIUM HYPOCHLORIDE
POWDER CHLORINE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder or crystal
COLOR	White or gray
ODOR	A characteristic chlorine odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Chlorine powder, within the available chlorine and chlorine odor judge this reason, it is a solid raw white and crystalline form. Chlorine powder, detergent primarily due to receive property and especially chlorine bleach active in the production, pool chemicals, water treatment, paper industry, textile and many other industrial areas is used as chlorine bleach.

CALCIUM CHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	10043 – 52 – 4
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Hygroscopic agent
TRADE NAME	Calcium chloride

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$CaCl_2$
MOLECULAR WEIGHT	110.99
PH RANGE	9
MELTING POINT (C)	772
BOILING POINT (C)	Not specified
DENSITY	2.150 g / cm3
SOLUBILITY	Soluble in water.

CALCIUM CHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Calcium chloride, is generally used as the electrolyte and salt; sports drinks and other beverages can be found in calcium chloride. Canned products preserve the structure of vegetables; pickles and similar products without increasing the sodium content of foods, salty taste to give used for confectionery products can be found in the calcium chloride.Cheese production is added to processed milk, to protect the natural balance between calcium and protein, calcium chloride can be used. Also; if the milk is of poor quality for cheese making, the clot becomes soft, which in addition to causing heavy losses to casein and fat loss, causes poor clotting; a fixed clotting time and the addition is made in order to obtain sufficient calcium chloride in clot firmness.

CALCIUM SILICATE

GENERAL DEFINITIONS

CAS NUMBER	1344 – 95 - 2
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Insulation agent
TRADE NAME	Calcium silicate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Ca_2SiO_3
MOLECULAR WEIGHT	116.16
PH RANGE	Not specified
MELTING POINT (C)	1540
BOILING POINT (C)	Not specified
DENSITY	Not specified
SOLUBILITY	Insoluble in water.

CALCIUM SILICATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Calcium silicate, as well as use in different industrial sector, in the food industry and, in particular, the dough baking soda in the production, as additives in table salt manufacturing and inhibiting agglomeration of the vanilla powder, spice mixes and in many food powder form production are used as additives anti-caking.

CALCIUM SULFATE

GENERAL DEFINITIONS

CAS NUMBER	7778 – 18 – 9
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	coagulant agent
TRADE NAME	Calcium sulfate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$Ca2SO_4$
MOLECULAR WEIGHT	136.14
PH RANGE	Not specified
MELTING POINT (C)	1450
BOILING POINT (C)	Not specified
DENSITY	2.960 g / cm3
SOLUBILITY	Partially soluble in water.

CALCIUM SULFATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Calcium sulfate dihydrate, known E 516 food code, $CaSO_4$ chemical formula and 136.14 g / mole with a molecular weight of white (which can be in different colors) and odorless raw material is a powder or crystal form. Calcium sulfate anhydrate, stabilizers, metal binding agent is used as a nutrient for the yeast bread. Also in the pharmaceutical sector and particularly finds use in the cosmetic and pharmaceutical industry.

CAPRYLYL CAPRYL GLUCOSIDE

GENERAL DEFINITIONS	
CAS NUMBER	68515 – 73 – 1 + 7732 – 18 – 5
GENERAL CATEGORY	Detergent and cosmetic ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Caprylyl capryl glucoside

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	11 – 12
MELTING POINT (C)	0
BOILING POINT (C)	100
DENSITY	1.15 g / cm3
SOLUBILITY	Soluble in water.

CAPRYLYL CAPRYL GLUCOSIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Yellowish
ODOR	Odorless
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Caprylyl capryl glucoside** is a naturally, and vegetable based surfactant derived from corn and natural oils. It is an excellent choice for use in household, industrial and institutional cleansers. **Caprylyl capryl glucoside** is not only highly compatible with alkaline builders (including NaOH), it will also exhibit hydrotropic properties in such systems where conventional nonionic and anionic surfactants have limited compatibility and solubility. Additionally, surfactants of this class have been found to be non-streaking and not leave a film on hard surfaces.

CARRAGEENAN

GENERAL DEFINITIONS	
CAS NUMBER	9000 – 07 - 1
GENERAL CATEGORY	Cosmetic and food ingredient
FACTORIAL CATEGORY	Thickener
TRADE NAME	Carrageenan

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Polymer
MOLECULAR WEIGHT	Polymer
PH RANGE	Not specified
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
DENSITY	Not specified
SOLUBILITY	Insoluble in water. Dispersible in water.

CARRAGEENAN

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Yellowish or white – off
ODOR	Odorless
AID MEASURES	It causes no irritation . In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Carrageenan is a good thickening agent. Therefore, consistency has the possibility of use in some cosmetic and detergent products wanted. Carrageenan, which aim to ensure that the milk pudding with cocoa particles suspended in the milk chocolate production and is used as a gelling agent in an aqueous gel. Also, ice making separation of the water and preventing the formation of large ice crystals is used as an inhibitor of discoloration in the colored cream. Purpose of use, ensure that the shelf stability of the product and to prepare canned sweet gel products are stored without cooling.

CARBOXYMETHYL CELLULOSE
CMC

GENERAL DEFINITIONS	
CAS NUMBER	9004 – 32 – 4
GENERAL CATEGORY	Cosmetic and food ingredient
FACTORIAL CATEGORY	Thickener
TRADE NAME	Carboxymethyl cellulose (CMC)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Polymer
MOLECULAR WEIGHT	Polymer
PH RANGE	Not specified
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
DENSITY	Not specified
SOLUBILITY	Partially soluble in water. Dispersible in water.

CARBOXYMETHYL CELLULOSE
CMC

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Yellowish or white – off
ODOR	Odorless
AID MEASURES	It causes no irritation . In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Carboxymethyl cellulose, CMC Na - CMC, are of commercial importance and produced a stabilized particle form. Easy to dissolve in water and is a hydrocolloid used for multiple purposes because of the Beam feature. Some can be used to thicken cosmetics and detergent products. It is also used in washing powder and detergent cleaning fluid mechanics. Ice cream, sorbet and frozen konfeksiyoner inhibitor of the formation of ice crystals in the product; cream, jelly and pudding in preventing the sineresis event; cakes and other bakery products increased volume of water the provider and protector; dietetic foods and are used as emulsifier and stabilizer in the beverage industry.

CARMOISINE

GENERAL DEFINITIONS	
CAS NUMBER	3567 – 69 – 9
GENERAL CATEGORY	Food ingredient
FOOD E CODE	E 122
TRADE NAME	Carmoisine

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{20}H_{12}N_2Na_2O_7S_2$
MOLECULAR WEIGHT	502.44
PH RANGE	6 – 8
MELTING POINT (C)	300
BOILING POINT (C)	Not specified
DENSITY	0.800 g / cm3
SOLUBILITY	Soluble in water.

CARMOISINE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Red
ODOR	Odorless
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Synthetic food dyes in food coloring product Carmoisine allowed in food codex, which are used in limited quantities in the classroom. Ice cream production, icy drinks, necessary in sodas, fruit soda is used in production. Also in the production of confectionery products, jellies, flavored drinks, chewing gum varieties in sugar and low-sugar, bakery products, confectionery etc. It is used in many food production. Food except pharmaceuticals, cosmetics, detergent products, car care products, in a small amount of agricultural products, cleaning products, colored stones, play dough, etc. are also used for coloring in contact with people of many products. Carmoisine gives red color and is soluble in water.

CARNAUBA WAX

GENERAL DEFINITIONS

CAS NUMBER	8015 – 86 – 9
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Wax
TRADE NAME	Carnauba wax

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Wax
MOLECULAR WEIGHT	Wax
PH RANGE	Not specified
MELTING POINT (C)	84
BOILING POINT (C)	Not specified
DENSITY	0.995 g / cm3
SOLUBILITY	Insoluble in water. Soluble in ethyl ether.

CARNAUBA WAX

PHYSICAL PROPERTIES AND USES

APPEARANCE	Waxy solid
COLOR	Yellow or green
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Carnauba wax is a vegetable-based oil derivatives. Sometimes found in yellow or green. Mainly including cosmetic, food, used as emulsion detergent and car care products in particular. Moreover, in manufacturing of cosmetic products, particularly lipstick, mascara, stick, and hair wax it is highly desirable in production. Apart from these, the skin care products and lotions used.

CETEARETH 11

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Emulsifier
TRADE NAME	Ceteareth 11

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	193.3
MELTING POINT (C)	55
BOILING POINT (C)	204.4
DENSITY	0.995 g / cm3
SOLUBILITY	Dispersible in water

CETEARETH 11

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	White
ODOR	Mild waxy
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	CETEARETH 11, density 0.995 g / cm 3 which, white and viscous liquid is a raw material form. CETEARETH 11 is generally used in the cosmetic industry and sometimes endüstrieyel products and detergent sectors. CETEARETH 11 production and particularly in cosmetics, skin care lotions and creams in hot processes in hair creams, hair care products as an emulsifier.

CETEARETH 20

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Emulsifier
TRADE NAME	Ceteareth 20

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	193.3
MELTING POINT (C)	55
BOILING POINT (C)	204.4
DENSITY	1.015 g / cm3
SOLUBILITY	Dispersible in water

CETEARETH 20

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	White
ODOR	Mild waxy
AID MEASURES	It causes no irritation more. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	CETEARETH 20, density 1.015 g / cm 3 which, white and viscous liquid is a raw material form. CETEARETH 20 is generally used in the cosmetic industry and sometimes endüstrieyel products and detergent sectors. CETEARETH 20 production and particularly in cosmetics, skin care lotions and creams in hot processes in hair creams, hair care products as an emulsifier.

CETRIMONIUM CHLORIDE

GENERAL DEFINITIONS	
CAS NUMBER	112 – 2 – 7
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Cationic
TRADE NAME	Cetrimonium chloride

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{19}H_{42}NCl$
MOLECULAR WEIGHT	320 g / mole
FLASH POINT (C)	Not specified
PH RANGE	6 – 7
BOILING POINT (C)	Not specified
DENSITY	1.070 g / cm3
SOLUBILITY	Soluble in water

CETRIMONIUM CHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and yellow
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation little. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Cetrimonium chloride**, which slightly viscous and liquid raw materials are often dominated by a cationic isopropyl alcohol smell. **Cetrimonium chloride**, is pretty much a raw material used in cosmetics. often used as an emulsifier in their hair cream and skin care cream.

CHOCOLATE BROWN HT

GENERAL DEFINITIONS	
CAS NUMBER	4553 – 89 – 3
GENERAL CATEGORY	Food ingredient
FOOD E CODE	E 155
TRADE NAME	Chocolate Brown HT

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C27H18N4Na2O9S2
MOLECULAR WEIGHT	652.99 g / mole
FLASH POINT (C)	Not specified
PH RANGE	4 – 6
BOILING POINT (C)	Not specified
DENSITY	0.70 g / cm3
SOLUBILITY	Soluble in water

CHOCOLATE BROWN HT

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	Red and brown
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Chocolate Brown HT**, paint with synthetic food dyes in food allowed in class products are used in limited quantities codex. Ice cream, iced drinks, candies, jellies, flavored drinks, chewing gum, baked goods, confectionery etc. are used in many food production. Food except pharmaceuticals, cosmetics, cleaning products, colored stones, play dough, etc. are also used for coloring in contact with people of many products Chocolate Brown HT, give a brown color and is soluble in water.

CINNAMYL ALCOHOL

GENERAL DEFINITIONS

CAS NUMBER	104 – 54 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Organic alcohol
TRADE NAME	Cinnamyl alcohol

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_9H_{10}O$
MOLECULAR WEIGHT	134.17 g / mole
FLASH POINT (C)	Not specified
MELTING POINT (C)	33
BOILING POINT (C)	250
DENSITY	1.04 g / cm3
SOLUBILITY	Slightly soluble in water. Soluble in alcohol.

CINNAMYL ALCOHOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Sweet Odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Cinnamyl alcohol is valuable in perfumery for its odor and fixative properties. It is a component of many flower compositions (lilac, hyacinth, and lily of the valley) and is a starting material for cinnamyl esters, several of which are valuable fragrance materials. In aromas, the alcohol is used forcinnamon notes and for rounding off fruit aromas. Cinnamon spice, floral, green and fermented with powdery balsamic nuances. Use in trace amounts.

CLIMBAZOLE

GENERAL DEFINITIONS	
CAS NUMBER	38083 – 17 – 9
GENERAL CATEGORY	cosmetic ingredient
FACTORIAL CATEGORY	Hair products production
TRADE NAME	Climbazole

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C15H17ClN2O2
MOLECULAR WEIGHT	292.17 g / mole
FLASH POINT (C)	Not specified
MELTING POINT (C)	94
BOILING POINT (C)	Not specified
DENSITY	1.320 g / cm3
SOLUBILITY	Slightly soluble in water.

CLIMBAZOLE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Climbazole, density 1.320 g / cm3, which, in crystalline form, colorless or close to white, is a solid raw material. Climbazole is generally a raw material commonly used in cosmetics. Especially hair care products, used extensively in shampoos for dandruff.

COCAMIDOPROPYL BETAINE

GENERAL DEFINITIONS

CAS NUMBER	61789 – 40 – 0 + 7647 – 14 – 5
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	surfactant
TRADE NAME	Cocamidopropyl betaine

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	5 – 7
MELTING POINT (C)	17
BOILING POINT (C)	100
DENSITY	1.010 g / cm3
SOLUBILITY	Soluble in water.

COCAMIDOPROPYL BETAINE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor slightly.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Betaine, intensive light and peculiar odor with a light, transparent and colorless liquid raw material is an active amphoteric surfactants. Betaine, as surface cleaning and particularly active in the cosmetics and detergents sector used in foaming shampoo and baby products. An extremely low irritation surfactant.

COCAMIDE DEA

GENERAL DEFINITIONS	
CAS NUMBER	68603 – 42 – 9 + 111 – 42 – 2
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	surfactant
TRADE NAME	Cocamide DEA

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	9 – 10
FLASHING POINT (C)	94
BOILING POINT (C)	100
DENSITY	0.950 g / cm3
SOLUBILITY	Soluble in water.

COCAMIDE DEA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and light yellow
ODOR	A characteristic oily odor slightly.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Cocamide DEA, idiosyncratic form of a dense liquid surfactant is a nonionic character is a smell of moving raw materials. Cocamide DEA, it is a surface active materials. It gives support to the main raw material in a mixture of surfactants. Thickener and foaming agent is very stable because of properties. Major fields of cosmetics and detergents are output. It also finds use in industrial cleaning facility area.

COCAMIDE MEA

GENERAL DEFINITIONS	
CAS NUMBER	68603 – 42 – 9 + 141 – 43 – 5
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	surfactant
TRADE NAME	Cocamide MEA

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	9 – 10
FLASHING POINT (C)	94
BOILING POINT (C)	100
DENSITY	0.950 g / cm3
SOLUBILITY	Soluble in water.

COCAMIDE
MEA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and light yellow
ODOR	A characteristic oily odor slightly.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Cocamide MEA, idiosyncratic form of a dense liquid surfactant is a nonionic character is a smell of moving raw materials. Cocamide MEA, it is a surface active materials. It gives support to the main raw material in a mixture of surfactants. Thickener and foaming agent is very stable because of properties. Major fields of cosmetics and detergents are output. It also finds use in industrial cleaning facility area.

CYCLOMETHICONE

GENERAL DEFINITIONS	
CAS NUMBER	541 – 02 – 6
GENERAL CATEGORY	cosmetic ingredient
FACTORIAL CATEGORY	Silicone active
TRADE NAME	Siloxane

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	Not specified
MELTING POINT (C)	-44
BOILING POINT (C)	211
DENSITY	0.950 g / cm3
SOLUBILITY	Slightly soluble in water.

CYCLOMETHICONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Cyclomethicone, a density of 0.950 g / cm 3 with the liquid form of colorless raw material. Cyclomethicone, the generally is a raw material commonly used in cosmetics. Especially hair care products, used extensively in shampoos for dandruff. It is also one of main raw materials in the production of hair serum. It also helps the pigment dispersion for some products.

CYCLOPENTASILOXANE

GENERAL DEFINITIONS	
CAS NUMBER	541 – 02 – 6
GENERAL CATEGORY	cosmetic ingredient
FACTORIAL CATEGORY	Silicone active
TRADE NAME	Cyclopentasiloxane

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{10}H_{30}O_5Si_5$
MOLECULAR WEIGHT	Not specified
FLASH POINT (C)	Not specified
MELTING POINT (C)	-44
BOILING POINT (C)	211
DENSITY	0.950 g / cm3
SOLUBILITY	Slightly soluble in water.

CYCLOPENTASILOXANE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Cyclopentasiloxane**, a density of 0.950 g / cm 3 with the liquid form of colorless raw material. **Cyclopentasiloxane**, the generally is a raw material commonly used in cosmetics. Especially hair care products, used extensively in shampoos for dandruff. It is also one of main raw materials in the production of hair serum. It also helps the pigment dispersion for some products.

C10 – 8EO
ALCOHOL ETHOXYLATE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Nonionic
TRADE NAME	Alcohol ethoxylate (C10 – 8EO)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	215
MELTING POINT (C)	30
PH RANGE	7
DENSITY	1.05 g / cm3
SOLUBILITY	Soluble in water.

C10 – 8EO
ALCOHOL ETHOXYLATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Colorless and clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Detergents definition of nonionic surfactants used in cleaners and other branches of industry. The main application areas, industrial, institutional and household detergents and cleaners are. These include anionic, cationic and nonionic surfactants can easily be combined with other active. Alkyl aryl sulphonates are fully compatible with ether sulfonates and sulfated and sulfonated product. Perfomance synergistic effects are obtained which allow the achievement of high level. Alcohol ethoxylates and n-alkyl-dimethyl-benzyl ammonium chloride, a mixture of products based cationic and other non-ionic surfactant and applied to low-foaming surfactant.

C10 – 3EO
ALCOHOL ETHOXYLATE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Nonionic
TRADE NAME	Alcohol ethoxylate (C10 – 3EO)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	200
MELTING POINT (C)	25
PH RANGE	6 - 7
DENSITY	1.05 g / cm3
SOLUBILITY	Soluble in water.

C10 – 3EO
ALCOHOL ETHOXYLATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Colorless and clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Detergents definition of nonionic surfactants used in cleaners and other branches of industry. The main application areas, industrial, institutional and household detergents and cleaners are. These include anionic, cationic and nonionic surfactants can easily be combined with other active. Alkyl aryl sulphonates are fully compatible with ether sulfonates and sulfated and sulfonated product. Perfomance synergistic effects are obtained which allow the achievement of high level. Alcohol ethoxylates and n-alkyl-dimethyl-benzyl ammonium chloride, a mixture of products based cationic and other non-ionic surfactant and applied to low-foaming surfactant.

C13 – 8EO
ALCOHOL ETHOXYLATE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Nonionic
TRADE NAME	Alcohol ethoxylate (C13 – 8EO)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	230
MELTING POINT (C)	27
PH RANGE	5 - 7
DENSITY	1.150 g / cm3
SOLUBILITY	Soluble in water.

C13 – 8EO
ALCOHOL ETHOXYLATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Colorless and clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Detergents definition of nonionic surfactants used in cleaners and other branches of industry. The main application areas, industrial, institutional and household detergents and cleaners are. These include anionic, cationic and nonionic surfactants can easily be combined with other active. Alkyl aryl sulphonates are fully compatible with ether sulfonates and sulfated and sulfonated product. Perfomance synergistic effects are obtained which allow the achievement of high level. Alcohol ethoxylates and n-alkyl-dimethyl-benzyl ammonium chloride, a mixture of products based cationic and other non-ionic surfactant and applied to low-foaming surfactant.

C 30 - 45
ALKYL METHICONE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Skin emollient
TRADE NAME	C 30 – 45 Alkyl methicone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	28
BOILING POINT (C)	91
PH RANGE	Not specified
DENSITY	1.020 g / cm3
SOLUBILITY	Insoluble in water.

C 30 - 45
ALKYL METHICONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Usually a skin emollient raw materials used in cosmetic products. In particular, sun care products, lotions and creams used in skin care and color cosmetics manufacturing. In products with a smooth, silky and wake-up feature to use a soft feeling to the fore. It also has a unique thickener.

DETERGENT FORMULATIONS ENCYCLOPEDIA

SOLVER

CHEM

PUBLICATIONS

SOLVER CHEM PUBLICATIONS

DETERGENT FORMULATIONS ENCYCLOPEDIA

D

DETERGENT

CHEMICALS

PROPERTIES

SOLVER CHEM PUBLICATIONS

DECYL GLUCOSIDE

GENERAL DEFINITIONS

CAS NUMBER	58846 – 77 – 8
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Surfactant – nonionic
TRADE NAME	Decyl glucoside

CHEMICAL PROPERTIES

CHEMICAL FORMULA	C16H32O6
MOLECULAR WEIGHT	320.43 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	100
PH RANGE	3 – 5
DENSITY	1.100 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

DECYL GLUCOSIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Decyl glucoside, density 1.100 g / cm 3 with the liquid form of colorless raw material. Decyl glycoside used quite cosmetic and detergent sectors. It is also used in the manufacture of certain industrial products. A non-ionic derivatives of irritation is extremely low. Baby products, bath products and hair care products available in easily. Desired work in the wide pH ayarlıg. Plenty of foam constitutes.

DECYL OLEATE

GENERAL DEFINITIONS	
CAS NUMBER	3687 – 46 – 5
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Skin lubricant agent
TRADE NAME	Decyl oleate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Not specified
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	0
BOILING POINT (C)	300
FLASH POINT (C)	150
DENSITY	0.860 g / cm3
SOLUBILITY	Insoluble in water.

DECYL OLEATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Oily
COLOR	Clear and pale yellow
ODOR	A characteristic odor slightly.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Decyl oleate, density of 0.860 g / cm 3, which, in liquid form, is a raw material liquid, transparent and pale yellow. Decyl oleate, quite cosmetic and detergent industry. It is also used in the manufacture of certain industrial products. Often used much in hand and face lotions and creams. Lubricating feature highlighted in use.

DECYLAMINE OXIDE

GENERAL DEFINITIONS	
CAS NUMBER	2016 – 57 – 1
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	Surfactant – nonionic
TRADE NAME	Decylamine oxide

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{10}H_{23}N$
MOLECULAR WEIGHT	157.30
MELTING POINT (C)	12 - 15
BOILING POINT (C)	216
PH RANGE	7 – 10
DENSITY	0.970 g / cm3
SOLUBILITY	Soluble in water.

DECYLAMINE OXIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Slightly viscous liquid.
COLOR	Clear and colorless
ODOR	A characteristic odor slightly.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Decylamine oxide, with its peculiar smell and consistency is a non-ionic surface active materials in liquid form. Decylamine oxide, detergents and foam stabilizers in the production of many industrial products are used as thickeners and cleaner.

DI C12 - 13
ALKYL MALATE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Skin emollient
TRADE NAME	Di C 12 - 13 Alkyl malate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	Not specified
BOILING POINT (C)	550
PH RANGE	Not specified
DENSITY	0.920 g / cm3
SOLUBILITY	Slightly soluble in water.

DI C12 - 13
ALKYL MALATE

PHYSICAL PROPERTIES
AND
USES

APPEARANCE	Viscous liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Di C12 - 13 alkyl malate, density 0.920 g / cm3, a viscous liquid form is a raw material liquid, transparent and light yellow. Di C12 - 13 alkyl malate, quite used in the cosmetic industry. Often used in the hair care products and skin care products pretty. hair and skin softening feature highlighted in use.

DIDECYL DIMETHYL AMMONIUM CHLORIDE

GENERAL DEFINITIONS	
CAS NUMBER	7173 – 51 – 5
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Disinfectant active
TRADE NAME	Didecyl dimethyl ammonium chloride

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{22}H_{48}ClN$
MOLECULAR WEIGHT	362.10 g / mole
MELTING POINT (C)	140
BOILING POINT (C)	Not specified
PH RANGE	6 – 9
DENSITY	0.920 g / cm3
SOLUBILITY	Soluble in water easily.

DIDECYL DIMETHYL AMMONIUM CHLORIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Didecyl dimethyl ammonium chloride, behave has a peculiar odor, liquid form, is a raw material in a transparent and pale yellow. Didecyl dimethyl ammonium chloride, detergent industry and in particular disinfectant products, industrial products and disinfectants in particular surgical tools, disinfectant used in swimming pools and many other related products.

DIETHANOLAMINE
DEA

GENERAL DEFINITIONS	
CAS NUMBER	111 – 42 – 2
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Alkaline agent
TRADE NAME	Diethanolamine (DEA)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_4H_{11}O_4$
MOLECULAR WEIGHT	105.40 g / mole
MELTING POINT (C)	28
BOILING POINT (C)	271
PH RANGE	11 – 12
DENSITY	1.097 g / cm3
SOLUBILITY	Soluble in water.

DIETHANOLAMINE
DEA

PHYSICAL PROPERTIES AND USES

APPEARANCE	liquid
COLOR	Clear and colorless
ODOR	A characteristic ammonia odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Diethanolamine (DEA)**, with its distinctive odor, viscous liquid form, transparent, colorless or slightly yellow color is a raw material. **Diethanolamine (DEA),** adjust pH in many industrial products, especially detergents and cleaning. Especially in the production of liquid detergent used in many oil and dirt solve solved formula. It is also used in some chemical raw materials into a water-based solvent products in the agricultural sector.

DIETHYLENE GLYCOL

GENERAL DEFINITIONS

CAS NUMBER	111 – 46 – 6
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Dehydrating agent
TRADE NAME	Diethylene glycol

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_4H_{10}O_3$
MOLECULAR WEIGHT	106.20 g / mole
FREEZING POINT (C)	-10
BOILING POINT (C)	245
FLASH POINT (C)	143
DENSITY	1.120 g / cm3
SOLUBILITY	Soluble in water easily.

DIETHYLENE GLYCOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Diethylene glycol** is used as a dehydrating agent for natural gas; a raw material for the production of plasticizers and polyester resins; a humectant; a textile lubricant and coupling agent; a solvent in textile dyeing and printing; a constituent of hydraulic fluids; a plasticizer for paper, cork and synthetic sponges; a solvent in printing inks; a raw material for the production of esters used as emulsifiers, demulsifiers, and lubricants; a selective solvent for aromatics in petroleum refining.

DIETHYLHYDROXYLAMINE
DEHA

GENERAL DEFINITIONS	
CAS NUMBER	3710 – 84 – 7
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Corrosion agent
TRADE NAME	DEHA

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_4H_{11}NO$
MOLECULAR WEIGHT	89.10 g / mole
MELTING POINT (C)	-26
BOILING POINT (C)	127
FLASH POINT (C)	Not specified
DENSITY	0.865 g / cm3
SOLUBILITY	Soluble in water easily.

DIETHYLHYDROXYLAMINE
DEHA

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and pale yellow
ODOR	A characteristic ammonia odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	DEHA, has a peculiar smell of ammonia, liquid form, is transparent and colorless raw material. DEHA is used as a corrosion inhibitor in water treatment chemicals as oxygen scavengers. Pasivizat is good. It is also used in various industrial fields.

DIMETHICONE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Silicone agent
TRADE NAME	Dimethicone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	Not specified
BOILING POINT (C)	232
FLASH POINT (C)	250
DENSITY	0.960 g / cm3
SOLUBILITY	Insoluble in water.

DIMETHICONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Light viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Dimethicone**, density 0.960 g / cm 3 with the liquid form is a material transparent and colorless liquid. **Dimethicone**, quite cosmetic and detergent industry. Also used in industrial products. Generally, personal care products, used in hair care products and skin care products, and particularly lotions and creams. Softness to the skin not used in products that feature highlighted in use.

DIOCTYL ETHER

GENERAL DEFINITIONS

CAS NUMBER	629 – 82 – 3
GENERAL CATEGORY	Industrial and cosmetic ingredient
FACTORIAL CATEGORY	Anti-static agent, lubricant, electrical insulator, water repellent
TRADE NAME	Dioctyl ether

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_{16}H_{34}O$
MOLECULAR WEIGHT	242.40 g / mole
MELTING POINT (C)	-7
BOILING POINT (C)	286
FLASH POINT (C)	135
DENSITY	0.806 g / cm3
SOLUBILITY	Insoluble in water.

DIOCTYL ETHER

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	DIOCTYL ETHER, an ether, can act as a base. They form salts with strong acids and addition complexes with Lewis acids. The complex between diethyl ether and boron trifluoride is an example. Ethers may react violently with strong oxidizing agents. In other reactions, which typically involve the breaking of the carbon-oxygen bond, ethers are relatively inert.

DIOCTYL DIMETHYL AMMONIUM CHLORIDE

GENERAL DEFINITIONS	
CAS NUMBER	5538 – 94 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Disinfectant agent
TRADE NAME	Dioctyl methyl ammonium chloride

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C22H48CIN
MOLECULAR WEIGHT	362.10 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	6 – 9
DENSITY	0.920 g / cm3
SOLUBILITY	Miscible with water.

DIOCTYL DIMETHYL AMMONIUM CHLORIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Clear and pale yellow
ODOR	A characteristic alcohol odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Dioctyl dimethyl ammonium chloride**, odorless and clear liquid form is the raw material. **Dioctyl dimethyl ammonium chloride**, detergents sector, and in particular disinfectant products used in industrial products and in particular for the disinfection of surgical tools in place surface disinfectant products and formulas.

DIOCTYL SODIUM SULFOSUCCINATE

GENERAL DEFINITIONS	
CAS NUMBER	577 – 11 – 7
GENERAL CATEGORY	Surfactant
FACTORIAL CATEGORY	Anionic
TRADE NAME	Dioctyl sodium sulfosuccinate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{20}H_{37}NaO_7S$
MOLECULAR WEIGHT	444.56 g / mole
MELTING POINT (C)	157
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	1.10 g / cm3
SOLUBILITY	Partially soluble in water.

DIOCTYL SODIUM SULFOSUCCINATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Creamy liquid
COLOR	White
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Dioctyl sodium sulfosuccinate**, and the oily layer, or cream form and a surfactant raw material to the anionic form of a solid structure with distinctive odor. **Dioctyl sodium sulfosuccinate**, cosmetic, detergent and emulsifier in the production of many industrial products are used as dispersants and wetting agent.

DIPROPYLENE GLYCOL

GENERAL DEFINITIONS	
CAS NUMBER	25265 – 71 – 8
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Essential oil agent
TRADE NAME	Dipropylene glycol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_6H_{14}O_3$
MOLECULAR WEIGHT	134.18 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	232
PH RANGE	Not specified
DENSITY	1.025 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

DIPROPYLENE GLYCOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	DPG; is a colorless, hygroscopic liquid, water, alcohol, esters, in all proportions with the organic solvent solution to form the majority and various vegetable oils and highly soluble in most organic solvents. Many sectors used as solvent DPG major industrial intermediates, unsaturated polyester resins, plasticizers, alkyd resins, cosmetics and urethane find application in the production of poyol and antifoaming agents are included as additives in industrial soaps and functional fluid formulations.

DISODIUM COCOAMPHODIACETATE

GENERAL DEFINITIONS	
CAS NUMBER	68650 – 39 – 5 + 7732 – 18 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Disodium cocoamphodiacetate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	-15
BOILING POINT (C)	100
PH RANGE	8 – 9
DENSITY	1.100 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

DISODIUM COCOAMPHODIACETATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Slightly viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor slightly.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Disodium Cocoamphodiacetate light is intense and unique with a light fragrance is transparent and colorless liquid amphoteric surface active materials. Disodium Cocoamphodiacetate as surface cleaning and particularly active in the cosmetics and detergents sector used in foaming shampoo and baby products. Co-surfactant materials can be said. An extremely low irritation surfactant.

DISODIUM LAURETH SULFOSUCCINATE

GENERAL DEFINITIONS	
CAS NUMBER	49754 – 59 – 4
GENERAL CATEGORY	Detergent and cosmetic ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Disodium laureth sulfosuccinate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	100
BOILING POINT (C)	100
PH RANGE	5 – 7
DENSITY	1.0 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

DISODIUM LAURETH SULFOSUCCINATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Slightly viscous liquid
COLOR	Clear and pale yellow
ODOR	A characteristic odor slightly.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Disodium laureth sulfosuccinate is a chemical found in hair and skin products. Quite simply, it's a de-greasing, emulsifier, and foaming agent used in many shampoos and other cleaning products much like SLS and SLES. It's not an organic substance, but it's also not like other chemicals either.

DMDM HYDANTOIN

GENERAL DEFINITIONS

CAS NUMBER	6440 – 58 – 0
GENERAL CATEGORY	Detergent and cosmetic ingredient
FACTORIAL CATEGORY	Protective
TRADE NAME	DMDM Hydantoin

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_{17}H_{12}N_2O_4$
MOLECULAR WEIGHT	188 g / mole
MELTING POINT (C)	220
BOILING POINT (C)	Not specified
PH RANGE	6 – 8
DENSITY	1.160 g / cm3
SOLUBILITY	Slightly soluble in water.

DMDM HYDANTOIN

PHYSICAL PROPERTIES AND USES

APPEARANCE	liquid
COLOR	Pale yellow
ODOR	A characteristic odor slightly.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	DMDM hydantoin, has a characteristic odor, a liquid form is a raw material liquid, transparent and yellow. DMDM Hydantoin, used as preservatives in many cosmetic products.

D – PANTHENOL

GENERAL DEFINITIONS	
CAS NUMBER	81 – 13 – 0
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Provitamin
TRADE NAME	D – Panthenol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_9H_{19}NO_4$
MOLECULAR WEIGHT	205.26 g / mole
MELTING POINT (C)	17 – 18
BOILING POINT (C)	120
PH RANGE	9 - 10
DENSITY	1.200 g / cm3
SOLUBILITY	Soluble in water.

D – PANTHENOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	liquid
COLOR	Clear and colorless
ODOR	A characteristic odor slightly.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	D - panthenol, density 1.200 g / cm 3 with the liquid form of colorless raw material. D - panthenol typically is a material widely used in cosmetics. Particularly in skincare and hair care products are used quite in. Provitamin B5 is used for the treatment of numerous problems, including particularly the hair dandruff. Observed that diaper rash cream, shower gel and used for the same purpose.

DOLOMITE

GENERAL DEFINITIONS	
CAS NUMBER	1332 – 58 – 7
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Filler agent
TRADE NAME	Dolomite

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$CaMg(CO_3)_2$
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	1770
BOILING POINT (C)	Not specified
PH RANGE	7 – 9
DENSITY	2.860 g / cm3
SOLUBILITY	Insoluble in water.

DOLOMITE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder or crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Dolomite, it finds use in many industrial areas, including industrial detergents in particular. In particular, the main raw material used in the scouring powder. Dolomite is the opportunity to use raw materials in their chemical structure as the main filler.

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EDTA
ETHYLENEDIAMINETETRAACETIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	60 – 00 – 4
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Sequestering
TRADE NAME	Edta

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C10H16N2O8
MOLECULAR WEIGHT	292.24 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	8 – 9
DENSITY	0.860 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

EDTA
ETHYLENEDIAMINETETRAACETIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Edta in powder form and does not smell, is a solid raw material in white. Edta, including detergent first, rather than due to the sequestering and chelating property is used in many industrial fields. In addition, the separation of metal ions, raw assist in degreasing baths, also find use in photography and the electrolysis process.

ESTERQUAT

GENERAL DEFINITIONS

CAS NUMBER	Mixture
GENERAL CATEGORY	Detergent and ındustrial ingredient
FACTORIAL CATEGORY	Cationic softening agent
TRADE NAME	Esterquat

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	6 – 8
DENSITY	1.070 g / cm3
SOLUBILITY	Soluble in cold water and hot water.

ESTERQUAT

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Paste
COLOR	Slightly yellow
ODOR	A characteristic alcohol odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Esterquat usually with isopropyl alcohol odor, which is quite hard or creamy, buttery is raw material having a cationic character. Esterquat used as the main raw material in the detergent industry and in particularly softeners.

ETHYL ALCOHOL

GENERAL DEFINITIONS	
CAS NUMBER	64 – 17 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent and disinfectant agent
TRADE NAME	Ethyl alcohol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_2H_5OH
MOLECULAR WEIGHT	46.06 g / mole
MELTING / FREEZING POINT (C)	-114.5
BOILING POINT (C)	78
FLASH POINT (C)	16.5
DENSITY	0.790 g / cm3
SOLUBILITY	Soluble in cold water.

ETHYL ALCOHOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic alcohol odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Ethyl alcohol, which itself has a fragrance liquid form, transparent and colorless solvent is an organic raw character. Ethyl alcohol, including detergent first, solvents and industrial products used as raw materials in some help. In degreaser, disinfect soil removal and surface cleaning products and surgical tools, disinfectant gel used for hand and face. In cosmetics, the cologne construction used in the manufacture of perfumes and personal care products.

ETHYL ACETATE

GENERAL DEFINITIONS

CAS NUMBER	141 – 78 – 6
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent agent
TRADE NAME	Ethyl acetate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_4H_8O_2$
MOLECULAR WEIGHT	88.10 g / mole
MELTING / FREEZING POINT (C)	-83
BOILING POINT (C)	77
FLASH POINT (C)	Not specified
DENSITY	0.902 g / cm3
SOLUBILITY	Soluble in cold water.

ETHYL ACETATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Ethyl Acetate is used as a solvent for chemical reactions. Because of its odor it is often used in cosmetics and its smell is associated with nail polishes. Additionally, it is used in confectionery, perfumes, and fruits because it evaporates at a fast rate, leaving but the scent of the perfume on the skin. Ethyl acetate is an effective poison for use in insect collector as its vapors are a respiratory tract irritant whose vapors can kill the insect quickly without destroying it, leaving it intact for study.

ETHYL PARABEN

GENERAL DEFINITIONS	
CAS NUMBER	120 – 47 – 8
GENERAL CATEGORY	Detergent and cosmetic ingredient
FACTORIAL CATEGORY	protective agent
TRADE NAME	Ethyl paraben

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_9H_{10}O_3$
MOLECULAR WEIGHT	166.18 g / mole
MELTING / FREEZING POINT (C)	116
BOILING POINT (C)	297
FLASH POINT (C)	Not specified
DENSITY	1.050 g / cm3
SOLUBILITY	Soluble in hot water and cold water.

ETHYL PARABEN

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	A preservative in the cosmetic and detergent industry is quite used. Although recently banned their usage in Europe and America are going to use me in some countries. In particular skin care products, used in hair care products and color cosmetics. It will have the opportunity to use periods in the detergent industry.

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FORMALDEHYDE

GENERAL DEFINITIONS	
CAS NUMBER	50 – 00 – 0 + 7732 – 18 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	protective agent
TRADE NAME	Formaldehyde solution

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	CH_2O
MOLECULAR WEIGHT	30.02 g / mole
MELTING / FREEZING POINT (C)	-15
BOILING POINT (C)	98
PH RANGE	3
DENSITY	1.080 g / cm3
SOLUBILITY	Soluble in water.

FORMALDEHYDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic pungent odor
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Formaldehyde, and a characteristic pungent odor, liquid form, a raw material is clear. Formaldehyde, used as a preservative in many sectors, including in particular detergent. It is forbidden to use as a preservative in recent years. It also found uses as raw materials in industrial areas and in particular resins.

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GLYCERINE

GENERAL DEFINITIONS	
CAS NUMBER	56 – 81 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Moisturizer agent
TRADE NAME	Glycerine

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_3H_5(OH)_3$
MOLECULAR WEIGHT	92.09 g / mole
MELTING / FREEZING POINT (C)	19
BOILING POINT (C)	290
PH RANGE	7
DENSITY	1.260 g / cm3
SOLUBILITY	Soluble in water.

GLYCERINE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Glycerine, non-odor, transparent and colorless, slightly viscous liquid is a raw material form. Glycerine, detergent and used in the cosmetics industry as well as moisturizing and protecting skin. Glycerin is also the main raw material of rubber polisher in auto care products. Also in the paper industry, rubber production is used in textile and many other industries.

GLYCERYL MONOOLEATE

GENERAL DEFINITIONS

CAS NUMBER	25496 – 72 – 4
GENERAL CATEGORY	Industrial and food ingredient
FACTORIAL CATEGORY	Emulsifiers and oilling agent
TRADE NAME	Glyceryl monooleate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_{21}H_{40}O_4$
MOLECULAR WEIGHT	356.54 g / mole
MELTING / FREEZING POINT (C)	Not specified
BOILING POINT (C)	200
FLASH POINT (C)	175
DENSITY	0.900 g / cm3
SOLUBILITY	Partially soluble in water.

GLYCERYL MONOOLEATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Oily liquid
COLOR	Clear and yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Glyceryl monooleate, a density of 0.900 g/cm3 which, oily and slightly viscous form, is a raw material liquid, transparent and yellow. Glyceryl monooleate is used in cosmetics and industrial sectors. It is usually the forefront in the use of the ability of lubricant and emulsified. Plenty includes Evita. Color cosmetics, skin care products and hair care products used in their production.

GLYCERYL MONOSTEARATE

GENERAL DEFINITIONS	
CAS NUMBER	31566 – 31 – 1
GENERAL CATEGORY	Industrial and food ingredient
FACTORIAL CATEGORY	Emulsifiers agent
TRADE NAME	Glyceryl monostearate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{21}H_{42}O_4$
MOLECULAR WEIGHT	358.63 g / mole
MELTING / FREEZING POINT (C)	55
BOILING POINT (C)	260
PH RANGE	6 – 7
DENSITY	0.970 g / cm3
SOLUBILITY	Insoluble in water.

GLYCERYL MONOSTEARATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or powder
COLOR	White and yellow
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Glycerol monostearate, odorless solid raw materials in powder form or a granular white or off-white color. Glycerol monostearate, are used as emulsifiers in certain detergent formulations. In addition, an oven, find use in bread and bakery products to shine. In addition, ice cream and desserts is used to emulsify fat.

GUAR GUM

GENERAL DEFINITIONS	
CAS NUMBER	9000 – 30 – 0
GENERAL CATEGORY	Industrial and food ingredient
FACTORIAL CATEGORY	Thickener
TRADE NAME	Guar gum

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Natural
MOLECULAR WEIGHT	Natural
MELTING / FREEZING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	5 – 6
DENSITY	1.200 g / cm3
SOLUBILITY	Insoluble in water. Disperible in water

GUAR GUM

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White off
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	GUAR GUM, ice cream production, product collection and used for thickening. It is a very important food thickener in ice cream production. GUAR GUM, except ice cream, sauces, the condiments used as a thickener in mayonnaise and ketchup. GUAR GUM, except the food sector, in industry and in the detergent industry in particular thickening gel in making fertilizer and organic agriculture is used as a gelling agent.

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HYDRAZINE

GENERAL DEFINITIONS	
CAS NUMBER	302 – 01 – 2
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Corrosion agent
TRADE NAME	Hydrazine

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	N_2H_4
MOLECULAR WEIGHT	32.05 g / mole
MELTING / FREEZING POINT (C)	1 – 2
BOILING POINT (C)	113.5
PH RANGE	Not specifeid
DENSITY	1.100 g / cm3
SOLUBILITY	Soluble in water.

HYDRAZINE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Hydrazine, which is a slightly peculiar smell, clear and colorless, is a liquid ingredient. Hydrazine, especially water treatment be used in many industrial fields. Good oxygen scavenging and therefore best used as a corrosion inhibitor.

HYDROFLUORIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	7664 – 39 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Strongest inorganic acid
TRADE NAME	Hydrofluoric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	HF
MOLECULAR WEIGHT	Aqueous mixture
MELTING / FREEZING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	1
DENSITY	1.150 g / cm3
SOLUBILITY	Soluble in water.

HYDROFLUORIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes an irritation strongly.
STORAGE	Keep cool and dry place
APPLICATIONS	Hydrochloric acid is the name given to the aqueous solution of hydrogen fluoride. Especially it used in the glass processing industry. Because glass abrasive effect, in glass bottles can not hide. It is a clear liquid identified as HF. Rather it should be considered in use. Opening severe mercury poisoning by inhalation, contact with the skin can cause deep burns. Hydrofluoric acid is also used in high-octane gasoline production. Also among the areas; Purification of aluminum and uranium, rust removal in laundry products is available as catalysts in the oil industry.

HYDROCHLORIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	7647 – 01 – 0
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Strongest mineral acid
TRADE NAME	Hydrochloric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	HCI
MOLECULAR WEIGHT	36.46
MELTING / FREEZING POINT (C)	-62
BOILING POINT (C)	108
PH RANGE	1
DENSITY	1.170 g / cm3
SOLUBILITY	Soluble in water.

HYDROCHLORIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes an irritation strongly.
STORAGE	Keep cool and dry place
APPLICATIONS	Hydrochloric acid is a kind of inorganic acids found use in a variety of fields. Acid concentration, it varies according to the work site is used. Hydrochloric acid pickling of steel, the formation of the organic compound is used in many industries to ensure the production of sub-fields and the pH stability of the inorganic materials in the industrial fields Hydrochloric acid is the main ingredient in the chemical. This acid is used in many small scale industries. Leather processing, cleaning, construction industry is one of them.

HYDROGEN PEROXIDE

GENERAL DEFINITIONS

CAS NUMBER	7722 – 84 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Active oxygen agent
TRADE NAME	Hydrogen peroxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	H_2O_2
MOLECULAR WEIGHT	34.01
MELTING / FREEZING POINT (C)	-10
BOILING POINT (C)	150
PH RANGE	2 – 3
DENSITY	1.135 g / cm3
SOLUBILITY	Soluble in water.

HYDROGEN PEROXIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Acharacteristic odor.
AID MEASURES	It causes an irritation strongly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Hydrogen peroxide**, no smell, the brown color is a solid raw crystal or powder form. **Hydrogen peroxide** is a very important incorporating active oxygen bleach. Including detergent bs, it finds use in many industrial fields. Cosmetics, textile, paper industry, the pharmaceutical industry and is used in many industries.

HYDROXYPROPYL METHYLCELLULOSE

GENERAL DEFINITIONS	
CAS NUMBER	9050 – 31 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Thickener
TRADE NAME	Hydroxypropyl methylcellulose

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING / FREEZING POINT (C)	Not spesified
BOILING POINT (C)	Not specified
PH RANGE	8 – 10
DENSITY	Not spesified
SOLUBILITY	Insoluble in water. Dispersıble in water.

HYDROXYPROPYL METHYLCELLULOSE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White or white – off
ODOR	Odorless
AID MEASURES	It causes no irritation . In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Hydroxypropyl methyl cellulose, one of the cold water soluble additives. Methyl cellulose solution is heated and the viscosity is increased from 50 to 55 ° C "is gelled. This gelation point may be raised relative to the hydroxypropyl groups (max 85 ° "ye) .Jelleş point, it modified using additives. With a plurality of electrolyte sucrose, glycerol, sorbitol and reducing the gelation point it has enhancing effect of ethanol and propylene glycol. Emulsifiers, film former, protective colloids, stabilizers, may be used as suspending agents and thickeners. Bakery products, mayonnaise, can be used in ice cream and other dairy products.

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ISOPROPYL ALCOHOL
IPA

GENERAL DEFINITIONS	
CAS NUMBER	67 – 63 – 0
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Isopropyl alcohol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_3H_8O
MOLECULAR WEIGHT	61.10 g / mole
MELTING POINT (C)	-88.5
BOILING POINT (C)	82.5
PH RANGE	Not specifeid
DENSITY	0.785 g / cm3
SOLUBILITY	Soluble in water.

ISOPROPYL ALCOHOL
IPA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic alcohol odor
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Isopropyl alcohol**, which has a smell its liquid form, transparent and colorless solvent is an organic raw character. **Isopropyl alcohol**, detergents and industrial degreasing cleaning products used as soil removal. In particular, clear glass production, surface cleaners, dirt and oils resolve in making auto car perfume, fragrance is used quite often in the manufacturing room. Water-based and used in the manufacture of auto brake pads spray.

ISOBUTYL ALCOHOL

GENERAL DEFINITIONS	
CAS NUMBER	78 – 83 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Isobutyl alcohol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_4H_{10}O$
MOLECULAR WEIGHT	74.12 g / mole
MELTING POINT (C)	-108
BOILING POINT (C)	107
PH RANGE	Not specifeid
DENSITY	0.800 g / cm3
SOLUBILITY	Slightly soluble in water.

ISOBUTYL ALCOHOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic alcohol odor
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Isobutyl alcohol, food flavorings used in the manufacturing process. In addition, as a solvent in paints and varnishes industry, the food industry in the manufacture of ink as a flavoring agent in the production of automotive paint and lacquer, the contribution of gasoline to reduce icing in the carburetor, as an additive for automotive paint cleaner, plastics, in rubber and other dispersions sees use as a plasticizer.

ISOPROPYL ACETATE

GENERAL DEFINITIONS	
CAS NUMBER	108 – 22 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Isopropyl acetate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_5H_{10}O_2$
MOLECULAR WEIGHT	102.13 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	89
PH RANGE	Not specifeid
DENSITY	0.870 g / cm3
SOLUBILITY	Soluble in water.

ISOPROPYL ACETATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Isopropyl acetate** is used as solvent in the food industry and especially the areas of food flavorings. Also oils for the manufacture of cellulose and used Iso propyl acetate solvent for printing inks, sees use in the cosmetics industry and the plastics industry.

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LINEAR ALKYLBENZENE SULFONIC ACID LABSA

GENERAL DEFINITIONS	
CAS NUMBER	27176 – 87 – 0
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Anionic surfactant
TRADE NAME	Linear alkylbenzene sulfonic acid (LABSA)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$CH_3(CH_2)_{11}C_6H_4SO_3H$
MOLECULAR WEIGHT	326.49 g / mole
MELTING POINT (C)	10
BOILING POINT (C)	315
PH RANGE	2
DENSITY	1.200 g / cm3
SOLUBILITY	Soluble in water.

LINEAR ALKYLBENZENE SULFONIC ACID
LABSA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Brown
ODOR	A characteristic acidic odor
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	LABSA, Linear Alkyl Benzene (LAB) 's result of reaction with SO3 (sulfonation) is formed. Nowadays, liquid, gel or powder detergent LABSA main surfaces of the process is used as the active ingredient. One of the main raw material of synthetic detergent industry. Laundry, dishwasher powder detergent, detergent gels, liquid soaps, cleaning powder, fatty soaps and so on. as. As mercerized and washing agents used in the textile industry.

LAKE
ALLURA RED AC

GENERAL DEFINITIONS	
CAS NUMBER	25956 – 17 – 6
GENERAL CATEGORY	Food ingredient
FACTORIAL CATEGORY	Food dye
TRADE NAME	Lake allura red AC

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{18}H_{14}N_2Na_2O_8S_2$
MOLECULAR WEIGHT	496.42 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	Not specified
SOLUBILITY	Insoluble in water. Soluble in oil

LAKE
ALLURA RED AC

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Red
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Synthetic food dyes which are oil-based or oil-soluble class **Lake allura red AC** allowed in food coloring in products are used in a limited amount of food codex. Ice cream production, icy drinks, needed in sparkling beverages, soda is used in fruit production. Also in the production of confectionery products, jellies, flavored drinks, sweetened and low-sugar chewing gum in kind, bakery products, confectionery, etc. it is used in many food production.Food except pharmaceuticals, cosmetics, detergent products, car care products, in a small amount of agricultural products, cleaning products, colored stones, play dough, etc. are also used for coloring in contact with people of many products **Lake allura red AC** give a bright red color and the food is fat-soluble paint.

LAKE AMARANTH

GENERAL DEFINITIONS	
CAS NUMBER	915 – 67 – 3
GENERAL CATEGORY	Food ingredient
FACTORIAL CATEGORY	Food dye
TRADE NAME	Lake amaranth

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	6
DENSITY	0.500 – 0.700 g / cm3
SOLUBILITY	Insoluble in water. Soluble in oil

LAKE AMARANTH

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	Bluish red
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Synthetic food dyes, which are used in limited quantities of class LAKE AMARANTH permitted products in the food codex. Food except pharmaceuticals, cosmetics, cleaning products, colored stones, play dough, etc. are also used for coloring many people in contact with the product. Which can be emulsified with the help of some emulsifiers can be used in water-based products. LAKE AMARANTH food dye red color and a fat soluble food dyes dispersed in oil.

LAKE PONCEAU 4R

GENERAL DEFINITIONS	
CAS NUMBER	2611 – 82 – 7
GENERAL CATEGORY	Food ingredient
FACTORIAL CATEGORY	Food dye
TRADE NAME	Lake ponceau 4R

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{20}H_{11}N_2Na_3O_{10}S_3$
MOLECULAR WEIGHT	604.46 g / mole
FOOD E CODE	E 124
BOILING POINT (C)	Not specified
PH RANGE	6
DENSITY	0.500 – 0.700 g / cm3
SOLUBILITY	Insoluble in water. Soluble in oil

LAKE
PONCEAU 4R

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder or granular
COLOR	Red
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Synthetic food dyes, which are used in limited quantities of class LAKE PONCEAU 4R permitted products in the food codex. Food except pharmaceuticals, cosmetics, cleaning products, colored stones, play dough, etc. are also used for coloring many people in contact with the product. Which can be emulsified with the help of some emulsifiers can be used in water-based products. LAKE PONCEAU 4R food dye red color and a fat soluble food dyes dispersed in oil.

LAURYL BETAINE POWDER

GENERAL DEFINITIONS	
CAS NUMBER	683 – 10 – 3
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Lauryl betaine powder

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C16H33NO2
MOLECULAR WEIGHT	271.44 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	1.200 g / cm3
SOLUBILITY	Soluble in water.

LAURYL BETAINE
POWDER

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	It is a hair and skin conditioner, a mild surface-active agent (surfactant) and works well in shampoo, shower gel or any cleansing product. Surfactants are part water-soluble and part oil-soluble, allowing the oil and water to become dispersed. Lauryl betaine is capable of carrying both a positive and a negative charge, making it a useful stabiliser. It improves quality and stability of foam, making it more durable in the presence of oil and dirt.The mildness of lauryl betaine (and its skin and hair conditioning properties) makes this an excellent ingredient to use across our range of shower gels, shampoos and bath products.

LAURYL BETAINE LIQUID

GENERAL DEFINITIONS	
CAS NUMBER	683 – 10 – 3
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Lauryl betaine liquid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C16H33NO2
MOLECULAR WEIGHT	271.44 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	100
PH RANGE	6 – 8
DENSITY	1.050 g / cm3
SOLUBILITY	Soluble in water.

LAURYL BETAINE
LIQUID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	It is a hair and skin conditioner, a mild surface-active agent (surfactant) and works well in shampoo, shower gel or any cleansing product. Surfactants are part water-soluble and part oil-soluble, allowing the oil and water to become dispersed. Lauryl betaine is capable of carrying both a positive and a negative charge, making it a useful stabiliser. It improves quality and stability of foam, making it more durable in the presence of oil and dirt. The mildness of lauryl betaine (and its skin and hair conditioning properties) makes this an excellent ingredient to use across our range of shower gels, shampoos and bath products.

LAURYL GLUCOSIDE

GENERAL DEFINITIONS	
CAS NUMBER	59122 – 55 – 3
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Lauryl glucoside

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C18H36O6
MOLECULAR WEIGHT	348.48 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	Not specified
PH RANGE	9.5
DENSITY	1.0 g / cm3
SOLUBILITY	Soluble in water.

LAURYL GLUCOSIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Lauryl glucoside is an non-ionic surfactant that can be used as a foaming agent, conditioner or emulsifier. Lauryl glucoside has excellent foaming capacity and good dermatological compatibility. Therefore it is suitable for use as a base surfactant or a co-surfactant in cosmetic surfactant cleansing preparations. **Products to Use in** • Shower Gel • Shampoo • Face Wash • Bath Foam

LAURYLDIMETHYLAMINE OXIDE

GENERAL DEFINITIONS	
CAS NUMBER	1643 – 20 – 5
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Lauramine oxide

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{14}H_{31}NO$
MOLECULAR WEIGHT	229.41 g / mole
FLASH POINT (C)	94
BOILING POINT (C)	100
PH RANGE	8
DENSITY	0.965 g / cm3
SOLUBILITY	Soluble in water.

LAURYLDIMETHYLAMINE OXIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	In cosmetics and personal-care products, Lauramine oxides are amine oxides that are used mostly in hair-care products as foam builders and stabilizers, viscosity enhancers, emollients, conditioners, emulsifiers, antistatic agents and wetting agents. Lauramine oxides are used mainly in hair-care products such as shampoos, hair rinses, tonics and hair-grooming aids.

LIQUID PARAFFIN

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Skin emollient
TRADE NAME	Liquid paraffin

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
FLASH POINT (C)	150
BOILING POINT (C)	350
PH RANGE	Not specified
DENSITY	0.845 g / cm3
SOLUBILITY	Insoluble in water. Soluble in petroleum oils.

LIQUID PARAFFIN

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid oily
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Liquid paraffin is a type of medicine called a laxative. It works by softening and lubricating the stools. This helps the stools to move more easily through the bowel. This medicine relieves constipation, making stools easier to pass. This means it can also be used to reduce pain associated with passing stools in people with conditions affecting the anus, such as piles (haemorrhoids) or anal fissure. It is not often used, as the liquid paraffin can sometimes leak from the bottom.

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MAGNESIUM HYDROXIDE SOLID

GENERAL DEFINITIONS	
CAS NUMBER	1309 – 42 – 8
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Magnesium source
TRADE NAME	Magnesium hydroxide powder

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$Mg(OH)_2$
MOLECULAR WEIGHT	58.32 g / mole
MELTING POINT (C)	350
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	2.360 g / cm3
SOLUBILITY	Slightly soluble in water.

MAGNESIUM HYDROXIDE
SOLID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Magnesium is a naturally occurring mineral. Magnesium is important for many systems in the body especially the muscles and nerves. Magnesium hydroxide also reduces stomach acid, and increases water in the intestines which may induce defecation. Magnesium hydroxide is used as a laxative to relieve occasional constipation (irregularity) and as an antacid to relieve indigestion, sour stomach, and heartburn.

MAGNESIUM HYDROXIDE LIQUID

GENERAL DEFINITIONS	
CAS NUMBER	1309 – 42 – 8 + 7732 – 18 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Magnesium source
TRADE NAME	Magnesium hydroxide liquid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	2100
BOILING POINT (C)	Not specified
PH RANGE	10 - 11
DENSITY	1.45 g / cm3
SOLUBILITY	Slightly soluble in water. Dispersible in water.

MAGNESIUM HYDROXIDE LIQUID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	blurry
ODOR	A characteristic odor
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Magnesium hydroxide, known for food code E 528 , gray and raw materials is a unique fragrant liquid form. Magnesium hydroxide, the water forms a suspension known as milk of magnesia. This suspension is used as an antacid and laxative. In addition, strong alkalis, used as acidity regulator. Rennet in cheese-enhancing and protecting the color of vegetables during canning process is used as a stabilizer.

MAGNESIUM CARBONATE

GENERAL DEFINITIONS	
CAS NUMBER	23389 – 33 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Food and medical agent
TRADE NAME	Magnesium carbonate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$MgCO_3$
MOLECULAR WEIGHT	84.3 g / mole
MELTING POINT (C)	662
BOILING POINT (C)	Not specified
PH RANGE	8
DENSITY	2.950 g / cm3
SOLUBILITY	Slightly soluble in water.

MAGNESIUM CARBONATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	$MgCO_3$ is also used in flooring, fireproofing, fire extinguishing compositions, cosmetics, dusting powder, and toothpaste. Other applications are as filler material, smoke suppressant in plastics, a reinforcing agent in neoprene rubber, drying agent, a laxative to loosen the bowels, and color retention in foods. In addition, high purity magnesium carbonate is used as antacid and as an additive in table salt to keep it free flowing. Magnesium carbonate is fortunately both very safe and extremely abundant. It can be acquired easily and cheaply and is relatively safe to handle as long as one is careful not to get it in the eyes, where it is an irritant.

MAGNESIUM CHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	7791 – 18 – 6
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Typical ionic halides
TRADE NAME	Magnesium chloride

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$MgCI_2.6H_2O$
MOLECULAR WEIGHT	203.30 g / mole
MELTING POINT (C)	118
BOILING POINT (C)	Not specified
PH RANGE	7
DENSITY	1.590 g / cm3
SOLUBILITY	Soluble in water.

MAGNESIUM CHLORIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Magnesium chloride** in water and soluble in alcohol. Cement, drilling and grinding wheels, the binding of chemical production and powder, magnesium metal and production of fire-proofing agent, as an additive in textile and paper making, using disinfectants, fire extinguishers and used in ceramics. **Magnesium chloride**, highways or ice in the ice against the rock salt instead of racking use in recent years have become widespread. In winter the liquid magnesium chloride before precipitation to dry before freezing the wet road surface or ground spraying ice or snow is prevented from sticking to each other and connected to the road. **Magnesium chloride** is used as icing inhibitors can be reduced corrosivity with other chemicals added to the solution

MAGNESIUM LAURETH SULFATE

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Magnesium laureth sulfate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	Not specified
BOILING POINT (C)	100
PH RANGE	6 – 8
DENSITY	1.050 g / cm3
SOLUBILITY	Partially soluble in water.

MAGNESIUM LAURETH SULFATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Generally, the main raw materials used in cosmetic and detergent products. Cosmetics; shower gels in the production of hygiene products, and wipes used in hair care formulations.

MAGNESIUM OXIDE

GENERAL DEFINITIONS

CAS NUMBER	1309 – 48 – 4
GENERAL CATEGORY	Industrial and food ingredient
FOOD E CODE	E 530
TRADE NAME	Magnesium oxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	MgO
MOLECULAR WEIGHT	40.30 g / cm3
MELTING POINT (C)	2800
BOILING POINT (C)	3600
PH RANGE	Not specified
DENSITY	3.600 g / cm3
SOLUBILITY	Slightly soluble in water.

MAGNESIUM OXIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Magnesium oxide, known for its food code E 530, with chemical formula MgO and 40.3 g / mole molecular weight of the white, odorless and which is a raw material in powder form. Magnesium oxide, are quite raw materials used in industrial fields, in the food industry, as anti-agglomeration agents and acidic agent.

MAGNESIUM SILICATE

GENERAL DEFINITIONS

CAS NUMBER	1343 – 88 – 0
GENERAL CATEGORY	Industrial and food ingredient
FOOD E CODE	E 553
TRADE NAME	Magnesium silicate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$MgO{:}SiO_2(x)H_2O$
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	1910
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	2.100 g / cm3
SOLUBILITY	Insoluble in water.

MAGNESIUM SILICATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder or crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Magnesium silicate, known by E 553 food code is an odorless white crystal and raw materials. Melting point is quite high. Insoluble in water. Magnesium silicate, as well as use in different industrial sector, in the food industry and, in particular, the dough baking soda in the production, as additives in table salt manufacturing and inhibiting agglomeration of the vanilla powder, spice mixes and in many food powder form production are used as additives anti-caking.

MAGNESIUM SULFATE MONOHYDRATE

GENERAL DEFINITIONS	
CAS NUMBER	14168 – 73 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Inorganic salt
TRADE NAME	Magnesium sulfate monohydrate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$MgSO_4.H_2O$
MOLECULAR WEIGHT	138.38 g / mole
MELTING POINT (C)	150
BOILING POINT (C)	Not specified
PH RANGE	Not specified
DENSITY	1.670 g / cm3
SOLUBILITY	Soluble in water.

MAGNESIUM SULFATE MONOHYDRATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Magnesium sulphate monohydrate, known for its food code E 518 , with MgSO4.H2O chemical formula and 138.38 g / mole molecular weight of the white, odorless and which is a raw material in powder form. Magnesium sulphate monohydrate is quite a raw material used in the industrial field and in the food industry, it has endurance-enhancing properties of nutritional supplements and canned.

MENTHOL

GENERAL DEFINITIONS	
CAS NUMBER	2216 – 51 – 5
GENERAL CATEGORY	Cosmetic and medicine ingredient
FACTORIAL CATEGORY	Mint oil
TRADE NAME	Menthol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{10}H_{20}O$
MOLECULAR WEIGHT	156.30 g / mole
MELTING POINT (C)	42
BOILING POINT (C)	212
PH RANGE	Not specified
DENSITY	0.890 g / cm3
SOLUBILITY	Slightly soluble in water. Soluble in gycerine and monopropylene glycol.

MENTHOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	A characteristic mint odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Levomenthol is a levo isomer of menthol, an organic compound made synthetically or obtained from peppermint or mint oils with flavoring and local anesthetic properties. When added to pharmaceuticals and foods,menthol functions as a fortifier for peppermint flavors. It also has a counterirritant effect on skin and mucous membranes, thereby producing a local analgesic or anesthetic effect.

METHYLISOTHIAZOLINONE

GENERAL DEFINITIONS	
CAS NUMBER	12672 – 54 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Protective agent
TRADE NAME	Methylisothiazolinone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_4H_5NOS.HCL$
MOLECULAR WEIGHT	151.56 g / mole
MELTING POINT (C)	48
BOILING POINT (C)	93
PH RANGE	Not specified
DENSITY	1.020 g / cm3
SOLUBILITY	Soluble in water

METHYLISOTHIAZOLINONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Methylisothiazolinone, density 1.020 g / cm 3 with powder or crystalline form, is a solid raw material in white. Methylisothiazolinone, cosmetics and fairly used in the industrial sector. Antibacterial, that are used as preservatives. Generally, all cosmetic products and especially in the production of shampoo, the solution used in the production of wet wipes and the skin care lotions and creams.

METHYL ALCOHOL

GENERAL DEFINITIONS

CAS NUMBER	67 – 56 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Methyl alcohol

CHEMICAL PROPERTIES

CHEMICAL FORMULA	CH_3OH
MOLECULAR WEIGHT	32.04 g / mole
MELTING POINT (C)	-97.5
BOILING POINT (C)	64.5
PH RANGE	Not specified
DENSITY	0.795 g / cm3
SOLUBILITY	Soluble in water

METHYL ALCOHOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic alcohol odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Methyl alcohol, It has a peculiar odor in liquid form, a transparent and colorless character is an organic solvent raw material. A poisonous alcohol. Methyl Alcohol, including in particular detergent, industrial products and solvents used as raw materials in some help. Generally, due to its low freezing point, used as anti-icing and ice cold pile in relief.

METHYL BENZOATE

GENERAL DEFINITIONS	
CAS NUMBER	93 – 58 – 3
GENERAL CATEGORY	Industrial and food ingredient
FACTORIAL CATEGORY	Ester agent
TRADE NAME	Methyl benzoate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C8H8O2
MOLECULAR WEIGHT	136.14 g / mole
MELTING POINT (C)	-13.5
BOILING POINT (C)	199
PH RANGE	Not specified
DENSITY	1.095 g / cm3
SOLUBILITY	Partially soluble in water

METHYL BENZOATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and light yellow
ODOR	A characteristic aodor.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Methyl benzoate, grape, a liquid synthetic molecule used in the production of fragrances such as phenol and camphor. cosmetic and detergent field and especially in personal care products used as a basis for fragrances in cosmetics. In the food industry it is used as an aid in some flavor.

METHYL ETHYL KETONE

GENERAL DEFINITIONS	
CAS NUMBER	78 – 93 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent
TRADE NAME	Methyl ethyl ketone

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_4H_8O
MOLECULAR WEIGHT	72.11 g / mole
MELTING POINT (C)	-86
BOILING POINT (C)	78.6
PH RANGE	Not specified
DENSITY	0.805 g / cm3
SOLUBILITY	Partially soluble in water

METHYL ETHYL KETONE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation slightly. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Methyl ethyl ketone, ketone-type active organic solvents. Phenolic alkyd, in systems containing vinyl resin in cellulosic system; used in paint removers and adhesives. It also has a professional cleaning and maintenance products for use in the food and beverage industry. Food is used as a carrier in food flavorings used in industry.

METHYL PARABEN

GENERAL DEFINITIONS

CAS NUMBER	99 – 76 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Preservative agent
TRADE NAME	Methyl paraben

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_4H_8O_3$
MOLECULAR WEIGHT	152.15 g / mole
MELTING POINT (C)	131
BOILING POINT (C)	270
PH RANGE	Not specified
DENSITY	0.910 g / cm3
SOLUBILITY	Soluble in water

METHYL PARABEN

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	A preservative in the cosmetic industry is quite used. Although recently banned their usage in Europe and America are going to use me in some countries. In particular, skin care products, used in hair care products and color cosmetics. It will have the opportunity to use periods in the detergent industry.

MONOETHANOLAMINE
MEA

GENERAL DEFINITIONS	
CAS NUMBER	141 – 43 – 5
GENERAL CATEGORY	Industrial detergent ingredient
FACTORIAL CATEGORY	Amino alcohol group
TRADE NAME	Monoethanolamine MEA

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C_2H_7NO
MOLECULAR WEIGHT	61.08 g / mole
MELTING POINT (C)	10.3
BOILING POINT (C)	170
PH RANGE	10
DENSITY	1.012 g / cm3
SOLUBILITY	Soluble in water

MONOETHANOLAMINE
MEA

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Clear and light yellow
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Monoethanolamine (MEA), with its distinctive odor, viscous liquid form, transparent, colorless or slightly yellow color is a raw material. Monoethanolamine (MEA), the pH adjustment in many industrial products, especially detergents and cleaning. Especially in the production of liquid detergent used in many oil and dirt solve solve formulas. It is also used in some chemical raw materials into a water-based solvent products in the agricultural sector.

MONOETHYENE GLYCOL

GENERAL DEFINITIONS	
CAS NUMBER	107 – 21 - 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Antifreeze agent
TRADE NAME	Monoethylene glycol (meg)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_2H_6O_2$
MOLECULAR WEIGHT	62 g / mole
MELTING POINT (C)	-12
BOILING POINT (C)	197
FLASH POINT (C)	111
DENSITY	1.120 g / cm3
SOLUBILITY	Soluble in water

MONOETHYLENE GLYCOL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Monoethylene glycol, which has its own smell, transparent and colorless, slightly viscous liquid is a raw form. Monoethylene glycol, automotive glass of water in the detergent industry, and especially car care products used as raw materials for production for the production of antifreeze and cars. Moreover, this raw material is used in many industrial fields.

MONOPROPYLENE GLYCOL

GENERAL DEFINITIONS	
CAS NUMBER	57 – 55 - 6
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Skin moisture agent
TRADE NAME	Monopropylene glycol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_3H_8O_2$
MOLECULAR WEIGHT	76.09 g / mole
MELTING POINT (C)	Not specified
BOILING POINT (C)	185
FLASH POINT (C)	104
DENSITY	1.030 g / cm3
SOLUBILITY	Soluble in water

MONOPROPYLENE GLYCOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes no irritation. In contact, wash plenty of water.
STORAGE	Keep cool and dry place
APPLICATIONS	Monopropylene glycol, are widely use in the formulation of the bakery product. Aromas and essences in the industry, medicine, and is used in cosmetics. (In the form of technical quality, it is an important material for polyurethane plastics and polyester resins. In the tobacco industry, is also used in the food industry for the lubrication of machine freezer). Mouth washer (mouthwash), tooth pastes, ointments, skin creams, shampoos and perfumes (too much even when diluted with water solutions containing propylene glycol usually remains clear) Emulsion formation as a preservative in cosmetics,Fragrances (essences) as a solvent for,as an extract in active agent to obtain extracts from natural sources, as a lubricant (eg for machinery used in the cosmetics industry) it is used.

DETERGENT FORMULATIONS ENCYCLOPEDIA

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DETERGENT FORMULATIONS ENCYCLOPEDIA



DETERGENT CHEMICALS PROPERTIES

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NITRIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	7697 – 37 – 2
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Strong acid
TRADE NAME	Nitric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	HNO_3
MOLECULAR WEIGHT	63.01 g / mole
MELTING POINT (C)	-42
BOILING POINT (C)	83
PH RANGE	1
DENSITY	1.510 g / cm3
SOLUBILITY	Soluble in water

NITRIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes an irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Nitric Acid is available in many areas of opportunity for industrial use. Among the major sectors, the paper industry, metal industry, textile industry and refinements can be counted. Nitric Acid, also in the production of fertilizers, explosives, also used in the pH adjustment and electro-polishing works.

NONYLPHENOL ETHOXYLATE
NP 6

GENERAL DEFINITIONS	
CAS NUMBER	127087 – 87 – 0
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Nonionic agent
TRADE NAME	Nonylphenol ethoxylate (NP: 6)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	-8
BOILING POINT (C)	293
PH RANGE	6 – 7
DENSITY	1.060 g / cm3
SOLUBILITY	Soluble in water

NONYLPHENOL ETHOXYLATE
NP: 6

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Nonylphenol ethoxylate (NP: 6), transparent, colorless and form liquid is a raw material with a distinctive odor. Nonylphenol ethoxylate (NP: 6), are used as nonionic surface active in all cleaning jobs. Good emulsifier, detergent industry, textile, car care products and many more are used as industrial cleaners and emulsifiers.

NONYLPHENOL ETHOXYLATE
NP 10

GENERAL DEFINITIONS	
CAS NUMBER	127087 – 87 – 0
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Nonionic agent
TRADE NAME	Nonylphenol ethoxylate (NP: 10)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	-8
BOILING POINT (C)	293
PH RANGE	6 – 7
DENSITY	1.060 g / cm3
SOLUBILITY	Soluble in water

NONYLPHENOL ETHOXYLATE
NP: 10

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	**Nonylphenol ethoxylate** (NP: 10), transparent, colorless and form liquid is a raw material with a distinctive odor. **Nonylphenol ethoxylate** (NP: 10), are used as nonionic surface active in all cleaning jobs. Good emulsifier, detergent industry, textile, car care products and many more are used as industrial cleaners and emulsifiers.

DETERGENT FORMULATIONS ENCYCLOPEDIA

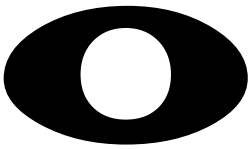
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DETERGENT FORMULATIONS ENCYCLOPEDIA



DETERGENT CHEMICALS PROPERTIES

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OCTYLDODECANOL

GENERAL DEFINITIONS	
CAS NUMBER	5333 – 42 – 6
GENERAL CATEGORY	Industrial and cosmetic ingredient
FACTORIAL CATEGORY	Skin emollient agent
TRADE NAME	Octyldodecanol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{20}H_{42}O$
MOLECULAR WEIGHT	298.56
MELTING POINT (C)	-23
BOILING POINT (C)	223
FLASH POINT (C)	188
DENSITY	0.840 g / cm3
SOLUBILITY	Insoluble in water

OCTYLDODECANOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid
COLOR	Clear and colorless
ODOR	A characteristic odor
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	**Octyldodecanol** a density of 0.840 g / cm 3, which, in the form of viscous liquid, transparent and light yellow liquid with a characteristic odor and is a raw material. **Octyldodecanol**, cosmetics and fairly used in the industrial sector. Emollient and moisturizing. Usually it is seen in quite used as hand and face creams and lotions.

OLEIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	112 – 80 – 1
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Fatty acid
TRADE NAME	Oleic acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C18H34O2
MOLECULAR WEIGHT	282.47 g / mole
MELTING POINT (C)	17
BOILING POINT (C)	286
FLASH POINT (C)	Not specified
DENSITY	0.840 g / cm3
SOLUBILITY	Insoluble in water. Soluble in methyl alcohol

OLEIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Viscous liquid oily.
COLOR	Clear and colorless
ODOR	A characteristic oily odor.
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	**Oleic acid** is used as raw material for producing soap. Textile and leather also help in the detergent manufacturing, ink manufacturing, amine as a lubricant agent is used in ceramics and paper industry

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PEA GREEN

GENERAL DEFINITIONS	
CAS NUMBER	2353 – 45 – 9
GENERAL CATEGORY	Industrial dye
FACTORIAL CATEGORY	Food color
TRADE NAME	Pea green color

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{37}H_{34}N_2Na_2O_{10}S_2$
MOLECULAR WEIGHT	582.66
MELTING POINT (C)	Not specified
PH RANGE	5 – 6
FLASH POINT (C)	Not specified
DENSITY	0.80 g / cm3
SOLUBILITY	Soluble in water

PEA
GREEN

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	Red
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Food coloring GREEN PEAR class food in which synthetic food dyes allowed in the codex products are used in limited amounts. Ice cream production, icy drinks, necessary in sodas, fruit soda is used in production. Also in the production of confectionery products, jellies, flavored drinks, chewing gum varieties in sugar and low-sugar, bakery products, confectionery etc. it is used in many food production. Food except pharmaceuticals, cosmetics, detergent products, car care products, in a small amount of agricultural products, cleaning products, colored stones, play dough, etc. are also used for coloring many people in contact with the product. GREEN PEA gives bright green color and is soluble in water.

PEG 400
POLYETHYLENE GLYCOL

GENERAL DEFINITIONS	
CAS NUMBER	25322 – 68 – 3
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Solvent and plasticizer agent
TRADE NAME	Peg 400 polyethylene glycol

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Not specified
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	8
PH RANGE	4.5 – 7.5
FLASH POINT (C)	238
DENSITY	1.125 g / cm3
SOLUBILITY	Soluble in water

PEG 400
POLYETHYENE GLYCOL

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	PEG 400, density 1.125 g / cm3, liquid form, is a solid raw material which is transparent and colorless and characteristic odor. PEG 400, cosmetics, detergents and fairly used in the industrial sector. Nonionic characteristic, cleaning solvent is used as thickener and emulsifier.

PEG 40
HYDROGENATED CASTOR OIL

GENERAL DEFINITIONS	
CAS NUMBER	61788 – 85 – 0
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	emulsifier
TRADE NAME	Peg 40 hydrogenated castor oil

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Not specified
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	Not specified
PH RANGE	6 – 7
FLASH POINT (C)	170
DENSITY	1.125 g / cm3
SOLUBILITY	Soluble in water

PEG 40
HYDROGENATED CASTOR OIL

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Yellow or brown
ODOR	A characterisitic odor.
AID MEASURES	It causes an irritation slightly.
STORAGE	Keep cool and dry place
APPLICATIONS	PEG 40 Hydrogenated Castor Oil, with its own peculiar smell, it is a solid cake or oily ingredient. PEG 40 Hydrogenated Castor Oil, in many industrial products, especially detergents and emulsifiers used in particular in the oil to emulsify.

PEG 40 STEARATE

GENERAL DEFINITIONS	
CAS NUMBER	9004 – 99 – 3
GENERAL CATEGORY	Cosmetic and detergent ingredient
FACTORIAL CATEGORY	emulsifier
TRADE NAME	Polyoxyethylene (40) stearate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Not specified
MOLECULAR WEIGHT	Not specified
MELTING POINT (C)	35 – 40
BOILING POINT (C)	212
FLASH POINT (C)	268
DENSITY	1.07 g / cm3
SOLUBILITY	Soluble in water

PEG 40 STEARATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder or granul
COLOR	White or white off
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	PEG 40 Stearate, density 1.070 g / cm 3 which the solid fat form, it is a white oily solid, and has a characteristic smell of ingredient. PEG 40 Stearate, quite cosmetic and industrial sectors. Good emulsifier. To this end, hair care products, hair şeklillendiric in the production of products used in the hand and facial creams and lotions.

PEG 75
ETOXYLATED LANOLIN

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Cosmetic ingredient
FACTORIAL CATEGORY	Emulsifier and wetting agent
TRADE NAME	Peg 75 Etoxylated lanolin

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Not specified
MOLECULAR WEIGHT	Not specified
PH RANGE	3.5 – 8
BOILING POINT (C)	Not specified
FLASH POINT (C)	Not specified
DENSITY	1.10 g / cm3
SOLUBILITY	Partially soluble in water

PEG 75
ETOXYLATED LANOLIN

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Yellow or amber
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation slightly.
STORAGE	Keep cool and dry place
APPLICATIONS	It can be used in clear liquid cosmetics and toiletries where viscosity must be controlled, such as in aftershaves and cleansing lotions. It minimises the defatting effect of detergents on skin and hair without impairing their cleansing ability. Ethoxylated lanolin derivatives are potent solubilisers of insoluble materials such as lanolin, lanolin oil and hydrocarbons. It is also used in stable low-viscosity micro emulsions of insoluble pharmacological compounds for spraying onto wounds or burns.

PHOSPHORIC ACID

GENERAL DEFINITIONS

CAS NUMBER	7664 – 38 – 2 + 7732 – 18 – 5
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Inorganic acid
TRADE NAME	Phosphoric acid

CHEMICAL PROPERTIES

CHEMICAL FORMULA	H_3PO_4
MOLECULAR WEIGHT	Mixture
PH RANGE	Acidic
BOILING POINT (C)	158
MELTING POINT (C)	21
DENSITY	1.10 g / cm3
SOLUBILITY	Soluble in water completely.

PHOSPHORIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Phosphoric acid, also known as orthophosphoric acid, is one of the most widely known and used acids. It is a mineral acid that contains three atoms of hydrogen, one atom of phosphorus, and four atoms of oxygen. Thus, the chemical formula of phosphoric acid is H3PO4. This is an important acid which has a wide range of uses. In the following sections, we will find out what all phosphoric acid is used for.

POLYSORBATE 20

GENERAL DEFINITIONS	
CAS NUMBER	9005 – 64 – 5
GENERAL CATEGORY	Detergent, food and cosmetic ingredient
FACTORIAL CATEGORY	Emulsifier
TRADE NAME	Polysorbate 20

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	C58H114O26
MOLECULAR WEIGHT	1227.77
PH RANGE	Not specified
BOILING POINT (C)	100
MELTING POINT (C)	Not specified
DENSITY	1.10 g / cm3
SOLUBILITY	Soluble in water.

POLYSORBATE 20

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Yellow
ODOR	A characteristic odor.
AID MEASURES	It causes no irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Polysorbate 20, cosmetic, especially food industry is used as a detergent and emulsifier in many industrial fields. Polysorbate 20, as in all the emulsifier, the water phase or the oil forming the oil phase and the water are emulsified additives in providing these products. in food, mayonnaise, sauces, oils in the cream, the flavors, and beer fikset as used in different proportions in the production of products such as malt.

POLYQUATERNIUM 7

GENERAL DEFINITIONS	
CAS NUMBER	Mixture
GENERAL CATEGORY	Detergent and cosmetic ingredient
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Polyquaternium 7

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	Not specified
BOILING POINT (C)	100
MELTING POINT (C)	Not specified
DENSITY	1.02 g / cm3
SOLUBILITY	Soluble in water.

POLYQUATERNIUM 7

PHYSICAL PROPERTIES AND USES

APPEARANCE	Viscous liquid
COLOR	Yellow
ODOR	Odorless.
AID MEASURES	It causes no irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Polyquaternium-7** is a clear, colorless to pale yellow liquid. In cosmetics and personal care products, it is used in a wide variety of products including shampoos, hair conditioners and other hair products, bath soaps and cleansing products. **Polyquaternium-7** prevents or inhibits the buildup of static electricity and dries to form a thin coating that is absorbed onto the hair shaft. Polyquaternium-7 also helps hair hold its style by inhibiting the hair's ability to absorb moisture.

POTASSIUM ACETATE

GENERAL DEFINITIONS

CAS NUMBER	127 – 8 - 2
GENERAL CATEGORY	Food ingredient
FACTORIAL CATEGORY	Potassium acetate salt
TRADE NAME	Potassium acetate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$C_2H_3KO_2$
MOLECULAR WEIGHT	98.14
PH RANGE	7 – 9
BOILING POINT (C)	Not specified
MELTING POINT (C)	292
DENSITY	1.60 g / cm3
SOLUBILITY	Partially soluble in water.

POTASSIUM ACETATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Crystal
COLOR	White
ODOR	Odorless.
AID MEASURES	It causes no irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Potassium acetate is a food preservative which is a white crystalline color and slight characteristic odor. Potassium acetate is known in the food industry with E 261 food code. PH range of 7-9 and having a melting point of 292 C., which is a water-soluble food preservatives comfortable. Potassium acetate, is used as a food preservative as well as an acidity regulator. Potassium acetate is also used in antifreeze and de-icing agent in detergent and industrial areas.

POTASSIUM HYDROXIDE

GENERAL DEFINITIONS

CAS NUMBER	1310 – 58 – 3
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Caustic potash
TRADE NAME	Potassium hydroxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	KOH
MOLECULAR WEIGHT	56.11
PH RANGE	13
BOILING POINT (C)	1384
MELTING POINT (C)	380
DENSITY	2.050 g / cm3
SOLUBILITY	Soluble in water.

POTASSIUM HYDROXIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or payet
COLOR	White
ODOR	Odorless.
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Potassium hydroxide, chemical compound with formula KOH. Pure potassium hydroxide forms white, deliquescent crystals. For commercial and laboratory use it is usually in the form of white pellets. A strong base, it dissolves readily in water, giving off much heat and forming a strongly alkaline, caustic solution It is commonly called caustic potash. It closely resembles sodium hydroxide in its chemical properties and has similar uses, e.g., in making soap, in bleaching, and in manufacturing chemicals, but is less widely used because of its higher cost.

POTASSIUM CHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	7447 – 40 – 7
GENERAL CATEGORY	Industrial ingredient
FACTORIAL CATEGORY	Potassium salt
TRADE NAME	Potassium chloride

CHEMICAL PROPERTIES

CHEMICAL FORMULA	KCI
MOLECULAR WEIGHT	74.55
PH RANGE	Not specified
BOILING POINT (C)	1420
MELTING POINT (C)	770
DENSITY	1.98 g / cm3
SOLUBILITY	Soluble in water.

POTASSIUM CHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or powder
COLOR	White
ODOR	Odorless.
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	**Drilling** water-based drilling mud potassium chloride at-drilling, used as stabilizers. It can also be used in oil production. **Electrolysis coating:** the acid bath, when provided with high voltage potassium chloride, can be done in less time than coating. Azo / textile paint on; potassium chloride is used together with sulfonic acids. **Heat treatment:** Soldering / for blasting and surface treatment materials that can not be used, heat treatment processes (especially aluminum sheet) **In metallurgy:** surface treatment, galvanizing, casting, smelting used. Furthermore, aluminum waste potassium chloride and sodium chloride (salt) is dissolved using so unnecessary waste removed. Aluminum waste becomes high in quality as a result.

POTASSIUM CHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	7778 – 80 – 5
GENERAL CATEGORY	Industrial and agriculture ingredient
E FOOD CODE	E 515
TRADE NAME	Potassium sulfate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	K_2SO_4
MOLECULAR WEIGHT	174.25
PH RANGE	13 – 13.5
BOILING POINT (C)	1689
MELTING POINT (C)	1067
DENSITY	2.66 g / cm3
SOLUBILITY	Partially soluble in water.

POTASSIUM SULFATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Crystal
COLOR	White
ODOR	Odorless.
AID MEASURES	It causes no irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Potassium Sulphate, most potatoes, tobacco, vegetables, fruit, citrus fruits, beans, corn, cotton, sugar beet and used in greenhouses. the leaves of the plant is not given enough potassium sulfate yellow or reddish brown spots appear. In irrigated agriculture and in rainy regions are common to potassium deficiency. poor in organic matter, potassium sulfate should be given the sandy soil. Acid is featured for many years in the soil acidity increases if used continuously. When used in the acid soil to be administered in an amount of lime to the soil. Also meets the needs of sulfur within the plant is found in sulfur.

POTASSIUM TRIPOLYPHOSPHATE

GENERAL DEFINITIONS

CAS NUMBER	Mixture
GENERAL CATEGORY	Industrial and detergent ingredient
FACTORIAL CATEGORY	Sequestering
TRADE NAME	Potassium tripolyphosphate (liquid)

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
PH RANGE	9 - 10
BOILING POINT (C)	1689
MELTING POINT (C)	630
DENSITY	1.10 g / cm3
SOLUBILITY	Soluble in water.

POTASSIUM TRIPOLYPHOSPHATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	Odorless.
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Potassium tripolyphosphate, no odor is a material transparent and colorless liquid. Potassium tripolyphosphate, including in particular detergent, ions are used as raw materials in industrial cleaning helps holder and many other areas. It also finds use in the field because of the water softening properties.

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SODIUM
ALPHA OLEFIN SULFONATE

GENERAL DEFINITIONS	
CAS NUMBER	68439 – 57 – 6 + 7732 – 18 – 5
GENERAL CATEGORY	Detergent and cosmetic ingredients
FACTORIAL CATEGORY	Surfactant
TRADE NAME	Sodium alpha olefin sulfonate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	Not specified
PH RANGE	5 – 6
BOILING POINT (C)	100
DENSITY	0.85 g / cm3
SOLUBILITY	Soluble in water

SODIUM
ALPHA OLEFIN SULFONATE

PHYSICAL PROPERTIES
AND
USES

APPEARANCE	Liquid
COLOR	Clear and colorless
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium alpha olefin sulfonate, yellow, light dense form and has a smell that its liquid anionic surfactant is a ingredient. Sodium alpha olefin sulfonate, foaming and cleaning power than that of a surfactant find use in many cosmetic and detergent products. Also gets a stable foam is a highly effective ingredient.

SODIUM HEGZAMETAPHOSPHATE

GENERAL DEFINITIONS

CAS NUMBER	10124 – 56 – 8
GENERAL CATEGORY	Detergent and industrial ingredients
FACTORIAL CATEGORY	sequestering
TRADE NAME	Sodium hegzametaphosphate (SHMP)

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Na6P6O18
MOLECULAR WEIGHT	611.77
MELTING POINT (C)	628
PH RANGE	Not specified
BOILING POINT (C)	1500
DENSITY	2.48 g / cm3
SOLUBILITY	Soluble in water

SODIUM HEGZAMETAPHOSPHATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	SHMP, powder or granular form and does not smell, is a solid raw material in white. SHMP, including in particular detergent, ions are used as raw materials in industrial cleaning helps holder and many other areas. It also finds use in the field because of the water softening properties.

SODIUM HYDROXIDE

GENERAL DEFINITIONS

CAS NUMBER	1310 – 73 – 2
GENERAL CATEGORY	Detergent and industrial ingredients
FACTORIAL CATEGORY	Strong alkaline
TRADE NAME	Sodium hydroxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	NaOH
MOLECULAR WEIGHT	39.99
MELTING POINT (C)	318
PH RANGE	14
BOILING POINT (C)	1390
DENSITY	2.100 g / cm3
SOLUBILITY	Soluble in water

SODIUM HYDROXIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Payet
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium hydroxide, sequins and beads in the form of a solid, odorless and white in an alkaline materials. Sodium hydroxide, including in particular detergent, paper industry, textile, chemical, used in car maintenance products and pool chemicals. usually uses in adjusting pH is alkaline product is quite a lot.

SODIUM HYPOCHLORIDE

GENERAL DEFINITIONS

CAS NUMBER	Mixture
GENERAL CATEGORY	Detergent and industrial ingredients
FACTORIAL CATEGORY	Bleaching agent
TRADE NAME	Sodium hypochloride

CHEMICAL PROPERTIES

CHEMICAL FORMULA	NaOCI
MOLECULAR WEIGHT	74.44
MELTING POINT (C)	-25
PH RANGE	11
BOILING POINT (C)	111
DENSITY	1.10 g / cm3
SOLUBILITY	Soluble in water

SODIUM HYPOCHLORIDE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Clear and yellow slightly
ODOR	A characteristic chlorine odor
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium hypochlorite, offers on-site chlorine and chlorine odor judge this reason, it is a ingredient in the form of liquid, transparent and pale yellow. Sodium Hypochlorite, to get detergent primarily due to chlorine bleach and most active feature in production, paper industry, textile and used in many industrial fields as chlorine bleach.

SODIUM CARBONATE

GENERAL DEFINITIONS	
CAS NUMBER	497 – 19 – 8
GENERAL CATEGORY	Detergent and industrial ingredients
FACTORIAL CATEGORY	Strong base agent
TRADE NAME	Sodium carbonate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$NaCO_3$
MOLECULAR WEIGHT	105.99
MELTING POINT (C)	851
PH RANGE	11 – 12
BOILING POINT (C)	Not specified
DENSITY	2.54 g / cm3
SOLUBILITY	Soluble in water

SODIUM CARBONATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Especially for the bleaching of linen cloth. Sodium carbonate ions into carbonate hardness that water away from the precipitating environment. In this way it is used as softener in the washing machine. The most important chemical used in glass production. Soda Blasting is raised to very high combined heat and cooled suddenly and glass is produced. The reactive dyes used in the textile industry are used to create the bond between dye and fiber, sodium carbonate. Food additive as acidity regulators, anti-caking agent and acts as stabilizer. Sherbet powder used in the production. Acts as a wetting agent in brick-making, in this way the need for less water while the extruded clay.

SODIUM LAURYL ETHER SULFATE

GENERAL DEFINITIONS	
CAS NUMBER	68585 – 34 – 2 + 7732 – 18 – 5
GENERAL CATEGORY	Detergent and cosmetic ingredients
FACTORIAL CATEGORY	Anionic surfactant
TRADE NAME	SLES

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Mixture
MOLECULAR WEIGHT	Mixture
MELTING POINT (C)	10 – 15
PH RANGE	7 – 7.5
BOILING POINT (C)	Not specified
DENSITY	1.05 g / cm3
SOLUBILITY	Soluble in water

SODIUM LAURYL ETHER SULFATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Gel
COLOR	Colorless pale yellow
ODOR	A characteristic odor.
AID MEASURES	It causes an irritation slightly. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Especially SLES detergent, shampoo, cosmetics, and in particular production, it finds application in the production of highly shower gel. Anionic surface-active agent in auto care products to be used for both the cleaning active as foaming agent. SLES, including in the textile and adhesives industries, the construction of drilling foam, foam concrete production and used in many industrial fields.

SODIUM LAURYL SULFATE

GENERAL DEFINITIONS	
CAS NUMBER	151 – 21 – 3
GENERAL CATEGORY	Detergent and cosmetic ingredients
FACTORIAL CATEGORY	Anionic surfactant
TRADE NAME	SLS

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$CH_3(CH_2)_{10}CH_2OOSOONa$
MOLECULAR WEIGHT	288.38
MELTING POINT (C)	204
PH RANGE	6 – 7
BOILING POINT (C)	Not specified
DENSITY	1.050 g / cm3
SOLUBILITY	Soluble in water

SODIUM LAURYL SULFATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	A characteristic odor
AID MEASURES	It causes no irritation
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium lauryl sulfate (SLS), the solid form is an anionic surfactant with a distinctive odor. Sodium lauryl ether sulphate (SLS), including especially cosmetics, detergents and cleaning products such as foam and is used in many car care products. both active in terms of being solid to the fore feature both high and foam.

SODIUM METASILICATE

GENERAL DEFINITIONS

CAS NUMBER	6834 – 92 – 0
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Builder agent
TRADE NAME	Sodium metasilicate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	Na2O3Si
MOLECULAR WEIGHT	122.06
MELTING POINT (C)	1090
PH RANGE	12 – 13
BOILING POINT (C)	Not specified
DENSITY	2.61 g / cm3
SOLUBILITY	Soluble in water

SODIUM METASILICATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium metasilicate is not a cleaning agent by itself. It is a VERY strong base. The pH of a 1% aqueous solution is about 13. In moist air it is corrosive to metals like zinc, aluminium, tin and lead, forming hydrogen gas. As a strong base, it reacts violently with acids. A major use is as a builder (a material that enhances or maintains the cleaning efficiency of a surfactant, principally by inactivating water hardness) in soaps and detergents. It may be used as part of a component in a chemical degreaser, where it would react with fatty acids(animal grease) to form a soap, which is then rinsed away. Sodium silicate, on the other hand, is an abrasive white powder frequently used in things like toothpaste. It doesn't clean by itself either.

SODIUM PERCARBONATE

GENERAL DEFINITIONS

CAS NUMBER	15630 – 89 – 4
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Oxygene bleaching agent.
TRADE NAME	Sodium percarbonate

CHEMICAL PROPERTIES

CHEMICAL FORMULA	$CNa_2O_3 \cdot 1.5H_2O_2$
MOLECULAR WEIGHT	157.01
MELTING POINT (C)	Not specified
PH RANGE	10 – 11
BOILING POINT (C)	Not specified
DENSITY	0.95 g / cm3
SOLUBILITY	Soluble in water

SODIUM PERCARBONATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder or granular
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash more water.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium percarbonate is a white crystalline or granular raw material form that does not smell. Sodium percarbonate is used as the main raw material in the production of filler in the powder detergent and bleach products. As bleaching paper and textile industry also finds the use of space.

SODIUM SULFATE

GENERAL DEFINITIONS	
CAS NUMBER	7757 – 82 – 6
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Oxidizing or reducing agents
TRADE NAME	Sodium sulfate

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	Na_2SO_4
MOLECULAR WEIGHT	142.04
MELTING POINT (C)	880
PH RANGE	7 – 8
BOILING POINT (C)	1100
DENSITY	2.670 g / cm3
SOLUBILITY	Soluble in water

SODIUM SULFATE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder or granular
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Sodium sulphate, is the most important use of the detergent industry is. Sodium sulfate, is included as filler in the formulation of powder and granular solid detergent. Detergents contain an average of about 16% sodium sulfate, but this rate is up to 40'y% in some formulas. Sodium sulfate used in the production rate of the detergent, affects the cost of the detergent and bleaching degrees. Natural sodium sulfate has an ideal filling function does not mold is inert and inexpensive. Diluents and provides good flow characteristics, colloidal properties of the cleaned soil will increase the effectiveness of detergent making stable. It also allows the realization of the best washing up the micelles by reducing the power of the critical concentration.

SODIUM TRIPOLYPHOSPHATE

GENERAL DEFINITIONS	
CAS NUMBER	7758 – 29 – 4
GENERAL CATEGORY	Industrial detergent ingredients
FACTORIAL CATEGORY	sequestrant and anticoagulant agent
TRADE NAME	Sodium tripolyphosphate (STPP)

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$Na_5P_3O_{10}$
MOLECULAR WEIGHT	367.85
MELTING POINT (C)	622
PH RANGE	7 – 8
BOILING POINT (C)	Not specified
DENSITY	2.52 g / cm3
SOLUBILITY	Soluble in water

SODIUM TRIPOLYPHOSPHATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	granular
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	STPP, powder or granular form and does not smell, is a solid raw material in white. STPP, including in particular detergent, ions are used as raw materials in industrial cleaning helps holder and many other areas. STPP, breaking from the hardness of the water and dirt remover laundry, there is binding feature of the site. STPP prevalent regular and compact laundry detergent (powder, liquid, gel and tablets), automatic dishwashing detergents (powders, liquids, gels and tablets) are used in toilet cleaners and surface cleaners. STPP is used in industrial and institutional cleaners. STPP also a water softener, as food preservatives, as peptizing agents, oil wells and cotton decoction is used as a clotting agent disrupting the release agent.

STEARIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	57 – 11 – 4
GENERAL CATEGORY	Industrial and cosmetic ingredients
FACTORIAL CATEGORY	Emulsifier and thickener agent
TRADE NAME	Stearic acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_{18}H_{36}O_2$
MOLECULAR WEIGHT	284.50
MELTING POINT (C)	69.4
FLASH POINT (C)	196
BOILING POINT (C)	350
DENSITY	0.950 g / cm3
SOLUBILITY	Insoluble in water. Soluble in diethyl ether and acetone

STEARIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Crystal or powder
COLOR	White or yellow
ODOR	Acharacteristic odor.
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Stearic acid, which is a slightly peculiar smell, raw white or pale yellow solid granular or powder form. Stearic acid, used as emulsifiers in cosmetics manufacture and consistency enhancer. Also in industrial areas and in particular the accelerator and softening agent in rubber production, as a hardening agent in the production of candles, as a raw material in the textile and help in curing agent is used as raw material in soap.

SULFAMIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	5329 – 14 – 6
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Cleaning agent
TRADE NAME	Sulfamic acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	H_3NSO_3
MOLECULAR WEIGHT	97.10
MELTING POINT (C)	205
PH RANGE	1 – 2
BOILING POINT (C)	Not specified
DENSITY	2.15 g / cm3
SOLUBILITY	Soluble in water

SULFAMIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes an irritation. In contact, wash plenty water.
STORAGE	Keep cool and dry place
APPLICATIONS	Sulfamic acid, non-odor, a solid raw material is white and crystalline format. Sulfamic acid industry rust remover is a material used for cleaning and descaling metal surfaces. Furthermore, the dye and will find use in the production of pigments. In the paper industry it is used as a chlorine stabilizer.

SULPHURIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	7664 – 93 – 9
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Strong acid
TRADE NAME	Sulphuric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	H_2SO_4
MOLECULAR WEIGHT	98.07
MELTING POINT (C)	10
PH RANGE	1
BOILING POINT (C)	337
DENSITY	1.84 g / cm3
SOLUBILITY	Soluble in water

SULPHURIC ACID

PHYSICAL PROPERTIES AND USES

APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	Odorless
AID MEASURES	In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Cover the irritated skin with an emollient. Cold water may be used. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.
STORAGE	Keep cool and dry place
APPLICATIONS	Sulfuric acid, water attracts many organic substances and form strong exothermic reactions. sulfuric acid dilution job to do this is very difficult and dangerous. Among the people known as battery acid sulfuric acid. Batteries and battery used in construction.

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TARTARIC ACID

GENERAL DEFINITIONS	
CAS NUMBER	526 – 83 – 0
GENERAL CATEGORY	Food and industrial ingredients
FACTORIAL CATEGORY	Organic acid
TRADE NAME	Tartaric acid

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_4H_4O_6$
MOLECULAR WEIGHT	150.085
MELTING POINT (C)	170
PH RANGE	2 – 3
BOILING POINT (C)	Not specified
DENSITY	1.790 g / cm3
SOLUBILITY	Soluble in water

TARTARIC ACID

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Tartaric acid, tartaric acid used in various food products such as confectionery products and beverages; The main grape in finding applications in a delicious beverage, grape jam and serves as tasty tart sweet confections acidity regulators with marmalade. In sour hard candy apples are used together with citric acid to give tart cherry and other flavors. However, in many applications, such as citric acid and tartaric acid show a swelling effect, which is monopotassium salt ingrediyent of cream of tartar it is used as baking powder. These emulsifiers in bakery outside use and protecting mission-tartaric acid, as excipients or carriers in the pharmaceutical industry, for cleaning and polishing metals finds silvered and application as chelating agents of metal color here in mirrors.

TITANIUM DIOXIDE

GENERAL DEFINITIONS

CAS NUMBER	13463 – 67 – 7
GENERAL CATEGORY	Food and industrial ingredients
FACTORIAL CATEGORY	Color white agent
TRADE NAME	Titanium dioxide

CHEMICAL PROPERTIES

CHEMICAL FORMULA	TiO_2
MOLECULAR WEIGHT	79.886
MELTING POINT (C)	1843
PH RANGE	Not specified
BOILING POINT (C)	2972
DENSITY	4.30 g / cm3
SOLUBILITY	Insoluble in water. Disperable in water.

TITANIUM DIOXIDE

PHYSICAL PROPERTIES AND USES

APPEARANCE	Powder
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Titanium dioxide synthetic food dyes in food coloring that class food products allowed in the codex are used in limited amounts. Ice cream production, icy drinks, necessary in sodas, fruit soda is used in production. Also in the production of confectionery products, jellies, flavored drinks, chewing gum varieties in sugar and low-sugar, bakery products, confectionery etc. it is used in many food production. Food except pharmaceuticals, cosmetics, detergent products, car care products, in a small amount of agricultural products, cleaning products, colored stones, play dough, etc. are also used for coloring many people in contact with the product.

TETRAPOTASSIUM PYROPHOSPHATE

GENERAL DEFINITIONS	
CAS NUMBER	7320 – 34 – 5
GENERAL CATEGORY	Industrial ingredients
FACTORIAL CATEGORY	Sequestering agent
TRADE NAME	Tetrakal

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$K_4O_7P_2$
MOLECULAR WEIGHT	330.34
MELTING POINT (C)	1090
PH RANGE	Not specified
BOILING POINT (C)	Not specified
DENSITY	2.35 g / cm3
SOLUBILITY	Partially soluble in water.

TETRAPOTASSIUM PYROPHOSPHATE

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Powder or crystal
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Tetrakal, no smell, the white color is a solid raw crystal or powder form. Tetrakal, several industrial areas and especially in cleaning products, including detergents in particular, holders used for ion and water softeners. Also find use in some fertilizer phosphate to meet the need in the agricultural field.

TRIETHANOLAMINE
TEA

GENERAL DEFINITIONS	
CAS NUMBER	102 – 71 – 6
GENERAL CATEGORY	Industrial and detergent ingredients
FACTORIAL CATEGORY	Alkaline and Ph regulator
TRADE NAME	Tea

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	$C_6H_{15}NO_3$
MOLECULAR WEIGHT	149.18
MELTING POINT (C)	21
PH RANGE	12 – 13
BOILING POINT (C)	338
DENSITY	1.125 g / cm3
SOLUBILITY	Soluble in water.

TRIETHANOLAMINE
TEA

PHYSICAL PROPERTIES AND USES	
APPEARANCE	Liquid
COLOR	Colorless and clear
ODOR	Odorless
AID MEASURES	It causes an irritation. Case of contact, wash plenty water.
STORAGE	Keep cool and dry place
APPLICATIONS	Triethanolamine (TEA), with its distinctive odor, viscous liquid form, transparent, colorless or slightly yellow color is a raw material. Triethanolamine (TEA), to adjust PH in many industrial products, especially detergents and cleaning. Especially in the production of liquid detergent used in many oil and dirt solve solve formulas. It is also used in some chemical raw materials into a water-based solvent products in the agricultural sector.

DETERGENT FORMULATIONS ENCYCLOPEDIA

SOLVER

CHEM

PUBLICATIONS

SOLVER CHEM PUBLICATIONS

DETERGENT FORMULATIONS ENCYCLOPEDIA

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DETERGENT CHEMICALS PROPERTIES

SOLVER CHEM PUBLICATIONS

UREA

GENERAL DEFINITIONS	
CAS NUMBER	57 – 13 – 6
GENERAL CATEGORY	Agricultural and industrial ingredients
FACTORIAL CATEGORY	Nitrogen source for fertilizers
TRADE NAME	Urea

CHEMICAL PROPERTIES	
CHEMICAL FORMULA	CH_4N_2O
MOLECULAR WEIGHT	60.06
MELTING POINT (C)	132
PH RANGE	6 – 7
BOILING POINT (C)	337
DENSITY	1.325 g / cm3
SOLUBILITY	Soluble in water

UREA

PHYSICAL PROPERTIES AND USES

APPEARANCE	Granular
COLOR	White
ODOR	Odorless
AID MEASURES	It causes no irritation.
STORAGE	Keep cool and dry place
APPLICATIONS	Urea, no smell, the white color is a solid raw crystal or granular form. Urea, detergent sector and industry the main raw material used as raw materials or help. Urea is used in making up the agricultural field, and urea formaldehyde resins.